Vector Calculus

Calc 2 → Vector Calc: carry-over cheat sheet

A consolidated lookup of every antiderivative from Calc 2 — one place to glance while the new material piles on. Every result carries a $+C$; the lone $\frac{1}{a}$ out front and the ln-flag are the usual misses.

Master integral table

Power & log

$$\int 1\,dx = x+C$$
$$\int x^n\,dx = \tfrac{x^{n+1}}{n+1}+C \quad (n \neq -1)$$
$$\int \tfrac{1}{x}\,dx = \ln|x|+C$$

Exponential

$$\int e^x\,dx = e^x+C$$
$$\int a^x\,dx = \tfrac{a^x}{\ln a}+C$$
$$\int e^{kx}\,dx = \tfrac{1}{k}e^{kx}+C$$

Trig

$$\int \sin x\,dx = -\cos x+C$$
$$\int \cos x\,dx = \sin x+C$$
$$\int \sec^2 x\,dx = \tan x+C$$
$$\int \csc^2 x\,dx = -\cot x+C$$
$$\int \sec x\tan x\,dx = \sec x+C$$
$$\int \csc x\cot x\,dx = -\csc x+C$$
$$\int \tan x\,dx = \ln|\sec x|+C$$
$$\int \sec x\,dx = \ln|\sec x+\tan x|+C$$

Inverse trig

$$\int \tfrac{1}{1+x^2}\,dx = \arctan x+C$$
$$\int \tfrac{1}{\sqrt{1-x^2}}\,dx = \arcsin x+C$$
$$\int \tfrac{1}{a^2+x^2}\,dx = \tfrac{1}{a}\arctan\tfrac{x}{a}+C$$
$$\int \tfrac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\tfrac{x}{a}+C$$

Tools you carry into vectors

- **Distance / Pythagoras**: $d=\sqrt{(\Delta x)^2+(\Delta y)^2}$ in 2D, add $+(\Delta z)^2$ in 3D. This *is* the magnitude formula in disguise.
- **Law of cosines**: $c^2=a^2+b^2-2ab\cos\theta$ — the seed of the dot-product angle formula.
- **Cosine quick values**: $\cos 0=1$, $\cos 30°=\frac{\sqrt{3}}{2}$, $\cos 45°=\frac{\sqrt{2}}{2}$, $\cos 60°=\frac{1}{2}$, $\cos 90°=0$.
- **Simplify roots**: $\sqrt{a^2b}=a\sqrt{b}$; keep magnitudes exact ($\sqrt{20}=2\sqrt{5}$) unless a decimal is asked.
- **Unit preview**: dividing any vector by its length gives a unit vector (length 1) — used constantly below.

vector basics (2D & 3D)

A **vector** has magnitude (length) and direction; a **scalar** is just a number. Written in **component form** $\vec{v}=\langle a,b\rangle$ (2D) or $\langle a,b,c\rangle$ (3D), or in **standard-basis form** $a\mathbf{i}+b\mathbf{j}$.

From two points (displacement)

The vector from $P=(x_1,y_1)$ to $Q=(x_2,y_2)$ is tip − tail:

$$\vec{PQ}=\langle x_2-x_1,\ y_2-y_1\rangle$$

A **position vector** is the special case from the origin: $\vec{OP}=\langle x_1,y_1\rangle$.

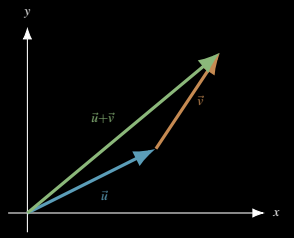
Example: $P(1,4)$, $Q(5,1)$: $\vec{PQ}=\langle 5-1,\ 1-4\rangle=\langle 4,-3\rangle$ — tip minus tail; reversing the order flips the vector.

Magnitude (length)

$\|\vec{v}\|=\sqrt{a^2+b^2}$ in 2D, $\sqrt{a^2+b^2+c^2}$ in 3D. It is the Pythagorean distance from tail to tip — never negative.

Algebra: scale, add, subtract

- **Scalar mult.** $k\vec{v}=\langle ka,kb\rangle$: stretches length by $|k|$; $k<0$ flips direction.
- **Add / subtract**: componentwise, $\vec{u}\pm\vec{v}=\langle u_1\pm v_1,\ u_2\pm v_2\rangle$.
- **Geometrically**: $\vec{u}+\vec{v}$ is tip-to-tail; $\vec{u}-\vec{v}$ points from $\vec{v}$'s tip to $\vec{u}$'s tip.



Example: $\vec{u}=\langle 1,2\rangle$, $\vec{v}=\langle 3,-1\rangle$: $2\vec{u}-3\vec{v}=\langle 2-9,\ 4+3\rangle=\langle -7,7\rangle$ — scale each first, then combine componentwise.

Unit vectors

A **unit vector** has length 1. Normalize any $\vec{v}$ by dividing by its length:

$$\hat{v}=\frac{\vec{v}}{\|\vec{v}\|}$$

The basis vectors $\mathbf{i}=\langle 1,0\rangle$, $\mathbf{j}=\langle 0,1\rangle$ are the unit vectors along the axes.

Worked: normalize $\vec{u}=\langle 3,-4\rangle$, then rescale

1. MAGNITUDE
$\|\vec{u}\|=\sqrt{9+16}=5$

↓

2. DIVIDE BY THE LENGTH
$\hat{u}=\frac{1}{5}\langle 3,-4\rangle=\langle\frac{3}{5},-\frac{4}{5}\rangle$; check: $\|\hat{u}\|=1$ ✓

↓

3. ANY LENGTH IN THAT DIRECTION
length 10 along $\vec{u}$: $10\hat{u}=\langle 6,-8\rangle$; opposite direction: $-\hat{u}$ *(normalize, then scale — the all-purpose move)*

Magnitude & angle → components

Length $\|\vec{v}\|$ at angle θ from the $+x$-axis decomposes as

$$\vec{v}=\langle\|\vec{v}\|\cos\theta,\ \|\vec{v}\|\sin\theta\rangle$$

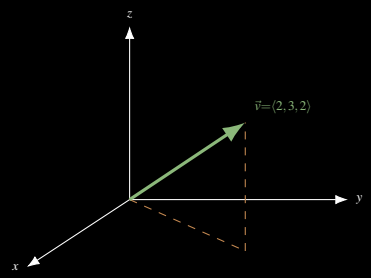
Example: length 20 at 60°: $\langle 20\cos 60°,\ 20\sin 60°\rangle=\langle 10,\ 10\sqrt{3}\rangle$ — the decomposition projectile motion leans on later.

3D vectors

Add a third axis z perpendicular to the xy-plane by the right-hand rule (fingers curl $x\to y$, thumb points $+z$). Points are (x,y,z); the axes split space into 8 **octants**.

Core formulas

- **Component form**: $\vec{v}=\langle a,b,c\rangle=a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$.
- **Magnitude**: $\|\vec{v}\|=\sqrt{a^2+b^2+c^2}$.
- **Distance** $P_1\to P_2$: $\sqrt{(\Delta x)^2+(\Delta y)^2+(\Delta z)^2}$.
- **Midpoint**: $\left(\frac{x_1+x_2}{2},\ \frac{y_1+y_2}{2},\ \frac{z_1+z_2}{2}\right)$ — average each coordinate.
- **Unit vector**: $\hat{v}=\vec{v}/\|\vec{v}\|$ — identical to 2D, just one more component.
- **Sphere**, center (h,k,l), radius r: $(x-h)^2+(y-k)^2+(z-l)^2=r^2$ — the distance formula, squared. Complete the square to find center/radius from an expanded form.
- **Coordinate planes**: $z=0$ is the xy-plane, $x=0$ the yz-plane, $y=0$ the xz-plane. A missing variable in an equation means that variable is unrestricted — $z=3$ is a whole horizontal plane, not a point.



Worked: $\vec{v}=\langle 1,-2,2\rangle$

1. MAGNITUDE
$\|\vec{v}\|=\sqrt{1+4+4}=3$

↓

2. NORMALIZE
$\hat{v}=\frac{1}{3}\langle 1,-2,2\rangle=\langle \frac{1}{3},-\frac{2}{3},\frac{2}{3}\rangle$

Same formula, different inputs — distance $(1,0,2)\to(4,4,4)$: $\sqrt{3^2+4^2+2^2}=\sqrt{29}$.

Worked: sphere from an expanded equation

$x^2+y^2+z^2-4x+6y-2z=11$:

1. GROUP BY VARIABLE
$(x^2-4x)+(y^2+6y)+(z^2-2z)=11$

↓

2. COMPLETE EACH SQUARE
$(x-2)^2-4+(y+3)^2-9+(z-1)^2-1=11$ *(half the coefficient, squared — then subtract it right back)*

↓

3. READ OFF
$(x-2)^2+(y+3)^2+(z-1)^2=25$: center $(2,-3,1)$, radius 5 *(radius $=\sqrt{25}$, not 25)*

Worked: sphere from a diameter's endpoints

$P(1,2,3)$ and $Q(5,2,-1)$ end a diameter:

1. CENTER = MIDPOINT
$\left(\frac{1+5}{2},\frac{2+2}{2},\frac{3-1}{2}\right)=(3,2,1)$

↓

2. RADIUS = HALF THE DISTANCE
$\|\vec{PQ}\|=\sqrt{16+0+16}=4\sqrt{2}\ \Rightarrow\ r=2\sqrt{2}$

↓

3. WRITE IT
$(x-3)^2+(y-2)^2+(z-1)^2=8$ *($r^2=(2\sqrt{2})^2=8$)*

dot product

Multiplies two vectors into a scalar (a number, not a vector). Two equivalent definitions:

$$\vec{u}\cdot\vec{v}=u_1v_1+u_2v_2+u_3v_3 \qquad \vec{u}\cdot\vec{v}=\|\vec{u}\|\,\|\vec{v}\|\cos\theta$$

The **algebraic** form (left) computes it. The **geometric** form (right) interprets it — θ is the angle between the vectors.

Properties

- **Commutative**: $\vec{u}\cdot\vec{v}=\vec{v}\cdot\vec{u}$.
- **Distributive**: $\vec{u}\cdot(\vec{v}+\vec{w})=\vec{u}\cdot\vec{v}+\vec{u}\cdot\vec{w}$.
- **Scalar**: $(k\vec{u})\cdot\vec{v}=k(\vec{u}\cdot\vec{v})$.
- **Self**: $\vec{v}\cdot\vec{v}=\|\vec{v}\|^2$, so $\|\vec{v}\|=\sqrt{\vec{v}\cdot\vec{v}}$.

Angle between vectors

Solve the geometric form for $\cos\theta$:

$$\cos\theta=\frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\,\|\vec{v}\|} \qquad \theta=\arccos\left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\|\vec{v}\|}\right)$$

The sign of $\vec{u}\cdot\vec{v}$ tells the story: $+$ acute, 0 right angle, $-$ obtuse.

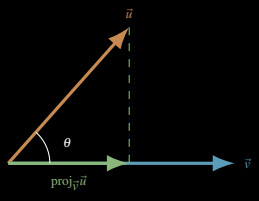
Orthogonality test

$\vec{u}\cdot\vec{v}=0 \iff \vec{u}\perp\vec{v}$ (perpendicular) — the fastest perpendicular check there is, no angle needed. E.g. $\langle 2,3\rangle\cdot\langle 3,-2\rangle=6-6=0$, so those two are perpendicular.

Projections

Drop $\vec{u}$ onto the line of $\vec{v}$:

- **Scalar projection** (a signed length): $\text{comp}_{\vec{v}}\vec{u}=\dfrac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}$.
- **Vector projection** (a vector along $\vec{v}$): $\text{proj}_{\vec{v}}\vec{u}=\dfrac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\,\vec{v}$.



Worked: the full battery — $\vec{u}=\langle 2,3\rangle$, $\vec{v}=\langle 4,-1\rangle$

1. DOT & LENGTHS
$\vec{u}\cdot\vec{v}=8-3=5$ $\quad\|\vec{u}\|=\sqrt{13},\ \|\vec{v}\|=\sqrt{17}$

↓

2. ANGLE
$\cos\theta=\frac{5}{\sqrt{221}}\approx 0.336\Rightarrow\theta\approx 70.3°$; the $+$ sign already said acute — not orthogonal

↓

3. BOTH PROJECTIONS
$\text{comp}_{\vec{v}}\vec{u}=\frac{5}{\sqrt{17}}\approx 1.21$ — a signed length; $\quad\text{proj}_{\vec{v}}\vec{u}=\frac{5}{17}\langle 4,-1\rangle=\langle\frac{20}{17},-\frac{5}{17}\rangle$ — a vector

↓

4. ORTHOGONAL DECOMPOSITION
$\vec{u}-\text{proj}_{\vec{v}}\vec{u}=\langle 2-\frac{20}{17},\ 3+\frac{5}{17}\rangle=\langle\frac{14}{17},\frac{56}{17}\rangle$: $\qquad\vec{u}=\langle\frac{20}{17},-\frac{5}{17}\rangle_{\parallel}+\langle\frac{14}{17},\frac{56}{17}\rangle_{\perp}$ *(check: $\langle 14,56\rangle\cdot\langle 4,-1\rangle=56-56=0$ ✓)*

Solve-for-unknown flavor: find k with $\langle 2,k\rangle\perp\langle 6,4\rangle$: $12+4k=0\Rightarrow k=-3$.

3D changes nothing: $\langle 1,2,2\rangle\cdot\langle 2,-2,1\rangle=2-4+2=0$ — perpendicular, no arccos needed.

Work (the physics application)

A constant force $\vec{F}$ moving an object along displacement $\vec{d}=\vec{PQ}$ does work

$$W=\vec{F}\cdot\vec{d}=\|\vec{F}\|\,\|\vec{d}\|\cos\theta$$

Only the component of force *along the motion* counts — that's the projection idea in disguise. Force $\perp$ motion does zero work.

Worked: 50 N at 30° drags a crate 20 m

1. GEOMETRIC FORM
$W=\|\vec{F}\|\,\|\vec{d}\|\cos\theta=50\cdot 20\cdot\cos 30°$

↓

2. EVALUATE
$1000\cdot\frac{\sqrt{3}}{2}=500\sqrt{3}\approx 866$ J

↓

3. COMPONENT-FORM CHECK
$\vec{F}=\langle 50\cos 30°,\ 50\sin 30°\rangle$, $\quad\vec{d}=\langle 20,0\rangle$: $\quad W=1000\cos 30°$ ✓— only the along-motion component did work

cross product (3D only)

Multiplies two 3D vectors into a new vector that is orthogonal to both inputs. Compare: dot → scalar. Cross → vector. 3D only — there is no 2D cross product (pad a 2D vector with a third component 0 if needed). It exists because "find a vector perpendicular to these two" is the most-reused move in this chapter — planes (next section) run on it.

$$\vec{u}\times\vec{v}=\langle u_2v_3-u_3v_2,\ -(u_1v_3-u_3v_1),\ u_1v_2-u_2v_1\rangle$$

Don't memorize those indices — nobody reliably can. Run the determinant process below every time. It regenerates the formula on demand.

What is it for? — four jobs

- **A** a vector **perpendicular to two others** $\Rightarrow \vec{u}\times\vec{v}$, then divide by its magnitude for a unit normal
- **B** **area** $\Rightarrow$ parallelogram $=|\vec{u}\times\vec{v}|$; triangle $=\frac{1}{2}|\vec{u}\times\vec{v}|$
- **C** **volume** of a parallelepiped $\Rightarrow |\vec{u}\cdot(\vec{v}\times\vec{w})|$ (triple scalar product)
- **D** **torque** $\Rightarrow \vec{r}\times\vec{F}$

All four run the same 3×3 determinant. Only what you do with the result changes. The cross product exists in 3D only — a 2D problem must be lifted with a z-component of 0.

Step 0: the 2×2 determinant (the only memorized piece)

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Criss-cross: down-right product minus up-right product. Every 3×3 reduces to three of these.

Setup: the 3×3 determinant

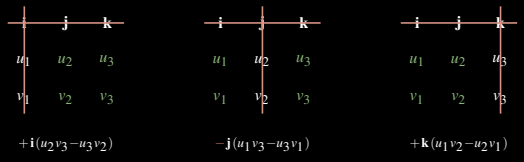
Row 1 = basis symbols, row 2 = the *first* vector, row 3 = the second:

$$\vec{u}\times\vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Row order matters: swapping rows 2 & 3 flips every sign of the answer.

Expand by minors: cross out row + column

Walk the top row. For each of $\mathbf{i},\mathbf{j},\mathbf{k}$: **cross out its row and its column** — the 4 survivors form its 2×2 **minor**. Attach alternating signs $+\,-\,+$: the middle ($\mathbf{j}$) term is subtracted.



$$\vec{u}\times\vec{v} = \mathbf{i}\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \mathbf{j}\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \mathbf{k}\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

Worked: $\langle1,2,3\rangle\times\langle4,5,6\rangle$

1. BUILD THE DETERMINANT (row 2 = first vector)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix}$$

↓

2. THREE MINORS, SIGNS $+-+$

$$\mathbf{i}\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} - \mathbf{j}\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} + \mathbf{k}\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}$$

↓

3. CRISS-CROSS EACH ($ad-bc$)

$\mathbf{i}(12-15) - \mathbf{j}(6-12) + \mathbf{k}(5-8)$

↓

4. RESULT

$-3\mathbf{i}+6\mathbf{j}-3\mathbf{k} = \langle -3,\,6,\,-3\rangle$ *(watch it: $-\mathbf{j}(-6)=+6\mathbf{j}$)*

↓

5. CHECK — dot with both originals

$\langle1,2,3\rangle\cdot\langle-3,6,-3\rangle = -3+12-9 = 0$ ✓
$\langle4,5,6\rangle\cdot\langle-3,6,-3\rangle = -12+30-18 = 0$ ✓

Step 5 is free insurance: the answer *must* dot to 0 against both inputs. If it doesn't, a minor or the middle minus went wrong.

Properties

- Anticommutative: $\vec{v}\times\vec{u} = -(\vec{u}\times\vec{v})$ — order matters, unlike the dot.
- $\vec{u}\times\vec{u} = \vec{0}$, and $\vec{u}\times\vec{0} = \vec{0}$.
- **Parallel test**: $\vec{u}\times\vec{v} = \vec{0} \iff \vec{u} \parallel \vec{v}$ — the cross-product twin of the dot's $\perp$ test.
- **Distributive**: $\vec{u}\times(\vec{v}+\vec{w}) = \vec{u}\times\vec{v} + \vec{u}\times\vec{w}$; scalars pull out: $(k\vec{u})\times\vec{v} = k(\vec{u}\times\vec{v})$.
- Not associative: $\vec{u}\times(\vec{v}\times\vec{w}) \neq (\vec{u}\times\vec{v})\times\vec{w}$ in general.
- $\vec{u}\cdot(\vec{u}\times\vec{v}) = 0$ and $\vec{v}\cdot(\vec{u}\times\vec{v}) = 0$ — orthogonal to both, always (that's the whole point).

Magnitude & direction

$$\|\vec{u}\times\vec{v}\| = \|\vec{u}\|\,\|\vec{v}\|\sin\theta$$

- Compare: dot uses $\cos\theta$, cross uses $\sin\theta$. So parallel vectors ($\theta=0$) give $\vec{0}$; perpendicular ones give the max length $\|\vec{u}\|\|\vec{v}\|$.
- **Right-hand rule**: point fingers along $\vec{u}$, curl toward $\vec{v}$; thumb points along $\vec{u}\times\vec{v}$. (This is why swapping the order flips the sign.)
- Unit vector $\perp$ both: $\dfrac{\vec{u}\times\vec{v}}{\|\vec{u}\times\vec{v}\|}$.

Worked: unit vector $\perp$ both, and the $\sin\theta$ angle

Using the flagship computation above ($\langle1,2,3\rangle\times\langle4,5,6\rangle = \langle-3,6,-3\rangle$):

1. NORMALIZE THE CROSS

$\|\langle-3,6,-3\rangle\| = \sqrt{9+36+9} = \sqrt{54} = 3\sqrt{6}$: $\quad \hat{n} = \dfrac{1}{3\sqrt{6}}\langle-3,6,-3\rangle = \dfrac{1}{\sqrt{6}}\langle-1,2,-1\rangle$

(the 3 factors out of the vector. The other unit normal is $-\hat{n}$)

↓

2. ANGLE FROM THE sin FORM

$\|\vec{u}\| = \sqrt{1+4+9} = \sqrt{14}$, $\quad \|\vec{v}\| = \sqrt{16+25+36} = \sqrt{77}$: $\quad \sin\theta = \dfrac{3\sqrt{6}}{\sqrt{14}\sqrt{77}} \approx 0.224 \Rightarrow \theta \approx 12.9^\circ$

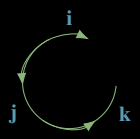
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3. CROSS-CHECK WITH THE DOT

$\vec{u}\cdot\vec{v} = 4+10+18 = 32$: $\quad \cos\theta = \frac{32}{\sqrt{14}\sqrt{77}} \approx 0.975 \Rightarrow 12.9^\circ$ ✓ — two independent routes, one angle

Basis circle

$$\mathbf{i}\times\mathbf{j} = \mathbf{k} \qquad \mathbf{j}\times\mathbf{k} = \mathbf{i} \qquad \mathbf{k}\times\mathbf{i} = \mathbf{j}$$



Going with the arrows: $+$ the next letter. Going against: negate it, for example, $\mathbf{j}\times\mathbf{i} = -\mathbf{k}$.

Areas (parallelogram & triangle)

Vectors $\vec{u},\vec{v}$ as adjacent sides:

$$A_{\text{parallelogram}} = \|\vec{u}\times\vec{v}\| \qquad A_{\text{triangle}} = \tfrac{1}{2}\|\vec{u}\times\vec{v}\|$$

Given points instead of vectors, build two edge vectors from a common corner first: $\vec{u} = Q-P$, $\vec{v} = R-P$. A fourth point adds nothing for a parallelogram — three corners already determine it, so the fourth is a consistency check, not extra data.

Worked: triangle $P(1,0,0)$, $Q(0,1,0)$, $R(0,0,1)$

1. TWO SIDE VECTORS FROM ONE VERTEX

tip − tail, both from P: $\quad \vec{PQ} = \langle-1,1,0\rangle$, $\quad \vec{PR} = \langle-1,0,1\rangle$

↓

2. CROSS THEM

$\vec{PQ}\times\vec{PR} = \mathbf{i}(1-0) - \mathbf{j}(-1-0) + \mathbf{k}(0+1) = \langle1,1,1\rangle$

↓

3. MAGNITUDE

$\|\langle1,1,1\rangle\| = \sqrt{3}$ — this is the parallelogram's area

↓

4. HALVE FOR THE TRIANGLE
$A=\frac{\sqrt{3}}{2}$

Keep these three points in mind — they thread through the whole plane section below.

Triple scalar product (volume)

$$\vec{u}\cdot(\vec{v}\times\vec{w})=\begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Same expansion process, just with u_1,u_2,u_3 on top instead of $\mathbf{i},\mathbf{j},\mathbf{k}$. $|\vec{u}\cdot(\vec{v}\times\vec{w})|$ = volume of the **parallelepiped** with edges $\vec{u},\vec{v},\vec{w}$; it equals $0 \iff$ the three vectors are coplanar (flat box).

Worked: volume of the parallelepiped on $\langle1,0,1\rangle$, $\langle1,1,0\rangle$, $\langle0,1,1\rangle$

1. STACK THE DETERMINANT
$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$

↓

2. EXPAND ALONG THE TOP
$1(1\cdot1-0\cdot1)-0+1(1\cdot1-1\cdot0)=1+1=2$

↓

3. VOLUME = ABSOLUTE VALUE
$V=|2|=2$ *(a negative determinant just means the trio is left-handed)*

Coplanarity test: rows $\langle1,2,3\rangle,\langle4,5,6\rangle,\langle7,8,9\rangle$ give $det=0 \Rightarrow$ coplanar — the "box" is flat, and that's the whole test.

Torque

Force $\vec{F}$ applied at the end of lever arm $\vec{r}$:

$$\vec{\tau}=\vec{r}\times\vec{F} \qquad \|\vec{\tau}\|=\|\vec{r}\|\,\|\vec{F}\|\sin\theta$$

Torque points along the axis of rotation (right-hand rule); pushing $\perp$ to the wrench ($\sin\theta=1$) is most effective.
Worked: a 0.3 m wrench, 200 N applied at 60° to the handle:
$\|\vec{\tau}\|=0.3\cdot200\cdot\sin60°=60\cdot\frac{\sqrt{3}}{2}=30\sqrt{3}\approx52$ N·m — the Work example's cross-product twin (sin where Work used cos).

lines & planes in space

Everything here needs exactly two ingredients: a point plus a direction. Lines use a **direction vector** $\vec{v}$ pointing *along* the line. Planes use a **normal vector** $\vec{n}$ pointing *perpendicular* to the plane.

Given this, want that — the lookup

you are given	the move
line through two points	direction $\vec{v}=P_2-P_1$
line $\perp$ to a plane	direction $\vec{v}=\vec{n}$
plane from a point + normal	$\vec{n}\cdot\langle x-x_0,y-y_0,z-z_0\rangle=0$
plane through three points	$\vec{n}=(P_2-P_1)\times(P_3-P_1)$
plane from a point + two vectors	$\vec{n}=\vec{v}_1\times\vec{v}_2$
plane $\perp$ to a line	$\vec{n}=\vec{v}$
distance, point to line	$\frac{\lvert\vec{PM}\times\vec{v}\rvert}{\lvert\vec{v}\rvert}$
distance, point to plane	$\frac{\lvert ax_0+by_0+cz_0-d\rvert}{\lvert\vec{n}\rvert}$
distance between skew lines	$\frac{\lvert(P_2-P_1)\cdot(\vec{v}_1\times\vec{v}_2)\rvert}{\lvert\vec{v}_1\times\vec{v}_2\rvert}$
angle between two planes	the angle between $\vec{n}_1$ and $\vec{n}_2$
line where two planes meet	direction $=\vec{n}_1\times\vec{n}_2$

THE WHOLE SECTION IN TWO OBJECTS
A **line** needs a point and a direction. A **plane** needs a point and a normal.
Every problem above is one question: which of those two am I missing, and what produces it?
The answer is almost always a **difference** of points, or a **cross product**.

Two nonparallel vectors span the plane's directions, so $\vec{v}_1\times\vec{v}_2$ is perpendicular to both and therefore normal to the plane. This is the three-point case with one step removed — given points, you build the two vectors first.

Lines — three equivalent forms

Through point $P_0(x_0,y_0,z_0)$ with direction $\vec{v}=\langle a,b,c\rangle$:
- **Vector**: $\vec{r}=\vec{r}_0+t\vec{v}$ ($\vec{r}_0=\langle x_0,y_0,z_0\rangle$; t slides the point along the line)
- **Parametric**: $x=x_0+ta$, $y=y_0+tb$, $z=z_0+tc$
- **Symmetric** (solve each for t): $\frac{x-x_0}{a}=\frac{y-y_0}{b}=\frac{z-z_0}{c}$

If a direction component is 0, that coordinate is frozen — write it separately, for example, $b=0$: $\frac{x-x_0}{a}=\frac{z-z_0}{c}$, $y=y_0$.
Through **two points** P,Q: direction $\vec{v}=\vec{PQ}$ (tip − tail), then use either point as P_0. The **segment** PQ only: $\vec{r}(t)=(1-t)\vec{p}+t\vec{q}$, $0\le t\le1$.

Worked: line through $(1,3,2)$ and $(4,2,6)$

1. DIRECTION = TIP − TAIL
$\vec{v}=\langle4-1,\ 2-3,\ 6-2\rangle=\langle3,-1,4\rangle$

↓

2. PARAMETRIC — EITHER POINT AS P_0
$x=1+3t,\quad y=3-t,\quad z=2+4t$

↓

3. SYMMETRIC — SOLVE EACH FOR t
$\frac{x-1}{3}=\frac{y-3}{-1}=\frac{z-2}{4}$

Worked (reverse): read a symmetric form

$\frac{x-5}{2}=\frac{y+1}{3}$, $z=4$: point $(5,-1,4)$ (flip the signs), direction $\langle2,3,0\rangle$ (the z-slot is the frozen coordinate — direction component 0, not "missing").
Parametric: $x=5+2t$, $y=-1+3t$, $z=4$.

Parallel, intersecting, or skew?

Skew = not parallel *and* never meet (only possible in 3D). Process:
1. Directions scalar multiples of each other? $\Rightarrow$ parallel.
2. Otherwise set the parametric forms equal — use different parameters (t and s): three equations. Solve two of them for t,s, then **check the third**.
3. Third equation holds $\Rightarrow$ intersecting (plug t back in for the point). Fails $\Rightarrow$ skew.

Worked: classify $L_1:(1,2,0)+t\langle1,1,1\rangle$ vs $L_2:(2,0,3)+s\langle1,-1,0\rangle$

1. PARALLEL CHECK
$\langle1,1,1\rangle$ vs $\langle1,-1,0\rangle$ — not scalar multiples: not parallel, keep going

↓

2. SET EQUAL — DIFFERENT PARAMETERS
$1+t=2+s,\qquad 2+t=-s,\qquad t=3$

↓

3. SOLVE TWO…
the z equation hands over $t=3$; then x: $4=2+s\Rightarrow s=2$

↓

4. … CHECK THE THIRD
y: $2+3=5$ vs $-s=-2$ — $5\neq-2$ fails $\Rightarrow$ skew *(had it held, plug $t=3$ into L_1 for the crossing point)*

Distance from a point to a line

Point M, line through P with direction $\vec{v}$:

$$d=\frac{\|\vec{PM}\times\vec{v}\|}{\|\vec{v}\|}$$

(parallelogram area ÷ base).

***Worked:** $M(1,1,3)$**, line through the origin with** $\vec{v}=\langle 1,1,0\rangle$*

1. BUILD $\vec{PM}$
$P=$ origin: $\vec{PM}=\langle 1,1,3\rangle$

↓

2. CROSS — RUN THE DETERMINANT
$$\vec{PM}\times\vec{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&1&3\\1&1&0\end{vmatrix}=\mathbf{i}(1\cdot 0-3\cdot 1)-\mathbf{j}(1\cdot 0-3\cdot 1)+\mathbf{k}(1\cdot 1-1\cdot 1)=\langle -3,3,0\rangle$$

↓

3. TWO MAGNITUDES
$\|\langle -3,3,0\rangle\|=\sqrt{9+9+0}=\sqrt{18}=3\sqrt{2}$, $\quad\|\vec{v}\|=\sqrt{1+1+0}=\sqrt{2}$

↓

4. DIVIDE
$d=\dfrac{3\sqrt{2}}{\sqrt{2}}=3$ *(the $\sqrt{2}$'s cancel — watch for it)*

Distance between skew lines

The two lines span parallel planes. The gap is the triple product of the connector against both directions:

$$d=\frac{|\vec{PQ}\cdot(\vec{v}_1\times\vec{v}_2)|}{\|\vec{v}_1\times\vec{v}_2\|}$$

Worked (the skew pair above): $\vec{PQ}=\langle 2-1,\,0-2,\,3-0\rangle=\langle 1,-2,3\rangle$. Cross the directions:

$$\vec{v}_1\times\vec{v}_2=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&1&1\\1&-1&0\end{vmatrix}=\mathbf{i}(0+1)-\mathbf{j}(0-1)+\mathbf{k}(-1-1)=\langle 1,1,-2\rangle$$

with $\|\langle 1,1,-2\rangle\|=\sqrt{1+1+4}=\sqrt{6}$. Then $\vec{PQ}\cdot(\vec{v}_1\times\vec{v}_2)=1-2-6=-7$:
$d=\dfrac{|-7|}{\sqrt{6}}=\dfrac{7\sqrt{6}}{6}\approx 2.86$.

Planes — point + normal

Through $P_0(x_0,y_0,z_0)$ with normal $\vec{n}=\langle a,b,c\rangle$:

- **Vector form**: $\vec{n}\cdot\vec{P_0P}=0$ — a point P is on the plane iff its offset from P_0 is $\perp\vec{n}$.
- **Scalar form**: $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$.
- **General form**: $ax+by+cz+d=0$ — the normal reads straight off the coefficients: $\vec{n}=\langle a,b,c\rangle$.

Plane $\perp$ to a line (normal = direction)

A plane perpendicular to a line uses the line's direction vector as its normal — pull it straight off the t-coefficients. (The dual trick also works: a line $\perp$ to a plane uses $\vec{n}$ as its *direction*.)
Worked: plane through $(3,-2,1)$, perpendicular to the line $x=1+4t$, $y=2-t$, $z=5+3t$.

1. PULL THE NORMAL OFF THE LINE
$\vec{n}=\langle 4,-1,3\rangle$ *(the t-coefficients. The line's own point $(1,2,5)$ is not used)*

↓

2. GRAB THE GIVEN POINT
$(x_0,y_0,z_0)=(3,-2,1)$

↓

3. SCALAR FORM $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$
$4(x-3)-1(y+2)+3(z-1)=0$ *(sign flip: $y-(-2)=y+2$)*

↓

4. DISTRIBUTE
$4x-12-y-2+3z-3=0$

↓

5. CLEAN UP
$4x-y+3z=17$

Check: plug the point back in: $4(3)-(-2)+3(1)=12+2+3=17$. ✓

Worked: the dual — line $\perp$ to a plane

Line through $(2,1,5)$ perpendicular to the plane $3x-y+2z=8$:

1. DIRECTION = THE PLANE'S NORMAL
$\vec{v}=\vec{n}=\langle 3,-1,2\rangle$ — read off the coefficients *(the plane's 8 plays no role)*

↓

2. GRAB THE GIVEN POINT
$P_0=(2,1,5)$

↓

3. WRITE THE LINE
$x=2+3t$, $\quad y=1-t$, $\quad z=5+2t$ *(mirror of the previous worked: there direction→normal, here normal→direction)*

Plane through three points (cross product!)

1. TWO IN-PLANE VECTORS
$P(1,0,0)$, $Q(0,1,0)$, $R(0,0,1)$: $\quad\vec{PQ}=\langle -1,1,0\rangle$, $\quad\vec{PR}=\langle -1,0,1\rangle$ *(tip − tail, both from P)*

↓

2. NORMAL = CROSS
$\vec{n}=\vec{PQ}\times\vec{PR}=\langle 1,1,1\rangle$ *(the triangle-area worked ran this same cross)*

↓

3. SCALAR FORM WITH ANY ONE POINT
$1(x-1)+1(y-0)+1(z-0)=0\ \Rightarrow\ x+y+z=1$

↓

4. CHECK ALL THREE
P: $1=1$; Q: $1=1$; R: $1=1$ ✓

Distance from a point to a plane

Point (x_1,y_1,z_1), plane $ax+by+cz+d=0$ (move everything to one side first):

$$D=\frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}}$$

Worked: point $(2,3,4)$ to the plane $x+y+z=1$

1. EVERYTHING TO ONE SIDE
$x+y+z-1=0$: $\quad a,b,c,d=1,1,1,-1$

↓

2. NUMERATOR & DENOMINATOR
$|2+3+4-1|=8$, $\qquad\|\vec{n}\|=\sqrt{3}$

↓

3. DIVIDE & RATIONALIZE
$D=\dfrac{8}{\sqrt{3}}=\dfrac{8\sqrt{3}}{3}\approx 4.62$

Distance between parallel planes — pick any point on one, run the formula against the other: $x+y+z=1$ vs $x+y+z=7$: point $(1,0,0)$ from the first: $D=\dfrac{|1-7|}{\sqrt{3}}=2\sqrt{3}$.

Point where a line meets a plane

Substitute the line's **parametric** x,y,z into the plane equation — three variables collapse into one equation in t. Solve for t, feed it back into the line.
If the t's cancel out: a contradiction ($0=5$) means the line is parallel to the plane (never meets it); an identity ($0=0$) means the line lies inside the plane (every t works). Quick pre-check: $\vec{v}\cdot\vec{n}=0\iff$ no single crossing point.
Worked: line $x=2+t$, $y=1-t$, $z=2t$ and the plane $x+y+z=1$ (the triangle plane again).

1. SUB THE LINE INTO THE PLANE
$(2+t)+(1-t)+(2t)=1$ *(every x,y,z replaced — only t survives)*

↓

2. SOLVE FOR t
$3+2t=1\Rightarrow t=-1$

↓

3. PLUG t BACK INTO THE LINE
$x=2-1=1$, $\quad y=1-(-1)=2$, $\quad z=2(-1)=-2\ \Rightarrow(1,2,-2)$

Check: put the point into the plane: $1+2-2=1$. ✓

Two planes: parallel, angle, intersection line

- **Parallel** $\iff$ normals are scalar multiples.
- **Angle between planes** = angle between normals: $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\,\|\vec{n}_2\|}$ (abs value $\Rightarrow$ the acute angle).
- **Line of intersection**: direction $= \vec{n}_1 \times \vec{n}_2$ (must lie in both planes, so $\perp$ both normals). For a point on it: set one variable to 0 and solve the two plane equations.

Worked (quick): $x+y+z=1$ and the xy-plane $z=0$: angle $\cos\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta \approx 54.7°$; direction $\langle 1,1,1\rangle \times \langle 0,0,1\rangle = \langle 1,-1,0\rangle$; point from $z=0,\ x+y=1$: $(1,0,0)$ — the line is exactly edge PQ of the triangle above. ✓

Worked (two slanted planes): $x+y+z=3$ and $x-y+2z=1$

1. THE TWO NORMALS
$\vec{n}_1 = \langle 1,1,1\rangle$, $\vec{n}_2 = \langle 1,-1,2\rangle$ — not multiples: they do intersect

↓

2. ANGLE
$\|\vec{n}_1\| = \sqrt{1+1+1} = \sqrt{3}$, $\|\vec{n}_2\| = \sqrt{1+1+4} = \sqrt{6}$, $\sqrt{3}\sqrt{6} = \sqrt{18} = 3\sqrt{2}$:
$\cos\theta = \dfrac{|1-1+2|}{3\sqrt{2}} \approx 0.471 \Rightarrow \theta \approx 61.9°$

↓

3. DIRECTION OF THE INTERSECTION
$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix} = \mathbf{i}(2+1) - \mathbf{j}(2-1) + \mathbf{k}(-1-1) = \langle 3,-1,-2\rangle$$

↓

4. A POINT — SET $z=0$, SOLVE THE 2×2
$x+y=3$, $x-y=1 \Rightarrow x=2,\ y=1$: point $(2,1,0)$

↓

5. ASSEMBLE
$\vec{r} = \langle 2,1,0\rangle + t\langle 3,-1,-2\rangle$ *(check: plug into both plane equations — t cancels in each)*

quadric surfaces

The 3D analog of a conic section: the graph of a second-degree equation in three variables,

$$Ax^2 + By^2 + Cz^2 + Gx + Hy + Iz + J = 0$$

(no cross terms like xy in this course). Just as every 2D conic is an ellipse, parabola, or hyperbola, every quadric is one of six standard surfaces (plus cylinders and degenerate cases). The whole topic is: **recognize which one**, then describe it with intercepts and traces.

Conic carry-over (the 2D shapes inside every trace)

Every cross-section of a quadric is a conic — recognize these on sight:

- $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$: **ellipse** (both squared, both $+$). Circle if $a=b$.
- $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$: **hyperbola** (both squared, one $-$).
- $y = ax^2$: **parabola** (only one variable squared).

Sign/degree pattern in 2D $\Rightarrow$ name in 3D — the same three tests, one dimension up.

Cylinders — the missing-variable rule

If one variable is absent from the equation, the surface is a **cylinder**: draw the 2D curve in its plane, then slide it parallel to the missing variable's axis.

- $x^2+y^2=4$ in 3D: circular cylinder along the z-axis (no z $\Rightarrow$ every height z gives the same circle).
- $z=y^2$: parabolic cylinder — a parabola sheet sliding along the x-axis.

Pitfall: in 3D, $x^2+y^2=4$ is *not* a circle — context decides. A missing variable means unrestricted, never zero — the same rule as the plane $z=3$ in the coordinate-planes note above.

Intercepts

Where the surface crosses a coordinate *axis*. On the x-axis, y and z are already 0 — so: set the other two variables $=0$, solve for the third. Answers are **points** $(x,0,0)$, $(0,y,0)$, $(0,0,z)$. Three independent mini-problems — never carry one axis's answer into the next.

Worked: intercepts of $\frac{x^2}{25} - \frac{y^2}{81} - z = 64$

1. X-INTERCEPTS (set $y=0$, $z=0$)
$\frac{x^2}{25} = 64 \Rightarrow x^2 = 1600 \Rightarrow x = \pm 40$ $\Rightarrow$ $(\pm 40,\ 0,\ 0)$

↓

2. Y-INTERCEPTS (set $x=0$, $z=0$)
$-\frac{y^2}{81} = 64 \Rightarrow y^2 = -5184$: no real solution $\Rightarrow$ no y-intercepts *(a complete answer — and a clue about the shape)*

↓

3. Z-INTERCEPTS (set $x=0$, $y=0$)
$-z = 64 \Rightarrow z = -64$ $\Rightarrow$ $(0,\ 0,\ -64)$

↓

4. ANSWER (points, with signs)
$(40,0,0)$, $(-40,0,0)$, $(0,0,-64)$ ✓

Pitfalls: $\sqrt{x^2} = \pm$ — squared variables give intercept pairs: $x^2 = 1600 \Rightarrow x = \pm 40$, not 400 (that's $1600/4$) and not 40^2 ($160{,}000$). And "no real solution" is a legitimate final answer — don't force a point to exist.

Traces — cross sections

Slice the surface with a plane parallel to a coordinate plane. The cut edge is a 2D curve called a **trace**. Process: set one variable equal to a constant k, then recognize the leftover 2D conic.

- $z=k$: horizontal slices (curves in the xy-plane at each height); $x=k$ and $y=k$: vertical slices.
- $k=0$ gives the traces *in* the coordinate planes — usually computed first.

Traces are how you sketch and how you verify a surface's identity: each of the six quadrics has a signature trace pattern (listed with each surface below).

Worked: traces of $z = \frac{x^2}{25} - \frac{y^2}{81} - 64$

The intercept example, solved for z.

1. TRACE $z=k$ (horizontal)
$\frac{x^2}{25} - \frac{y^2}{81} = k + 64$: both squared, opposite signs $\Rightarrow$ hyperbolas (opening along x when $k > -64$, along y when $k < -64$)

↓

2. TRACE $x=k$
$z = -\frac{y^2}{81} + \left(\frac{k^2}{25} - 64\right)$: one variable squared $\Rightarrow$ parabola, opening *down* in y

↓

3. TRACE $y=k$
$z = \frac{x^2}{25} - \left(\frac{k^2}{81} + 64\right)$: parabola, opening *up* in x

↓

4. READ THE PATTERN
hyperbolas + up-parabolas + down-parabolas $\Rightarrow$ hyperbolic paraboloid (saddle) ✓
(up along x, down along y — hence the missing y-intercepts)

The six standard quadrics

All shown centered at the origin; a,b,c stretch the axes. Swapping which variable is "special" rotates the surface onto a different axis — the patterns are what to memorize, not the letters.

Group 1 — all three variables squared:

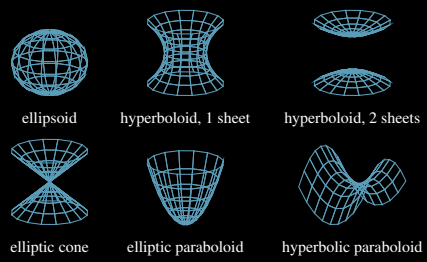
- **Ellipsoid**: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$ — all $+$, $=1$. Football / sphere shape (sphere when $a=b=c$). All traces are ellipses. Intercepts $(\pm a,0,0)$, $(0,\pm b,0)$, $(0,0,\pm c)$.
- **Hyperboloid of one sheet**: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$ — two $+$, one $-$. Cooling-tower shape, one connected piece. Axis runs along the minus variable. Traces: ellipses $\perp$ axis, hyperbolas along it.

- **Hyperboloid of two sheets**: $-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ — one $+$, two $-$. Two separate bowls opening away from each other along the plus variable. The gap shows up as "no real solution" when slicing between the sheets.
- **Elliptic cone**: $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$ — same signs as one sheet but $=0$. Double ice-cream cone meeting at the origin. Axis along the minus variable. Traces $\perp$ axis: ellipses. Through the axis: pairs of crossing lines.

Group 2 — one variable first-power (paraboloids):

- **Elliptic paraboloid**: $z=\frac{x^2}{a^2}+\frac{y^2}{b^2}$ — squares *same* sign. Bowl / satellite dish opening along the linear variable (up if $+$, down if $-$). Traces: ellipses $\perp$ axis, parabolas along it.
- **Hyperbolic paraboloid**: $z=\frac{x^2}{a^2}-\frac{y^2}{b^2}$ — squares *opposite* signs. Saddle / Pringle. Traces: hyperbolas $\perp$ axis, parabolas along it — one family up, one down.

Gallery



Identification — the decision routine

Run these checks in order on *any* second-degree equation:

1. **Variable missing?** $\Rightarrow$ cylinder (slide the 2D curve). Stop.
2. **Tidy it first**: everything squared on one side, divide so the constant is 1 (or recognize $=0$). If the RHS is -1, multiply through by -1 before counting signs.
3. **All three squared?** Count the minus signs: $0\Rightarrow$ ellipsoid, $1\Rightarrow$ 1 sheet, $2\Rightarrow$ 2 sheets. **Mnemonic**: # of sheets = # of minus signs. If the equation is $=0$ instead of $=1$: cone.
4. **One variable linear (first power)?** $\Rightarrow$ paraboloid. Squares *same* sign $\Rightarrow$ elliptic (bowl); *opposite* $\Rightarrow$ hyperbolic (saddle).
5. **Find the axis**: for hyperboloids and cones, the axis is the "odd sign out" variable. For paraboloids, the axis is the linear variable.

Worked: identify $4x^2+y^2+z^2=16$

1. ALL SQUARED, RHS NOT 1 — DIVIDE BY 16
$\frac{x^2}{4}+\frac{y^2}{16}+\frac{z^2}{16}=1$

↓

2. COUNT MINUSES: 0
ellipsoid, semi-axes $a=2$, $b=c=4$ (long directions y,z) ✓

Worked: identify $z^2=x^2+\frac{y^2}{9}$

1. GATHER
$x^2+\frac{y^2}{9}-z^2=0$ — all squared, $=0$ — not a hyperboloid

↓

2. NAME + AXIS
elliptic cone, axis along z (the odd sign out) ✓

Worked: identify $x^2-\frac{y^2}{4}+\frac{z^2}{25}=1$ ***and slice it at*** $y=4$

1. ALL SQUARED, $=1$, ONE MINUS
hyperboloid of one sheet, axis along y

↓

2. TRACE $y=4$
$x^2+\frac{z^2}{25}=1+\frac{16}{4}=5$: an ellipse — slices $\perp$ the axis are ellipses that grow as $|y|$ grows (the cooling-tower flare) ✓

Worked: the RHS $=-1$ ***trap — identify*** $x^2-y^2+z^2=-1$

1. RHS IS -1 — MULTIPLY THROUGH
$-x^2+y^2-z^2=1$ *(every sign flips. Counting minuses before this step names the wrong surface)*

↓

2. COUNT MINUSES: 2
hyperboloid of two sheets, axis along y (the lone $+$) *(counting the original's single minus would have said "one sheet" — wrong)*

Worked: missing variable — identify $y=x^2$ ***in space***

1. VARIABLE MISSING
no z anywhere $\Rightarrow$ cylinder: the 2D parabola slides along the z-axis

↓

2. NAME IT
parabolic cylinder — a sheet, not a solid. The missing variable is the slide direction

Worked: shifted center — complete the square

$x^2+y^2-z^2-2x+4y=4$:

1. GROUP BY VARIABLE
$(x^2-2x)+(y^2+4y)-z^2=4$

↓

2. COMPLETE EACH SQUARE
$(x-1)^2-1+(y+2)^2-4-z^2=4$ *(same move as the sphere worked earlier)*

↓

3. TIDY & DIVIDE
$(x-1)^2+(y+2)^2-z^2=9 \quad\rightarrow\quad \frac{(x-1)^2}{9}+\frac{(y+2)^2}{9}-\frac{z^2}{9}=1$

↓

4. NAME + AXIS + CENTER
one minus $\Rightarrow$ hyperboloid of one sheet, axis parallel to z, centered $(1,-2,0)$ ✓

Common pitfalls

- $x^2=c\Rightarrow x=\pm\sqrt{c}$ — intercepts on squared variables come in pairs. Dropping the $-$ loses half the answer.
- No real solution = no intercept — write that, don't invent a point. It's geometry: the surface misses that axis.
- Each axis is its own problem — zero the other two fresh each time. Never substitute one axis's intercept into the next computation.
- $=1$ vs $=0$: same signs, different surface — hyperboloid (1 sheet) vs cone. Check the constant before naming.
- Divide before counting: the minus-sign count only works with RHS $=+1$. $=16$? Divide. $=-1$? Multiply by -1 (flips every sign).
- z vs z^2: first power $\Rightarrow$ paraboloid family. Squared $\Rightarrow$ ellipsoid/hyperboloid/cone family. The single most common misread.

vector-valued functions (space curves)

A function that takes a scalar t in and hands a vector back:

$$\vec{r}(t)=\langle f(t),g(t),h(t)\rangle=f(t)\,\mathbf{i}+g(t)\,\mathbf{j}+h(t)\,\mathbf{k}$$

f,g,h are ordinary functions — the **component functions**. As t runs, the tip of the position vector $\vec{r}(t)$ traces a **plane curve** (2 components) or **space curve** (3). Think of t as time and $\vec{r}(t)$ as a flight path.

Evaluate & domain — componentwise

- **Evaluate**: plug t into each slot. The output is a vector.
- **Domain** = the intersection of the component domains — every component must be defined.

Worked: $\vec{r}(t)=\left\langle \sqrt{t+2},\ \ln(3-t),\ \frac{1}{t}\right\rangle$

- $\sqrt{t+2}$: $t\geq -2$ $\quad$ $\ln(3-t)$: $t<3$ $\quad$ $\frac{1}{t}$: $t\neq 0$
- Intersect all three: $[-2,0)\cup(0,3)$

Limits & continuity — componentwise

$$\lim_{t\to a}\vec{r}(t)=\left\langle \lim_{t\to a}f(t),\ \lim_{t\to a}g(t),\ \lim_{t\to a}h(t)\right\rangle$$

Take each limit separately — every Calc-1 rule (and the fundamental limits) applies per slot. $\vec{r}$ is **continuous** at a $\iff$ every component is continuous at a.

Worked: $\lim_{t\to 0}\left\langle \frac{\sin t}{t},\ \cos t,\ t^2+3\right\rangle=\langle 1,\ 1,\ 3\rangle$ *(first slot is the fundamental limit).*

Sketching: table of t-values + arrows

Pick t-values, compute the tip points, connect them in order of increasing t, and mark arrows — the **orientation** of the curve. Before grinding a table, check the library below for a recognizable pattern.

Worked: $\vec{r}(t)=\langle t,\ t^2-1\rangle$:

t	-2	-1	0	1	2
point	$(-2,3)$	$(-1,0)$	$(0,-1)$	$(1,0)$	$(2,3)$

Connect left-to-right (increasing t), arrows rightward — a parabola traversed left to right; the elimination below confirms $y=x^2-1$.

Parameterization library

- **Line** through P_0, direction $\vec{v}$: $\vec{r}(t)=\vec{r}_0+t\vec{v}$ — same as the last section.
- **Circle**, radius a, counterclockwise: $\vec{r}(t)=\langle a\cos t,\ a\sin t\rangle$, $0\le t\le 2\pi$.
- **Ellipse** $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$: $\vec{r}(t)=\langle a\cos t,\ b\sin t\rangle$.
- **Graph of** $y=F(x)$: $\vec{r}(t)=\langle t,\ F(t)\rangle$ — let x itself be the parameter.
- **Helix**: $\langle a\cos t,\ a\sin t,\ ct\rangle$ — a circle in x,y plus a steady climb in z.

Why circles work: $x^2+y^2=a^2\cos^2 t+a^2\sin^2 t=a^2$ — the Pythagorean identity eats the t.

Worked: parameterize — the reverse direction

1. SEGMENT $A(1,2,3)\to B(4,0,5)$

$\vec{r}(t)=(1-t)\langle 1,2,3\rangle+t\langle 4,0,5\rangle$

$=\langle 1-t+4t,\ 2-2t+0,\ 3-3t+5t\rangle=\langle 1+3t,\ 2-2t,\ 3+2t\rangle,\quad 0\le t\le 1$

($t=0$ at A, $t=1$ at B — check both)

↓

2. GRAPH WITH AN ORIENTATION

$y=x^2$ from $(2,4)$ *back* to $(0,0)$: x must decrease: $\vec{r}(t)=\langle 2-t,\ (2-t)^2\rangle$, $0\le t\le 2$

(plain $\langle t,t^2\rangle$ runs the wrong way)

↓

3. CURVE OF INTERSECTION

cylinder $x^2+y^2=4$ meets plane $z=3-y$: circle handles two coordinates, plane hands over the third: $\vec{r}(t)=\langle 2\cos t,\ 2\sin t,\ 3-2\sin t\rangle$

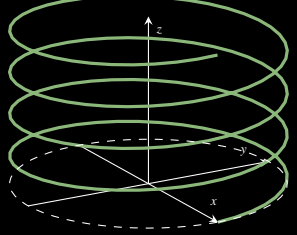
Eliminating the parameter

To identify a curve, turn $\vec{r}(t)$ back into an x-y equation:

- $\langle t,\ t^2-1\rangle$: substitute $x=t$ into y: $y=x^2-1$ (a parabola).
- $\langle 2\cos t,\ 2\sin t\rangle$: don't solve for t — square and add: $x^2+y^2=4$ (circle, radius 2).

Trig components $\Rightarrow$ reach for an **identity**; polynomial components $\Rightarrow$ **substitute**. **3D flavor**: for the helix $\langle 2\cos t,2\sin t,t/8\rangle$ below, squaring-and-adding the first two slots gives $x^2+y^2=4$ — the curve lives on that cylinder, climbing it at rate $\frac{1}{8}$.

helix $\vec{r}(t)=\langle 2\cos t,\ 2\sin t,\ t/8\rangle$



Do two curves meet? — intersect against collide

Two different questions, and the wording decides which one is asked:

A **Intersect** — the curves pass through the same point, at possibly different parameter values. Use two letters, t and s.

B **Collide** — the same point at the same parameter value. Set $t=s$, so one letter, and it must hold in every component.

THE PROCESS

1. **Write two letters** — $\vec{r}_1(t)$ and $\vec{r}_2(s)$. Reusing t silently asks the collide question.
2. **Set the components equal** — that is three equations.
3. **Solve two** of them for t and s.
4. **Check the third.** If it fails, the curves do not intersect.
5. **Report the point**, not the parameters — substitute back.

Step 4 is the whole problem. Two equations in two unknowns almost always produce candidates; the third equation is what rejects the impostors.

Worked: $\vec{r}_1(t)=\langle t,\ t^2-1,\ t^3\rangle$ and $\vec{r}_2(s)=\langle s-1,\ s-2,\ s^2-3\rangle$

1–2. TWO LETTERS, THREE EQUATIONS

$$t=s-1,\qquad t^2-1=s-2,\qquad t^3=s^2-3$$

↓

3. SOLVE THE FIRST TWO

from the first: $s=t+1$. Substitute into the second:

$$t^2-1=(t+1)-2=t-1\ \Rightarrow\ t^2=t\ \Rightarrow\ t(t-1)=0\ \Rightarrow\ t=0 \text{ or } t=1$$

two candidates: $(t,s)=(0,1)$ and $(t,s)=(1,2)$

↓

4. CHECK THE THIRD — ONE CANDIDATE DIES

$(t,s)=(0,1)$: $t^3=0$ against $s^2-3=1-3=-2$. $0\neq -2\Rightarrow$ rejected

$(t,s)=(1,2)$: $t^3=1$ against $s^2-3=4-3=1$. holds ✓

↓

5. REPORT THE POINT

$\vec{r}_1(1)=\langle 1,\ 0,\ 1\rangle$ and $\vec{r}_2(2)=\langle 1,\ 0,\ 1\rangle$ ✓

the curves intersect at $(1,\ 0,\ 1)$

Do they collide? Set $t=s$ in the first equation: $t=t-1$, so $0=-1$ — impossible. They never collide: the curves cross the same point of space, but the second curve arrives one unit of parameter later.

The aha: had you written t in both, the first equation alone would have said "no solution" and you would have reported no intersection — the wrong answer to the question that was asked.

calculus of vector-valued functions

Everything here is componentwise: a vector-valued function is just ordinary Calc-1 functions riding in slots, and every operation — derivative, integral, limit — applies the usual Calc-1 rules to each slot separately. The vector wrapper adds no new calculus; you do the same move three times in parallel and keep the answers in $\langle\ \rangle$ brackets.

THE PROCESS — WORK COMPONENTWISE

1. **Differentiate or integrate each component** on its own. The vector structure never interferes.
2. **Add the constant vector $\vec{C}$** after every integration.
3. **Use products carefully** — the dot and cross rules follow the ordinary product rule, but the cross product does not commute, so keep the order.

Everything here is Calc 1, run three times in parallel. The only new content is the product rules and the fact that $\vec{C}$ is a vector.

Derivative — componentwise

$$\vec{r}\,'(t)=\langle f'(t),\ g'(t),\ h'(t)\rangle$$

Geometric meaning (why the derivative matters): $\vec{r}\,'(t)$ is the tangent vector to the curve at the point $\vec{r}(t)$, pointing in the direction of motion (increasing t). Interpretations stack:

- $\vec{r}(t)$ = position $\to\ \vec{r}\,'(t)$ = velocity $\to\ \vec{r}\,''(t)$ = acceleration
- **speed** $=\|\vec{r}\,'(t)\|$ — a scalar, the magnitude of velocity.

Every Calc-1 rule applies per slot: power, product, quotient, chain, and the common derivatives ($\sin\to\cos$, $\cos\to-\sin$, $e^t\to e^t$, $\ln t\to 1/t$) — full statements in the **Derivative Rules** tables back in the Calculus section. The usual slip is the chain rule — an inside function must multiply in its derivative:

- $\frac{d}{dt}[\cos(5t)] = -\sin(5t)\cdot 5 = -5\sin(5t)$
- $\frac{d}{dt}[e^{3t}] = e^{3t}\cdot 3 = 3e^{3t}$
- $\frac{d}{dt}[(t^2+1)^4] = 4(t^2+1)^3\cdot 2t$

Worked: $\vec{r}(t) = \langle t^2,\ \sin t,\ e^{3t}\rangle$

1. EASY SLOTS
$t^2 \to 2t$ (power) $\qquad \sin t \to \cos t$

↓

2. THE CHAIN-RULE SLOT
$e^{3t} \to 3e^{3t}$ *(inside $3t$ multiplies its derivative in — the slot where points die)*

↓

3. ASSEMBLE
$\vec{r}\,'(t) = \langle 2t,\ \cos t,\ 3e^{3t}\rangle$

Higher derivatives: differentiate again, componentwise — $\vec{r}\,'' = \langle f'', g'', h''\rangle$.

Unit tangent vector $\vec{T}(t)$

The tangent vector rescaled to length 1 — the direction of motion with the speed stripped off. Same normalize-by-dividing move as the unit vectors above, now applied to the derivative:

$$\vec{T}(t) = \frac{\vec{r}\,'(t)}{\|\vec{r}\,'(t)\|}$$

Two flavors of the question — know which you're asked:

- **General formula** (answer in terms of t): compute $\vec{r}\,'(t)$, then $\|\vec{r}\,'(t)\|$ as a function of t, divide. Simplify the radical if a Pythagorean identity collapses it (common with sin/cos slots).
- **At a specific $t=a$**: substitute first, then normalize — you normalize a concrete vector of numbers instead of a function. Normalizing first forces you to simplify a t-dependent radical you never needed — more algebra, more error surface.

Worked: $\vec{r}(t) = \langle t^2,\ 2t,\ 1\rangle$, find $\vec{T}(1)$.

1. DIFFERENTIATE
$\vec{r}\,'(t) = \langle 2t,\ 2,\ 0\rangle$

↓

2. SUBSTITUTE FIRST
$\vec{r}\,'(1) = \langle 2,\ 2,\ 0\rangle$

↓

3. MAGNITUDE
$\|\langle 2,2,0\rangle\| = \sqrt{4+4+0} = \sqrt{8} = 2\sqrt{2}$

↓

4. DIVIDE
$\vec{T}(1) = \frac{1}{2\sqrt{2}}\langle 2,2,0\rangle = \langle \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ 0\rangle$ *(rationalized: $\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0\rangle$ — same value)*

Worked: the general-formula flavor — helix $\vec{r} = \langle 2\cos t,\ 2\sin t,\ t\rangle$

1. DIFFERENTIATE
$\vec{r}\,'(t) = \langle -2\sin t,\ 2\cos t,\ 1\rangle$

↓

2. MAGNITUDE AS A FUNCTION OF t
$\|\vec{r}\,'\| = \sqrt{4\sin^2 t + 4\cos^2 t + 1} = \sqrt{4+1} = \sqrt{5}$ *(the Pythagorean identity collapses the radical — constant speed)*

↓

3. DIVIDE — STILL IN t
$\vec{T}(t) = \frac{1}{\sqrt{5}}\langle -2\sin t,\ 2\cos t,\ 1\rangle$ *(now any t is a plug-in away)*

Worked: tangent line to a space curve

Line tangent to $\vec{r}(t) = \langle t^2, 2t, 1\rangle$ at $t=1$ — the direction is already computed above:

1. POINT — EVALUATE $\vec{r}$
$\vec{r}(1) = \langle 1^2,\ 2(1),\ 1\rangle = (1,2,1)$

↓

2. DIRECTION — EVALUATE $\vec{r}\,'$
$\vec{r}\,'(1) = \langle 2,2,0\rangle$

↓

3. LINE FORMS FROM THE LINES SECTION
$x = 1+2s, \quad y = 2+2s, \quad z = 1$ *(new parameter s — t is spoken for. Frozen z written separately)*

Derivatives of dot & cross products

Both get product rules that mirror the ordinary one — derivative of the first times the second, plus the first times the derivative of the second:

$$\frac{d}{dt}[\vec{u}\cdot\vec{v}] = \vec{u}\,'\cdot\vec{v} + \vec{u}\cdot\vec{v}\,'$$

$$\frac{d}{dt}[\vec{u}\times\vec{v}] = \vec{u}\,'\times\vec{v} + \vec{u}\times\vec{v}\,'$$

The dot version outputs a scalar; the cross version a vector. Order matters in the cross version: $\vec{u}\times\vec{v} = -(\vec{v}\times\vec{u})$, so $\vec{u}$ stays the left factor in *both* terms — never swap within a term.

Two valid methods — they must agree:

- **Method A — combine, then differentiate**: expand the dot/cross explicitly first, then take an ordinary derivative. Best for the dot product (a single scalar function).
- **Method B — product rule**: apply the formula with $\vec{u}, \vec{u}\,', \vec{v}, \vec{v}\,'$. Required when the components aren't given — only values at a point are (for example, $\vec{u}(2), \vec{u}\,'(2), \vec{v}(2), \vec{v}\,'(2)$). Often cleaner for cross, where expanding symbolically gets messy.

Worked (dot, both ways): $\vec{u} = \langle t, t^2\rangle$, $\vec{v} = \langle \cos t, \sin t\rangle$

1. METHOD A — COMBINE, THEN DIFFERENTIATE
$\vec{u}\cdot\vec{v} = t\cos t + t^2\sin t$: product rule per term: $(\cos t - t\sin t) + (2t\sin t + t^2\cos t)$

↓

2. METHOD B — BUILD THE FOUR PIECES
$\vec{u}\,' = \langle 1, 2t\rangle, \quad \vec{v}\,' = \langle -\sin t, \cos t\rangle$

↓

3. APPLY THE PRODUCT RULE
$\vec{u}\,'\cdot\vec{v} + \vec{u}\cdot\vec{v}\,' = (\cos t + 2t\sin t) + (-t\sin t + t^2\cos t)$

↓

4. COMPARE
collect: both give $\cos t + t\sin t + t^2\cos t$ ✓ — a mismatch means an arithmetic bug

Worked (cross — where Method B earns its keep)

$\vec{u} = \langle t, 0, t^2\rangle$, $\vec{v} = \langle 0, t, 1\rangle$:

1. THE FOUR PIECES
$\vec{u}\,' = \langle 1, 0, 2t\rangle, \quad \vec{v}\,' = \langle 0, 1, 0\rangle$

↓

2. TWO CROSSES — KEEP $\vec{u}$ ON THE LEFT
$$\vec{u}\,'\times\vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2t \\ 0 & t & 1 \end{vmatrix} = \mathbf{i}(0-2t^2) - \mathbf{j}(1-0) + \mathbf{k}(t-0) = \langle -2t^2, -1, t\rangle$$

$$\vec{u}\times\vec{v}\,' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ t & 0 & t^2 \\ 0 & 1 & 0 \end{vmatrix} = \mathbf{i}(0-t^2) - \mathbf{j}(0-0) + \mathbf{k}(t-0) = \langle -t^2, 0, t\rangle$$

↓

3. ADD
$\langle -2t^2 - t^2,\ -1+0,\ t+t\rangle = \langle -3t^2, -1, 2t\rangle$

↓

4. CHECK BY METHOD A
$\vec{u}\times\vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ t & 0 & t^2 \\ 0 & t & 1 \end{vmatrix} = \mathbf{i}(0-t^3) - \mathbf{j}(t-0) + \mathbf{k}(t^2-0) = \langle -t^3, -t, t^2\rangle$; differentiate per slot: $\langle -3t^2, -1, 2t\rangle$ ✓

Values-at-a-point flavor (the exam form — components never given): given $\vec{u}(2) = \langle 1,3\rangle$, $\vec{u}\,'(2) = \langle 2,0\rangle$, $\vec{v}(2) = \langle 4,-1\rangle$, $\vec{v}\,'(2) = \langle 1,5\rangle$:

$$\frac{d}{dt}[\vec{u}\cdot\vec{v}]\Big|_{t=2} = \vec{u}\,'(2)\cdot\vec{v}(2) + \vec{u}(2)\cdot\vec{v}\,'(2) = (8+0) + (1+15) = 8+16 = 24$$

Method A is impossible here — Method B is the only road.

Special cases worth starring:

- $\frac{d}{dt}[\vec{r}\cdot\vec{r}] = 2\vec{r}\cdot\vec{r}\,'$ — the dot-product power rule.

- If $\|\vec{r}(t)\|$ is **constant** (motion on a sphere), then $\vec{r}\cdot\vec{r}'=0$. The "show that" in one line: $\vec{r}\cdot\vec{r}=\|\vec{r}\|^2=c^2$ is constant, so differentiating gives $2\vec{r}\cdot\vec{r}'=0$ — velocity $\perp$ position.

Integrals — componentwise

Integrate each slot separately. Every Calc-2 tool applies per component — u-sub, the $\frac{1}{k}$ rule, the ln-flag, the arctan/arcsin forms, parts. A single component might need any of them. The master integral table at the top of this section carries straight over.

- **Indefinite**: $\int \vec{r}(t)\,dt=\left\langle \int f\,dt,\ \int g\,dt,\ \int h\,dt\right\rangle+\vec{C}$. The constant is a vector $\vec{C}=\langle C_1,C_2,C_3\rangle$, one constant per slot — not a lone scalar tacked on the end. Load-bearing for the motion problems below.
- **Definite**: FTC per slot, no $\vec{C}$ — each component evaluates to a number. The result is a concrete vector.

Worked:

- $\int\langle 2t,\ \cos t,\ e^t\rangle\,dt=\langle t^2,\ \sin t,\ e^t\rangle+\vec{C}$
- $\int_0^1\langle 2t,\ 3t^2\rangle\,dt=\left\langle t^2\big|_0^1,\ t^3\big|_0^1\right\rangle=\langle 1,\ 1\rangle$
- $\int\langle \sin(2t),\ e^{5t}\rangle\,dt=\left\langle -\frac{1}{2}\cos(2t),\ \frac{1}{5}e^{5t}\right\rangle+\vec{C}$ *(the $\frac{1}{k}$ rule per slot)*

Drawing/choosing the tangent vector at $t=a$

Three facts pin the arrow $\vec{r}'(a)$:

- **Tail** sits at the point $\vec{r}(a)$ on the curve — compute $\vec{r}(a)$ to know where.
- **Tangent** — along the curve, not into or away from it.
- **Forward** — points in the direction of increasing t (the orientation of travel).

Multiple choice: eliminate by two tests — wrong base point (arrow not touching $\vec{r}(a)$) → reject. Wrong direction (backward, or sticking out of the curve) → reject. The survivor is tangent, at the right point, pointing forward.
Sign trick: the signs of $\vec{r}'(a)$ alone often decide — $\langle +,-\rangle$ points right-and-down. Match the picture.

Common pitfalls

- Chain rule per slot: $\cos(5t)$, e^{3t} need the inside's derivative multiplied in — the most common derivative slip.
- Substitute before normalizing: for $\vec{T}(a)$ at a specific $t=a$, plug in first, then divide by the magnitude.
- Cross-product derivative keeps $\vec{u},\vec{v}$ order — anticommutative. Never swap factors within a term.
- $+\vec{C}$ is a constant vector $\langle C_1,C_2,C_3\rangle$, not a scalar.
- Speed vs velocity: speed $\|\vec{r}'\|$ is a scalar. Velocity $\vec{r}'$ is a vector — don't confuse the two.
- Method A and Method B must agree — a mismatch means an arithmetic error to hunt.

arc length

Distance traveled along the curve. One fact generates every formula: arc length is the integral of speed, and speed $=\|\vec{r}'(t)\|$.

The formula

$$L=\int_a^b\|\vec{r}'(t)\|\,dt=\int_a^b\sqrt{[x'(t)]^2+[y'(t)]^2+[z'(t)]^2}\,dt$$

- **2D**: drop the $[z']^2$ term — that's the entire difference.
- **Graph** $y=f(x)$ (the Calc-2 flavor): parameterize as $\langle t,\ f(t)\rangle$, so $x'=1$ supplies the "1+":

$$L=\int_a^b\sqrt{1+[f'(x)]^2}\,dx$$

One formula, not two.

The routine — same four moves every time

Differentiate componentwise → square & sum the slots → simplify under the root → integrate.

- Never integrate the raw root. Problems are rigged so the radicand collapses to a constant (trig slots — Pythagorean identity) or a perfect square ($(A+B)^2$ — check: middle term $=2AB$). If it won't collapse, hunt the missed identity.
- **"Set up only"** ⇒ stop at the integral.

Worked (constant collapse): $\langle\cos t,\sin t,t\rangle$ on $[0,2\pi]$

1. DIFFERENTIATE
$\vec{r}'(t)=\langle-\sin t,\ \cos t,\ 1\rangle$

↓

2. SQUARE & SUM
$\sin^2t+\cos^2t+1=2$ *(Pythagorean identity eats the t)*

↓

3. INTEGRATE
$L=\int_0^{2\pi}\sqrt{2}\,dt=2\sqrt{2}\,\pi$

Constant radicand ⇒ $L=\sqrt{\text{const}}\times(b-a)$ — no antiderivative needed. Any helix $\langle a\cos t,a\sin t,ct\rangle$ collapses the same way, to a^2+c^2.

Worked (perfect square): $\langle t^2,2t,\ln t\rangle$ on $[1,3]$

1. DIFFERENTIATE
$\vec{r}'(t)=\langle 2t,\ 2,\ 1/t\rangle$

↓

2. SQUARE & SUM
$(2t)^2+(2)^2+(1/t)^2=4t^2+4+\dfrac{1}{t^2}$

↓

3. SPOT THE SQUARE
$4t^2+4+\frac{1}{t^2}=\left(2t+\frac{1}{t}\right)^2$ *(check: $2\cdot2t\cdot\frac{1}{t}=4$ = the middle term ✓)*

↓

4. DROP THE ROOT
$\sqrt{\left(2t+\frac{1}{t}\right)^2}=\left|2t+\frac{1}{t}\right|=2t+\frac{1}{t}$ *(positive on $[1,3]$, so the bars drop)*

↓

5. INTEGRATE
$\int_1^3\left(2t+\frac{1}{t}\right)dt=t^2+\ln t\Big|_1^3=(9+\ln 3)-(1+0)=8+\ln 3$

Worked ($y=f(x)$ flavor): $y=\frac{2}{3}x^{3/2}$ on $[0,3]$

1. DIFFERENTIATE
$f'(x)=x^{1/2}$

↓

2. BUILD $1+[f']^2$
$1+\left(x^{1/2}\right)^2=1+x$ *(the curve is chosen so this comes out clean — trust it)*

↓

3. INTEGRATE (u-sub $u=1+x$)
$\int_0^3\sqrt{1+x}\,dx=\frac{2}{3}(1+x)^{3/2}\Big|_0^3=\frac{2}{3}\left(4^{3/2}-1^{3/2}\right)=\frac{2}{3}(8-1)=\frac{14}{3}$

2D sanity check: circle $\langle 3\cos t,3\sin t\rangle$ on $[0,2\pi]$: radicand $=9$, $L=\int_0^{2\pi}3\,dt=6\pi$ — the circumference, as it must be.

"Set up only" looks like this: $\vec{r}=\langle t,t^2,t^3\rangle$ on $[0,1]$: $L=\int_0^1\sqrt{1+4t^2+9t^4}\,dt$ — nothing collapses, so the integral *is* the answer; don't force it.

Arc length function & reparametrization

Make the upper bound variable: $s(t)=\int_0^t\|\vec{r}'(u)\|\,du$ — "distance traveled so far." Invert it (t in terms of s) to **reparametrize by arc length**: the curve traversed at unit speed, where the parameter *is* the distance.

Worked: reparametrize the helix $\langle\cos t,\sin t,t\rangle$

1. ARC LENGTH FUNCTION
speed is the constant $\sqrt{2}$ (worked above): $s(t)=\int_0^t\sqrt{2}\,du=\sqrt{2}t$

↓

2. INVERT
$t=s/\sqrt{2}$

↓

3. SUBSTITUTE BACK
$\vec{r}(s)=\left\langle \cos\frac{s}{\sqrt{2}},\ \sin\frac{s}{\sqrt{2}},\ \frac{s}{\sqrt{2}}\right\rangle$ *(check: $\|\vec{r}\,'(s)\|=1$ — unit speed, one meter of s per meter of curve)*

Common pitfalls

- The integrand is speed, a scalar — integrate $\|\vec{r}\,'\|$, not the vector $\vec{r}\,'$.
- $\sqrt{(\cdot)^2}=|\cdot|$ — confirm the inside is positive on the interval before dropping the bars.
- An un-integrable root means a missed identity or perfect square — go back and simplify.
- Read the ask: **"write"** stops at the integral; **"find the length"** evaluates fully.

motion in space

The componentwise calculus above, applied to physics. The whole topic is one chain:

$$\text{position } \vec{s}(t) \xrightarrow{\ d/dt\ } \text{velocity } \vec{v}(t) \xrightarrow{\ d/dt\ } \text{accel. } \vec{a}(t)$$

reversed by integration: $\vec{a} \xrightarrow{\ \int\ } \vec{v} \xrightarrow{\ \int\ } \vec{s}$. All three are vectors; **speed** $=\|\vec{v}(t)\|$ is the lone scalar.

Which direction are you travelling? — two jobs

A **given position** $\vec{r}(t) \Rightarrow$ differentiate down: $\vec{v}=\vec{r}\,'$, $\vec{a}=\vec{r}\,''$, speed $=|\vec{v}|$

B **given acceleration** plus initial conditions $\Rightarrow$ integrate up: $\vec{v}=\int \vec{a}\,dt$, then $\vec{r}=\int \vec{v}\,dt$. Each integration adds a constant vector, fixed by $\vec{v}(0)$ and $\vec{r}(0)$.

The constant of integration is a vector, not a scalar. Three components, three conditions.

Speed is a scalar, velocity is a vector. Speed is $|\vec{v}|$, and a magnitude discards direction — so an object can have constant speed while its velocity changes every instant.

Worked: the differentiate direction — given $\vec{s}(t)=\langle t^2,\ t^3-3t\rangle$

1. DIFFERENTIATE DOWN THE CHAIN
$\vec{v}(t)=\langle 2t,\ 3t^2-3\rangle$ $\qquad$ $\vec{a}(t)=\langle 2,\ 6t\rangle$

↓

2. SPEED AT $t=1$
$\vec{v}(1)=\langle 2,0\rangle$: $\|\vec{v}(1)\|=2$ — moving purely horizontally at that instant

↓

3. EVER AT REST?
needs both slots zero at once: $2t=0 \Rightarrow t=0$, but $3t^2-3=0 \Rightarrow t=\pm 1$ — no common t $\Rightarrow$ never at rest

↓

4. MOVING STRAIGHT VERTICALLY?
x-velocity $=0$: $t=0$ (and there $\vec{v}(0)=\langle 0,-3\rangle$ — straight down) ✓

Recover $\vec{v}(t)$ and $\vec{s}(t)$ from $\vec{a}(t)$

Integrate up the chain, solving each constant vector from an initial condition. Order matters: finish $\vec{v}(t)$ (solve $\vec{C}_1$) *before* integrating to $\vec{s}(t)$. Then answer the ask — recovery is the setup, not the answer.

Worked: a particle has $\vec{a}(t)=\langle 6t,\ e^t\rangle$ with $\vec{v}(0)=\langle 1,3\rangle$, $\vec{s}(0)=\langle 0,2\rangle$. Find its **position** and **speed** at $t=1$.

1. INTEGRATE $\vec{a}\to\vec{v}$
$\vec{v}(t)=\int\langle 6t,e^t\rangle\,dt+\vec{C}_1=\langle 3t^2,\ e^t\rangle+\vec{C}_1$

↓

2. SOLVE $\vec{C}_1$ FROM $\vec{v}(0)$
$\langle 0,1\rangle+\vec{C}_1=\langle 1,3\rangle \Rightarrow \vec{C}_1=\langle 1,2\rangle$, so $\vec{v}(t)=\langle 3t^2+1,\ e^t+2\rangle$

↓

3. INTEGRATE $\vec{v}\to\vec{s}$
$\vec{s}(t)=\int\langle 3t^2+1,\ e^t+2\rangle\,dt+\vec{C}_2=\langle t^3+t,\ e^t+2t\rangle+\vec{C}_2$

↓

4. SOLVE $\vec{C}_2$ FROM $\vec{s}(0)$
$\langle 0,1\rangle+\vec{C}_2=\langle 0,2\rangle \Rightarrow \vec{C}_2=\langle 0,1\rangle$, so $\vec{s}(t)=\langle t^3+t,\ e^t+2t+1\rangle$

↓

5. POSITION AT $t=1$ (vector)
$\vec{s}(1)=\langle 1+1,\ e+2+1\rangle=\langle 2,\ e+3\rangle \approx \langle 2,5.72\rangle$

↓

6. SPEED AT $t=1$ (scalar)
$\vec{v}(1)=\langle 4,\ e+2\rangle$; $\|\vec{v}(1)\|=\sqrt{4^2+(e+2)^2}\approx 6.19$

Note step 2: $\vec{C}_1 \neq \vec{v}(0)$ here — the antiderivative $\langle 3t^2, e^t\rangle$ is $\langle 0,1\rangle$ at $t=0$, not $\vec{0}$. Solve the constant explicitly every time; the shortcut $\vec{C}_1=\vec{v}(0)$ only happens to work when the antiderivative vanishes at $t=0$ (for example, constant $\vec{a}$).

Projectile motion (gravity)

The recovery above with $\vec{a}$ fixed by physics: $\vec{a}(t)=\langle 0,-g\rangle$, $g=9.8$ m/s^2 or 32 ft/s^2. Nothing pushes sideways after launch, so horizontal velocity never changes — all the action is vertical. Launch from height h_0 at speed v_0, angle θ, always lands on the shape:

$$\vec{v}(t)=\langle v_0\cos\theta,\ v_0\sin\theta-gt\rangle$$

$$\vec{s}(t)=\left\langle (v_0\cos\theta)\,t,\ h_0+(v_0\sin\theta)\,t-\tfrac{1}{2}gt^2\right\rangle$$

Decompose the launch speed by the angle first — $\vec{v}(0)=\langle v_0\cos\theta, v_0\sin\theta\rangle$; using raw v_0 in a slot is the classic error.

Match the ask to the move:

- **Time of flight**: y-slot of $\vec{s}$ = ground height (usually 0); quadratic in t — positive root.
- **Range**: landing time into the x-slot of $\vec{s}$.
- **Max height**: y-slot of $\vec{v}=0 \to t_{\text{peak}} \to$ into the y-slot of $\vec{s}$.
- **Speed**: $\|\vec{v}(t)\|$ — both slots, then magnitude. **Velocity** is the vector itself.

Worked: ground launch — $v_0=20$ m/s, $\theta=30°$, $h_0=0$

1. DECOMPOSE
$\vec{v}(0)=\langle 20\cos 30°,\ 20\sin 30°\rangle=\langle 10\sqrt{3},\ 10\rangle$

↓

2. BUILD $\vec{v},\vec{s}$
$\vec{v}(t)=\langle 10\sqrt{3},\ 10-9.8t\rangle$, $\quad$ $\vec{s}(t)=\langle 10\sqrt{3}t,\ 10t-4.9t^2\rangle$

↓

3. FLIGHT TIME (y-pos $=0$)
$10t-4.9t^2=0 \Rightarrow t(10-4.9t)=0 \Rightarrow t=10/4.9\approx 2.04$ s *($t=0$ = launch)*

↓

4. RANGE (x-pos at landing)
$10\sqrt{3}\,(2.04)\approx 35.3$ m

↓

5. MAX HEIGHT (y-vel $=0$)
$10-9.8t=0 \Rightarrow t\approx 1.02$ s; $\quad 10(1.02)-4.9(1.02)^2\approx 5.1$ m

Worked: launch from a height — $v_0=20$ m/s, $\theta=30°$, $h_0=10$ m

Same set-up, but $h_0 \neq 0$ means the landing equation no longer factors — quadratic formula, keep the positive root.

1. BUILD $\vec{v},\vec{s}$
$\vec{v}(t)=\langle 10\sqrt{3},\ 10-9.8t\rangle$, $\quad$ $\vec{s}(t)=\langle 10\sqrt{3}t,\ 10+10t-4.9t^2\rangle$

↓

2. FLIGHT TIME (quadratic)
$4.9t^2-10t-10=0 \Rightarrow t=\dfrac{10\pm\sqrt{100+196}}{9.8}$;
positive root: $t\approx 2.78$ s *($t\approx -0.74$ discarded)*

↓

3. RANGE
$10\sqrt{3}\,(2.78)\approx 48.1$ m

↓

4. IMPACT SPEED
$\vec{v}(2.78)=\langle 10\sqrt{3},\ 10-9.8(2.78)\rangle\approx\langle 17.3,-17.2\rangle$
$\|\vec{v}\|=\sqrt{17.3^2+17.2^2}\approx 24.4$ m/s

Step 4 is the "how fast does it hit" ask: a scalar — plug the landing time into $\vec{v}$, then take the magnitude. The negative y-slot just says it's moving down.

Worked (inverse): what v_0 at $45°$ lands 40 m away?

1. DERIVE THE RANGE IN SYMBOLS
flight time from $y=0$: $v_0\sin\theta\, t-\frac{1}{2}gt^2=0 \Rightarrow t\left(v_0\sin\theta-\frac{1}{2}gt\right)=0 \Rightarrow t=\frac{2v_0\sin\theta}{g}$; into the x-slot: $R=v_0\cos\theta\cdot\frac{2v_0\sin\theta}{g}=\frac{2{v_0}^2\sin\theta\cos\theta}{g}=\frac{{v_0}^2\sin 2\theta}{g}$ *(double-angle identity)*

↓

2. SET EQUAL TO THE GIVEN
$45° \Rightarrow \sin 2\theta=1$: $\frac{{v_0}^2}{9.8}=40$

↓

3. SOLVE FOR THE UNKNOWN
${v_0}^2=392 \Rightarrow v_0=14\sqrt{2}\approx 19.8$ m/s *(same equations, solved for a different letter)*

Worked: will it clear the wall?

The ground-launch shot above ($v_0=20$, $\theta=30°$) meets a 4 m wall at $x=30$ m:

1. TIME OF ARRIVAL AT THE WALL
$10\sqrt{3}\,t=30 \Rightarrow t=\frac{30}{10\sqrt{3}}=\frac{3}{\sqrt{3}}=\sqrt{3}\approx 1.73$ s

↓

2. HEIGHT AT THAT TIME
$y=10(1.73)-4.9(3)=17.3-14.7=2.6$ m

↓

3. COMPARE
$2.6<4 \Rightarrow$ hits the wall *(location → time → height — always through t)*

Horizontal launch, US units ($\theta=0$, $g=32$): ball rolls off a 64 ft cliff at 10 ft/s: $\vec{s}(t)=\langle 10t,\ 64-16t^2\rangle$; lands $64-16t^2=0 \Rightarrow t=2$ s, 20 ft out — with $\theta=0$ the x-slot is just v_0t.

Word problems — translation dictionary

No new formulas, just translation. English → math:

- "starts at / initial position" → $\vec{s}(0)$; "launched at / initial velocity" → $\vec{v}(0)$ (decompose if speed + angle given)
- "under gravity / free fall" → $\vec{a}=\langle 0,-g\rangle$
- "how long until it hits the ground" → y-position $=0$, solve for t; "how far away does it land" → landing t into x-position
- "highest point" → y-velocity $=0$
- "how fast" → speed $\|\vec{v}\|$ (a number); "which direction / velocity vector" → the vector $\vec{v}(t)$

Approach: pull out the givens → match the ask to a move above → integrate, solve the constants → answer as the right type (scalar or vector) with units.

Common pitfalls

- Constants of integration are vectors — solve $\vec{C}_1,\vec{C}_2$ per component, explicitly, every time.
- Landing = y-position 0; max height = y-velocity 0 — the two most-confused conditions. Don't swap them.
- Gravity is negative and vertical: $\langle 0,-g\rangle$ — sign and slot both matter.
- Take the positive time root — negative times are unphysical.
- Speed is $\|\vec{v}\|$ (scalar); velocity and acceleration are vectors — answer in the type the question wants.

functions of several variables

A **function of two variables** $z=f(x,y)$ takes a point (x,y) of the plane in and hands one number back. The **domain** is now a region of the xy-plane (not an interval); the **range** is still a set of real numbers. Evaluate by substituting both coordinates: $f(x,y)=x^2+3y \Rightarrow f(2,-1)=4-3=1$.

Domain — the same three blockers as Calc 1

Ban exactly what breaks: **even roots** need $(\cdot)\geq 0$, **logs** need $(\cdot)>0$, **denominators** need $\neq 0$. The inequality now describes a 2D region — sketch it: solid boundary curve for $\geq/\leq$, dashed for a strict inequality.

Worked: domain & range of $f(x,y)=\sqrt{9-x^2-y^2}$

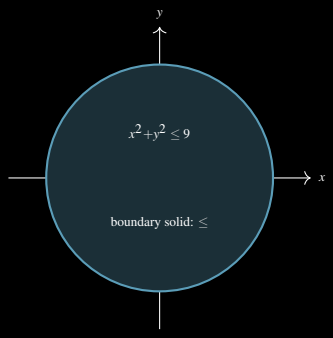
1. BAN THE BREAKAGE
even root needs $9-x^2-y^2\geq 0$

↓

2. REARRANGE INTO A KNOWN SHAPE
$x^2+y^2\leq 9$: the closed disk, center $(0,0)$, radius 3 *(radius $=\sqrt{9}=3$, not 9)*

↓

3. RANGE FROM THE EXTREMES
biggest at the center: $f(0,0)=\sqrt{9}=3$; smallest on the boundary circle: $\sqrt{0}=0$ ⇒ range $=[0,3]$



More domains, same move: $g(x,y)=\ln(y-x^2)$: $y>x^2$ — strictly *above* the parabola, dashed boundary. $h(x,y)=\frac{1}{x-y}$: the whole plane minus the line $y=x$.

Graphs — the surface $z=f(x,y)$

The graph of f is the set of points $(x,y,f(x,y))$: a surface floating over the domain, one height z above each input point. Sketch route: identify the surface from the quadric section, then confirm with traces ($x=k$, $y=k$, $z=k$).

- **Plane**: $f(x,y)=ax+by+c$ — linear in x,y, always a plane.
- **Elliptic paraboloid**: $f(x,y)=x^2+y^2$ — the bowl.
- **Hemisphere**: $f(x,y)=\sqrt{9-x^2-y^2}$ — the *upper half* of the sphere $x^2+y^2+z^2=9$; the root's output is ≥ 0, so $z\geq 0$.
- **Cone, top half**: $f(x,y)=\sqrt{x^2+y^2}$.

A full sphere is not a graph: it fails the vertical line test — two z's over one (x,y), but a function produces one height per point. Solving $x^2+y^2+z^2=9$ for z honestly splits it: $z=\pm\sqrt{9-x^2-y^2}$, two hemispheres, two functions.

Level curves & contour maps

Set the *output* to a constant: the **level curve** at value c is the curve $f(x,y)=c$, drawn flat in the xy-plane. It is the $z=c$ trace of the surface, projected down — a topographic map of the graph. Several c's together form a **contour map**. Read steepness from the spacing: tight rings = steep, wide rings = flat.

Worked: level curves of $f(x,y)=x^2+y^2$ at $c=1,2,3,4$

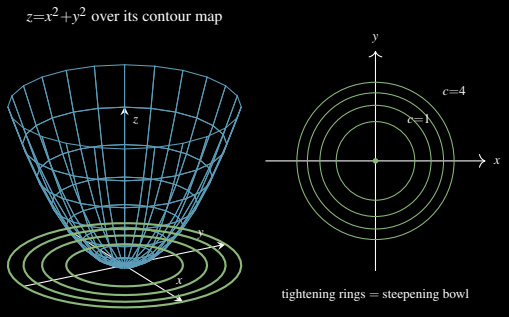
1. SET $f=c$
$x^2+y^2=c$: a circle of radius $\sqrt{c}$ for any $c>0$

↓

2. DRAW EACH c
radii $\sqrt{1}$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{4}=1,\ 1.41,\ 1.73,\ 2$ — concentric circles that crowd together going outward: the bowl steepens

↓

3. EDGE CASES
$c=0$: the single point $(0,0)$ (bottom of the bowl). $c<0$: empty — a c below the range has no curve; don't invent one.



Worked: non-circular contours — $f=x^2-y^2$ (the saddle's map)

1. SET $f=c$
$x^2-y^2=c$: hyperbolas — which way they open depends on the sign of c

↓

2. THE THREE FAMILIES
$c>0$: open along x; $c<0$: open along y; $c=0$: the crossed lines $y=\pm x$

↓

3. READ THE SURFACE OFF THE MAP
two hyperbola families meeting at crossed lines = the saddle's signature — contrast the bowl's concentric circles *(reverse skill: given a contour map, name the surface)*

Three or more variables — level surfaces

$w=f(x,y,z)$ takes a point of *space* in. Its graph would need 4 axes, so you can't draw it. The drawable object is the **level surface**: $f(x,y,z)=c$, a surface in 3D — always a friend from the quadric section.

- $f=x^2+y^2+z^2$: level surfaces $x^2+y^2+z^2=c$ — concentric spheres of radius $\sqrt{c}$ for $c>0$; a single point at $c=0$; empty for $c<0$.
- $f=x^2+y^2-z^2$: $c>0 \Rightarrow$ hyperboloid of one sheet; $c=0 \Rightarrow$ cone; $c<0 \Rightarrow$ two sheets — one function morphing through three quadrics as c crosses 0.
- **Domain logic is unchanged**: $f(x,y,z)=\dfrac{1}{\sqrt{z-x^2-y^2}}$ needs $z>x^2+y^2$ — the solid region above a paraboloid, boundary excluded.

General pattern: a function of n variables assigns one output to each n-tuple, and its **level sets** live one dimension down from the graph — curves for 2 variables, surfaces for 3.

Common pitfalls

- Domain is a region, not an interval — answer with an inequality *and* a sketch. Boundary solid for $\geq$, dashed for $>$.
- Level curves live in the xy-plane — flat, no z-axis on the drawing. The 3D slice at $z=c$ is a **trace**; its shadow on the floor is the level curve.
- c outside the range $\Rightarrow$ empty level curve — check before drawing ($x^2+y^2=-1$ has no points).
- Graph $\neq$ solid: $z=f(x,y)$ is an infinitely thin surface, one height per point. A full sphere needs *two* functions.
- 2 vars $\rightarrow$ level *curves*; 3 vars $\rightarrow$ level *surfaces* — you draw one dimension down from the input count.

multivariable limits & continuity

$\lim\limits_{(x,y)\to(a,b)} f(x,y)=L$ means f gets arbitrarily close to L as (x,y) closes in on (a,b) — along every possible path. In Calc 1 a point had only two sides (left/right); a point of the plane has infinitely many approach paths (lines, parabolas, spirals…), and all of them must agree for the limit to exist.

Which job is this? — three outcomes

A **substitute and it works** $\Rightarrow$ that is the limit. Always try this first.

B **show the limit does NOT exist** $\Rightarrow$ find two paths giving different values. One counterexample finishes it.

C **prove the limit DOES exist** $\Rightarrow$ paths can never prove existence. Use the squeeze theorem or convert to polar.

B and C need opposite tools, so decide which you are doing before you start. Trying paths for hours cannot settle case C, and a squeeze argument is wasted effort in case B.

Limit laws — same sheet as Calc 1

Every one-variable limit law carries over unchanged — the only edit is the point: $x \to a$ becomes $(x,y) \to (a,b)$. Below, every $\lim$ means $\lim\limits_{(x,y)\to(a,b)}$, with $\lim f=L$ and $\lim g=M$ both assumed to exist.

- **Constant**: $\lim c=c$
- **Identity**: $\lim x=a$, $\lim y=b$
- **Constant Multiple**: $\lim[c\cdot f]=cL$
- **Sum/Diff**: $\lim[f\pm g]=L\pm M$
- **Product**: $\lim[f\cdot g]=LM$
- **Quotient**: $\lim\frac{f}{g}=\frac{L}{M}$, if $M\neq 0$
- **Power**: $\lim[f]^n=L^n$
- **Root**: $\lim\sqrt[n]{f}=\sqrt[n]{L}$ ($L>0$ when n even)
- **Composition**: g continuous at $L \Rightarrow \lim g(f(x,y))=g(L)$

The **squeeze theorem** carries over the same way: $g\leq f\leq h$ near (a,b) and $\lim g=\lim h=L \Rightarrow \lim f=L$. The fine print matters more now: each law *requires* the pieces' limits to exist — with infinitely many paths, that's exactly what fails in the DNE examples below, and no law applies to them.

Evaluating — substitute first

The limit laws (sum, product, quotient, powers, composition) all carry over, and polynomials/rational functions are continuous wherever defined — so step 1 is always: plug the point in.

- Get a number $\Rightarrow$ done, that's the limit:
  $\lim\limits_{(x,y)\to(2,-1)} \dfrac{x^2y+2y}{x-y}=\dfrac{-4-2}{2+1}=-2.$
- $\dfrac{\text{nonzero}}{0} \Rightarrow$ blow-up: limit does not exist, immediately.
- $\dfrac{0}{0} \Rightarrow$ indeterminate — more work: try algebra (factor, conjugate); if nothing cancels, suspect DNE and hunt for two disagreeing paths.

Worked: $0/0$ rescued by algebra

$$\lim_{(x,y)\to(1,1)} \frac{x^2-y^2}{x-y}$$

1. SUBSTITUTE
$\frac{1-1}{1-1}=\frac{0}{0}$ — indeterminate; don't stop, don't declare DNE

↓

2. FACTOR & CANCEL
$\dfrac{(x-y)(x+y)}{x-y}=x+y$ *(valid on the approach: points near $(1,1)$ with $x\neq y$)*

↓

3. SUBSTITUTE AGAIN
$\lim\limits_{(x,y)\to(1,1)}(x+y)=2$ ✓

Conjugate flavor, same move — multiply by the conjugate, the top becomes xy, cancel:

$$\lim_{(x,y)\to(0,0)} \frac{\sqrt{xy+1}-1}{xy}\cdot\frac{\sqrt{xy+1}+1}{\sqrt{xy+1}+1}=\lim_{(x,y)\to(0,0)}\frac{1}{\sqrt{xy+1}+1}=\frac{1}{2}$$

The two-path test — proving a limit does NOT exist

If two different paths into (a,b) give two different values, the limit does not exist. Substituting a path's equation turns the problem into a one-variable limit. Standard path menu: the axes ($y=0$, $x=0$), lines through the point ($y=mx$), power curves ($y=x^2$, $x=y^2$).

The test only ever proves nonexistence. Two — or two hundred — agreeing paths prove *nothing*: agreement on every straight line can still hide a disagreeing parabola (example below).

Worked: show $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ does not exist

1. SUBSTITUTE
$\frac{0}{0}$ — indeterminate. Nothing factors ⇒ try paths

↓

2. PATH 1 — THE x-AXIS ($y=0$)
$f(x,0)=\frac{0}{x^2}=0 \longrightarrow 0$

↓

3. PATH 2 — THE LINE $y=x$
$f(x,x)=\frac{x^2}{2x^2}=\frac{1}{2} \longrightarrow \frac{1}{2}$

↓

4. COMPARE
$0 \neq \frac{1}{2}$ ⇒ the limit does not exist ✓

Riding in on a general line $y=mx$: $f=\frac{mx^2}{x^2+m^2x^2}=\frac{m}{1+m^2}$ — the "limit" depends on the slope you approach along, which is exactly what nonexistence looks like.

Worked: lines are not enough — $\frac{xy^2}{x^2+y^4}$

1. TRY EVERY LINE $y=mx$
$f(x,mx)=\frac{x(mx)^2}{x^2+(mx)^4}=\frac{m^2x^3}{x^2+m^4x^4}=\frac{m^2x}{1+m^4x^2} \to 0$ — all lines agree *($\div x^2$ top and bottom)*

↓

2. DON'T CONCLUDE
agreement proves nothing — hunt a sneakier path

↓

3. RIDE THE PARABOLA $x=y^2$
$\frac{y^2\cdot y^2}{y^4+y^4}=\frac{1}{2}$

↓

4. COMPARE
$0 \neq \frac{1}{2}$ ⇒ DNE — only disagreement is ever conclusive ✓

Worked: proving a limit EXISTS — $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2}$

The dual problem: paths can never *prove* existence — a squeeze (or polar) can:

1. BOUND THE FRACTION
$\frac{x^2}{x^2+y^2} \le 1$ always, so $\left|\frac{x^2y}{x^2+y^2}\right| \le |y|$

↓

2. SQUEEZE
$0 \le |f| \le |y| \to 0$ as $(x,y)\to(0,0)$ ⇒ $\lim=0$ ✓

↓

3. POLAR ROUTE — SAME VERDICT
$x=r\cos\theta$, $y=r\sin\theta$: $f=\frac{r^3\cos^2\theta\sin\theta}{r^2}=r\cos^2\theta\sin\theta$, and $|f| \le r \to 0$ *(a surviving r factor kills the θ-dependence — that's what existence looks like)*

Continuity

Same three-part checklist as Calc 1, now at a point of the plane. f is **continuous** at (a,b) iff:
1. $f(a,b)$ is defined,
2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists (every path agrees),
3. the limit equals $f(a,b)$.

Continuous catalog: polynomials everywhere; rational functions wherever the denominator $\neq 0$; roots, trig, $e^{(\cdot)}$, ln on their domains. And sums, products, and compositions of continuous functions stay continuous.

Continuity questions thus reduce to domain questions: $f(x,y)=\frac{x-y}{x^2+y^2-9}$ is continuous everywhere *except* the circle $x^2+y^2=9$.

Piecewise worked: $f=\frac{x^2y}{x^2+y^2}$ for $(x,y) \neq (0,0)$, $f(0,0)=0$: the squeeze above gave $\lim=0=f(0,0)$ ⇒ continuous at the origin. Had the definition said $f(0,0)=1$: limit exists but misses the value — discontinuous (the removable flavor).

Common pitfalls

- $0/0$ does not mean DNE — it means *undecided*: algebra first, paths second.
- The two-path test cannot prove a limit exists — agreement proves nothing. To establish existence, use substitution/algebra (or a squeeze).
- Paths must pass through the target point — $y=x+1$ is useless for a limit at the origin.
- nonzero$/0$ is not indeterminate — it is an immediate DNE (the function blows up).
- After substituting a path, the limit is one-variable — finish it with Calc 1 tools.

partial derivatives

A **partial derivative** is an ordinary derivative taken in one input direction at a time: differentiate with respect to one variable while freezing every other variable as a constant.

$$f_x=\frac{\partial f}{\partial x}=\lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h} \qquad f_y=\frac{\partial f}{\partial y}=\lim_{h\to 0}\frac{f(x,y+h)-f(x,y)}{h}$$

All Calc 1 rules (product, quotient, chain) apply unchanged — the frozen variable just rides along like the number 7 would.

Worked: both partials of $f(x,y)=x^2y^3+\sin(xy)$

1. f_x — FREEZE y
x^2y^3: constant y^3 times $x^2 \to 2xy^3$. $\sin(xy)$: chain rule, inner derivative $\frac{\partial}{\partial x}(xy)=y \to y\cos(xy)$. ⇒ $f_x=2xy^3+y\cos(xy)$

↓

2. f_y — FREEZE x
x^2y^3: constant x^2 times $y^3 \to 3x^2y^2$. $\sin(xy)$: inner derivative is now $x \to x\cos(xy)$. ⇒ $f_y=3x^2y^2+x\cos(xy)$

↓

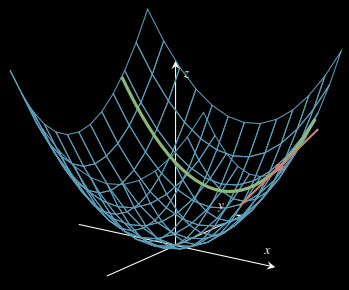
3. VALUES — PLUG THE POINT IN LAST
at $(0,3)$: $f_x(0,3)=0+3\cos 0=3$, $f_y(0,3)=0+0\cdot\cos 0=0$ ✓

Differentiate first, evaluate second — substituting the point before differentiating turns f into a constant and every "derivative" into 0.

Interpretation — slope of a trace

$f_x(a,b)$ is the slope of the surface in the x-direction at (a,b): slice the surface with the vertical plane $y=b$, get the trace curve $z=f(x,b)$, and $f_x(a,b)$ is that curve's ordinary Calc 1 slope at $x=a$. Likewise $f_y(a,b)$ slices with $x=a$. Units: output units per unit of that input, other input pinned — for a temperature plate $T(x,y)$, $T_x(a,b)$ is degrees gained per meter stepped in the x-direction. Sign reads as always: positive = increasing in that direction.

$f_x(1,1)$ on $z=x^2+y^2$



$z=x^2+1$: the $y=1$ trace; tangent at $(1,1,2)$, slope $f_x(1,1)=2x\big|_{x=1}=2$.

Interpreting the value *— units and sign*

Exams ask what a partial derivative means, not just what it equals. Two facts answer every such question:

READING ANY RATE

Units $=\dfrac{\text{units of the output}}{\text{units of the input that varies}}$ — read them straight off the problem statement.

Sign = increasing or decreasing as that one input grows.

f_x holds y fixed. It describes motion parallel to the x-axis and says nothing about any other direction.

Worked (in context): temperature on a plate

$T(x,y)=99-x^2+xy+3y$ gives the temperature in °C at the point (x,y), with x and y measured in metres east and north. Interpret both partials at $(2,1)$.

1. THE PARTIALS, THEN THE VALUES

$T_x=-2x+y, \quad T_y=x+3$

$T_x(2,1)=-4+1=-3 \qquad T_y(2,1)=2+3=5$

↓

2. ATTACH THE UNITS

output is °C, inputs are metres, so both rates are in °C per metre.

↓

3. SAY IT IN WORDS — SIGN CARRIES THE MEANING

$T_x(2,1)=-3$: moving east from $(2,1)$, the temperature falls about 3 °C per metre.

$T_y(2,1)=5$: moving north, it rises about 5 °C per metre.

("about" is honest — the rate is exact only at the instant you leave the point.)

Neither number describes walking northeast. A partial derivative only ever looks along an axis. For any other direction you need a directional derivative — worked below, on this same plate.

Three or more variables

Same rule, more freezes: for $f(x,y,z)$, each partial holds the *two* other variables constant.

Worked: all three partials of $f=x^2y-yz^3$

1. f_x — FREEZE y,z

x^2y: constant y times $x^2 \to 2xy$; yz^3: no x at all $\to 0$. $\Rightarrow$ $f_x=2xy$

↓

2. f_y — FREEZE x,z

both terms are (constant)$\cdot y$: $f_y=x^2-z^3$ *(the frozen z^3 survives as a coefficient)*

↓

3. f_z — FREEZE x,y

only $-yz^3$ has z: $f_z=-3yz^2$

Second partials — there are four

Differentiate twice, choosing a variable each time: pure $f_{xx}=(f_x)_x$, $f_{yy}=(f_y)_y$; mixed $f_{xy}=(f_x)_y$, $f_{yx}=(f_y)_x$. Read the notation in the right direction: subscripts left-to-right (f_{xy} = "x first, then y"); Leibniz right-to-left ($\frac{\partial^2 f}{\partial y\,\partial x}$ = *also* "x first, then y") — the two notations name the same thing in opposite reading order.

Worked: all four second partials of $f(x,y)=x^3y^2-3xy$

1. FIRST PARTIALS

$f_x=3x^2y^2-3y \qquad f_y=2x^3y-3x$

↓

2. PURE SECONDS

$f_{xx}=(f_x)_x=6xy^2 \qquad f_{yy}=(f_y)_y=2x^3$

↓

3. MIXED SECONDS

$f_{xy}=(f_x)_y=6x^2y-3 \qquad f_{yx}=(f_y)_x=6x^2y-3$

↓

4. CHECK

$f_{xy}=f_{yx}$ ✓ *(Clairaut says they had to match)*

Clairaut's theorem — mixed partials agree

If f_{xy} and f_{yx} are both continuous on a disk around a point, then $f_{xy}=f_{yx}$ there. Every function in this course qualifies, so the theorem doubles as a free error check: compute both mixed partials — if they disagree, a differentiation slipped somewhere.

Common pitfalls

- Frozen $\neq$ erased: $\frac{\partial}{\partial x}(xy)=y$ — the constant multiplier survives. But $\frac{\partial}{\partial x}(y^3)=0$ — no x at all.
- Chain rule still fires: $\frac{\partial}{\partial x}\sin(xy)=y\cos(xy)$ — the inner derivative is the frozen variable.
- Notation direction: $\frac{\partial^2 f}{\partial y\,\partial x}=f_{xy}$, not f_{yx} (Clairaut usually bails you out, but quote the right one).
- Evaluate after differentiating, never before — a substituted point is a constant with derivative 0.
- $f_x(a,b)$ is a slope in a vertical slice (y pinned at b), not a rate along a diagonal — each partial owns exactly one axis direction.

tangent planes & linear approximation

Calc 1's tangent line, leveled up: at (a,b) the surface $z=f(x,y)$ has one height and two slopes, and the flat object carrying all of that is a **tangent plane**. Build it like point–slope form, once per direction — start at the height, add slope $\times$ run for each input:

$$z=f(a,b)+f_x(a,b)\,(x-a)+f_y(a,b)\,(y-b)$$

Compare Calc 1's $y=f(a)+f'(a)(x-a)$ — same skeleton, one more term. Rearranged, the plane's **normal vector** is $\langle f_x(a,b),\, f_y(a,b),\, -1\rangle$.

How is the surface written? — two forms, one machine

A **a graph,** $z=f(x,y) \Rightarrow z=f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$

B **a level surface,** $F(x,y,z)=c \Rightarrow \nabla F$ is the normal: $\nabla F\cdot\langle x-x_0, y-y_0, z-z_0\rangle=0$

A is B in disguise. Put the graph in level-surface form as $F=f(x,y)-z$, and $\nabla F=\langle f_x, f_y, -1\rangle$ — which is the coefficient pattern in A. Use B whenever z cannot be isolated, as in $x^2+4y^2+z^2=18$. The level-surface case is worked in the gradient section below.

Evaluate the partials at the point before building the plane. Leaving f_x as a formula produces a surface, not a plane.

Worked: tangent plane to $z=x^2+y^2$ *at* $(1,2)$

1. HEIGHT

$f(1,2)=1+4=5$ — the plane must pass through $(1,2,5)$

↓

2. SLOPES — EVALUATE AT THE POINT

$f_x=2x \;\to\; f_x(1,2)=2 \qquad f_y=2y \;\to\; f_y(1,2)=4$ *(numbers, not formulas, go in the plane)*

↓

3. ASSEMBLE

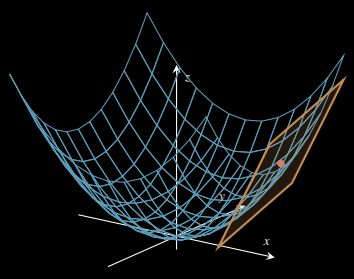
$z=5+2(x-1)+4(y-2)$

↓

4. SIMPLIFY & CHECK

distribute: $z=5+2x-2+4y-8$; collect: $z=2x+4y-5$; plug the point back: $2+8-5=5$ ✓ *(the plane must contain the point of tangency)*

same recipe at $(1,1)$: tangent plane $(1,1,2)$ on $z=x^2+y^2$



$z=2+2(x-1)+2(y-1)$ at $(1,1)$ — the f_x-trace tangent and the $D_{\vec{u}}$ tangent from the nearby figures both lie inside this plane: the plane is what all the direction-slopes sweep out.

Linear approximation — the plane as a calculator

Near (a,b) the surface hugs its tangent plane, so the plane's height estimates the function:

$$f(x,y) \approx L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

The move: base point = a nearby *nice* point (exact values by hand); target = the ugly point the problem actually asks about.

Worked: estimate $\sqrt{3.06^2+3.92^2}$ by hand

1. NAME f AND PICK THE NICE BASE
$f(x,y)=\sqrt{x^2+y^2}$; the 3-4-5 triangle sits next door: $(a,b)=(3,4)$, $f(3,4)=5$

↓

2. PARTIALS AT THE BASE
chain rule on the root: $f_x = \frac{1}{2}(x^2+y^2)^{-1/2}\cdot 2x = \frac{x}{\sqrt{x^2+y^2}}$; at $(3,4)$: $\frac{3}{\sqrt{9+16}} = \frac{3}{5} = 0.6$; likewise $f_y = \frac{4}{5} = 0.8$

↓

3. BUILD L
$L(x,y) = 5 + 0.6(x-3) + 0.8(y-4)$

↓

4. PLUG THE UGLY POINT
$L(3.06, 3.92) = 5 + 0.6(0.06) + 0.8(-0.08) = 5 + 0.036 - 0.064 = 4.972$

Exact value: $4.97293\ldots$ — the aha: two small multiplications replaced a square root, and the error only shows up in the 4th digit. That's the whole point of the tangent plane.

Differentials — error propagation

Rename the runs $dx = x-a$, $dy = y-b$; the plane's *rise* is the **total differential**:

$$dz = f_x\,dx + f_y\,dy \qquad \Delta z \approx dz$$

dz estimates the change in f when the inputs wiggle — the tool for measurement error.

Worked: area error — $A=xy$, $x=30\pm0.1$, $y=20\pm0.1$

1. PARTIALS
$A_x = y = 20, \qquad A_y = x = 30$

↓

2. TOTAL DIFFERENTIAL
$dA = 20(0.1) + 30(0.1) = 5$ — worst-case area error ± 5

↓

3. SANITY CHECK VS THE TRUE CHANGE
$30.1 \cdot 20.1 = 605.01$: $605.01 - 600 = 5.01$ — dz nailed it ✓

Percent-error flavor: for products/powers, relative errors add with their exponents: $V=\pi r^2 h$ with r off by 2% and h by 1%: $\frac{dV}{V} = 2\frac{dr}{r} + \frac{dh}{h} = 2(2\%) + 1\% = 5\%$ *(take* ln*, then differentiate — logarithmic differentiation's day job)*.

Differentiability — when the plane is trustworthy

f is **differentiable** at (a,b) when the tangent-plane error dies faster than the distance to (a,b) — that is, the plane is a genuinely good local stand-in. Practical test: if f_x and f_y exist and are continuous near the point, f is differentiable there (everything in this course). Differentiable ⇒ continuous. Partials merely existing is not enough: $\frac{xy}{x^2+y^2}$ has both partials at the origin yet isn't even continuous there (the two-path example) — no tangent plane exists.

Common pitfalls

- Numbers, not formulas, in the plane: evaluate f_x, f_y *at* (a,b) before assembling — a leftover $2x$ in the plane equation is the classic slip.
- Subtract the base point: the runs are $(x-a)$ and $(y-b)$ — sign errors here shift the whole plane.
- dz is the change, not the value: new value $\approx f(a,b) + dz$; don't report dz alone when asked for the estimate of f.
- Local only: L degrades as you leave (a,b) — pick the nice base point *adjacent* to the target, not the nicest one anywhere.
- Check step 4 always: the plane must reproduce $f(a,b)$ at the point — free error catch.

multivariable chain rule

Calc 1 chained *one* inner function: $\frac{d}{dt} f(g(t)) = f'(g(t))\,g'(t)$. Now $z=f(x,y)$ where x and y *both* depend on t: z changes because x moves it *and* because y moves it, and the total rate is the sum of one chain-rule product per route from z down to t:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

Which tree? — three shapes

A **one independent variable** (x,y both depend on t) ⇒ a total derivative: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$

B **two independent variables** (x,y depend on u,v) ⇒ one partial per independent variable, each a full sum

C **implicit** — no explicit formula, only $F(x,y)=0$ ⇒ $\frac{dy}{dx} = -\frac{F_x}{F_y}$

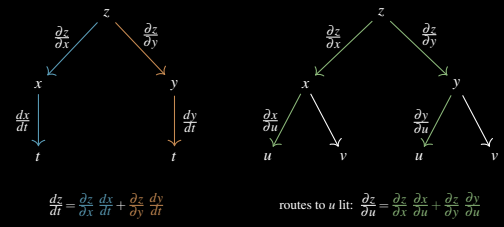
THE PROCESS
1. **Draw the tree.** Put the output on top and the independent variables at the bottom.
2. **Count the paths** from the output down to the variable you differentiate by.
3. **Multiply along each path**, then **add the paths**.
4. **Substitute** the inner functions only at the end.

Build the formula before you compute anything. The tree tells you how many terms to expect, so a missing term becomes visible immediately.

Tree diagrams — build the formula before computing

Draw the dependency tree and the formula assembles itself: multiply down each route, add the routes.

1. **Top**: the dependent variable (z). **Middle**: the intermediates (x,y). **Bottom**: the independent variable(s) (t; or u,v).
2. **Label each edge** with the derivative of its upper node with respect to its lower node. Count the inputs to pick the letter: upper node depends on *one* variable ⇒ d; on *several* ⇒ ∂.
3. For a target derivative, trace every route from the top to that bottom variable: multiply the edge labels along a route, then add the routes.



The urge to cancel — wrong as algebra, useful as a check

The notation begs you to cancel: it *looks* like the ∂x's in $\frac{\partial z}{\partial x}\frac{dx}{dt}$ should cancel to leave $\frac{dz}{dt}$. Partials are not fractions and never cancel — if they did, *each* term would equal $\frac{dz}{dt}$ on its own and the formula would collapse to the nonsense $\frac{dz}{dt} = \frac{dz}{dt} + \frac{dz}{dt}$. Each $\frac{\partial z}{\partial x}$ is a single limit taken with the other variable frozen. The top and bottom don't split apart.

But the would-cancel pattern is a legal bookkeeping check: in a correctly built term the middles match — the variable under the first ∂ reappears on top of the next factor, so every route reads $z \to$ intermediate $\to t$ with no gaps, and each intermediate shows up in exactly one term. If the pattern *wouldn't* cancel cleanly, a term is mis-written. Verify with it. Never derive with it.

Worked: $\frac{dz}{dt}$ ***for*** $z=x^2y, \quad x=t^2, \quad y=t^3$

1. PARTIALS OF THE OUTER
$\frac{\partial z}{\partial x}=2xy \qquad \frac{\partial z}{\partial y}=x^2$

↓

2. DERIVATIVES OF THE INNER
$\frac{dx}{dt}=2t \qquad \frac{dy}{dt}=3t^2$

↓

3. MULTIPLY EACH ROUTE, ADD
$\frac{dz}{dt}=(2xy)(2t)+(x^2)(3t^2)$

↓

4. SUBSTITUTE $x(t), y(t)$ BACK
$=(2t^2t^3)(2t)+(t^4)(3t^2)=4t^6+3t^6=7t^6$ *(the answer must become in t only)*

↓

5. CHECK — SUBSTITUTE FIRST INSTEAD
$z=(t^2)^2t^3=t^7 \;\Rightarrow\; \frac{dz}{dt}=7t^6$ ✓

The aha: when the pieces are simple you can always substitute first and differentiate with Calc 1 — the chain rule *must* agree. It earns its keep when substitution is ugly or when $x(t), y(t)$ are only known numerically.

Worked: riding a level curve — $z=x^2+y^2, \quad x=\cos t, \quad y=\sin t$

1. ROUTES
$\frac{dz}{dt}=(2x)(-\sin t)+(2y)(\cos t)$

↓

2. SUBSTITUTE
$=-2\cos t\sin t+2\sin t\cos t=0$

The path $(\cos t, \sin t)$ walks the unit circle — a level curve of the bowl — so the height never changes. **Preview**: the gradient section explains this as ∇f being perpendicular to the direction of motion.

Two independent variables

Same tree, wider bottom row: $z=f(x,y)$ with $x=x(u,v)$, $y=y(u,v)$. Two targets, two routes each (all derivatives now ∂ — every node has several inputs somewhere below):

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \qquad \frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

Worked: $z=x^2+xy, \quad x=u+v, \quad y=u-v$

1. ALL SIX EDGE LABELS
$\frac{\partial z}{\partial x}=2x+y, \;\; \frac{\partial z}{\partial y}=x \qquad \frac{\partial x}{\partial u}=1, \;\; \frac{\partial x}{\partial v}=1, \;\; \frac{\partial y}{\partial u}=1, \;\; \frac{\partial y}{\partial v}=-1$

↓

2. ROUTES TO u
$\frac{\partial z}{\partial u}=(2x+y)(1)+(x)(1)=3x+y$

↓

3. ROUTES TO v
$\frac{\partial z}{\partial v}=(2x+y)(1)+(x)(-1)=x+y$

↓

4. SUBSTITUTE BACK
$\frac{\partial z}{\partial u}=3(u+v)+(u-v)=4u+2v \qquad \frac{\partial z}{\partial v}=(u+v)+(u-v)=2u$

↓

5. CHECK — SUBSTITUTE FIRST
$z=(u+v)^2+(u+v)(u-v)=2u^2+2uv \;\Rightarrow\; z_u=4u+2v, \; z_v=2u$ ✓

Generalized chain rule

Any number of intermediates, any number of independents: for $w=f(x_1,\ldots,x_m)$ with each x_i a function of $t_1,\ldots,t_n$,

$$\frac{\partial w}{\partial t_j}=\frac{\partial w}{\partial x_1}\frac{\partial x_1}{\partial t_j}+\frac{\partial w}{\partial x_2}\frac{\partial x_2}{\partial t_j}+\cdots+\frac{\partial w}{\partial x_m}\frac{\partial x_m}{\partial t_j}$$

Term count = number of intermediate variables — one term per route. Draw the tree when in doubt. A missing branch is the classic miss.

Worked: three intermediates — $w=xyz, \quad x=t, \; y=t^2, \; z=t^3$

1. THREE ROUTES
$\frac{dw}{dt}=yz\cdot 1+xz\cdot 2t+xy\cdot 3t^2$

↓

2. SUBSTITUTE BACK
$(t^2)(t^3)(1)+(t)(t^3)(2t)+(t)(t^2)(3t^2)=t^5+2t^5+3t^5=6t^5$

↓

3. CHECK — SUBSTITUTE FIRST
$w=t\cdot t^2\cdot t^3=t^6 \Rightarrow \frac{dw}{dt}=6t^5$ ✓ *(three routes, three terms — drop one and the check fails)*

Worked: chain-rule related rates — the exam's favorite costume

A cylinder's radius grows at 2 cm/s while its height shrinks at 3 cm/s. How fast is the volume changing at $r=5$, $h=10$?

1. NAME THE CHAIN
$V=\pi r^2h$ with $r(t)$, $h(t)$: $\quad \frac{dV}{dt}=\frac{\partial V}{\partial r}\frac{dr}{dt}+\frac{\partial V}{\partial h}\frac{dh}{dt}$

↓

2. PARTIALS
$V_r=2\pi rh, \qquad V_h=\pi r^2$

↓

3. PLUG THE SNAPSHOT — VALUES LAST
$\frac{dV}{dt}=2\pi(5)(10)(2)+\pi(25)(-3)=200\pi-75\pi=125\pi \text{ cm}^3/\text{s}$ *(shrinking h enters as -3 — signs carry the physics)*

***Interpreting the value — a chain rule rate is* per unit time**

$\frac{dz}{dt}$ differs from a partial in one way that decides its units: the input is time, so the answer is

$$\frac{dz}{dt} \text{ in } \frac{\text{units of } z}{\text{unit of time}}$$

One number, even though two inputs move at once. Each term of the sum is one input's contribution; the total is what the moving object actually experiences.

Worked (in context): the bug on the plate

The same plate, $T(x,y)=99-x^2+xy+3y$ in °C with x, y in metres. A bug crawls along $x(t)=2+t, \quad y(t)=1+2t$, with t in seconds. How fast is the temperature changing for the bug at $t=0$?

1. WHAT THE BUG'S MOTION SUPPLIES
at $t=0$ the bug is at $(2,1)$; $\quad x'(t)=1$ and $y'(t)=2$, so it moves 1 m/s east and 2 m/s north

↓

2. THE PARTIALS THERE — ALREADY COMPUTED ABOVE
$T_x(2,1)=-3, \quad T_y(2,1)=5$ *(both in °C per metre)*

↓

3. CHAIN RULE — ONE TERM PER INPUT
$$\frac{dT}{dt}=T_x\frac{dx}{dt}+T_y\frac{dy}{dt}=(-3)(1)+(5)(2)=-3+10=7$$

↓

4. UNITS, THEN THE SENTENCE
$\frac{\text{°C}}{\text{m}}\cdot\frac{\text{m}}{\text{s}}=\frac{\text{°C}}{\text{s}}$ *(the metres cancel — that cancellation is the chain rule)*
At $t=0$ the bug feels the temperature rising at 7 °C per second.

Read the two terms: heading east costs -3, heading north gains $+10$. The northward motion wins, so the bug warms up even though it is walking into a

cooling direction. A term with the wrong sign is not an error — it is one influence losing to another.

Implicit differentiation — the one-line upgrade

Write the equation as $F(x,y)=0$ (move everything to one side). Differentiating both sides with respect to x by the chain rule (y depends on x) *builds* the formula:

$$F_x\cdot 1+F_y\cdot\frac{dy}{dx}=0 \quad\Longrightarrow\quad \frac{dy}{dx}=-\frac{F_x}{F_y} \qquad (F_y\neq 0)$$

For a surface $F(x,y,z)=0$ defining z implicitly, the same move gives

$$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z} \qquad \frac{\partial z}{\partial y}=-\frac{F_y}{F_z} \qquad (F_z\neq 0)$$

Worked: slope of $x^2+y^2=25$ at $(3,-4)$

1. SET $F=0$
$F(x,y)=x^2+y^2-25$

↓

2. TWO PARTIALS
$F_x=2x \qquad F_y=2y$

↓

3. FORMULA
$$\frac{dy}{dx}=-\frac{F_x}{F_y}=-\frac{2x}{2y}=-\frac{x}{y}$$

↓

4. EVALUATE
at $(3,-4)$: $-\dfrac{3}{-4}=\dfrac{3}{4}$ ✓

The aha: this is the entire Calc 1 implicit-differentiation grind (differentiate term-by-term, collect $\frac{dy}{dx}$, solve) compressed into two partials and a ratio.

Worked (surface flavor): $\frac{\partial z}{\partial x}$ on $x^2+y^2+z^2=14$ at $(1,2,3)$

1. SET $F=0$
$F=x^2+y^2+z^2-14$

↓

2. THE TWO PARTIALS THAT MATTER
$F_x=2x, \qquad F_z=2z$ *(F_y only matters for $\partial z/\partial y$)*

↓

3. RATIO & EVALUATE
$$\frac{\partial z}{\partial x}=-\frac{2x}{2z}=-\frac{x}{z} \;\rightarrow\; \text{at } (1,2,3):\ -\tfrac{1}{3}$$
(check the point is on the surface: $1+4+9=14$ ✓)

Common pitfalls

- d vs ∂: one independent variable at the bottom ⇒ $\frac{dz}{dt}$; several ⇒ $\frac{\partial z}{\partial u}$. Count inputs, don't guess.
- One term per intermediate variable — a dropped route is the #1 chain-rule error. Draw the tree.
- Never cancel the ∂'s — partials aren't fractions (canceling would prove $\frac{dz}{dt}=2\frac{dz}{dt}$). The matching-middles pattern is only a check that each term chains correctly.
- Finish in the independent variable: after the routes are added, substitute $x(t),y(t)$ back so the answer contains only t (or u,v) — or plug in the numeric point.
- The implicit formula has a minus sign: $-F_x/F_y$, from moving the $F_y\frac{dy}{dx}$ term across. Forgetting it flips every slope.
- Set $F=0$ first — the formula reads partials off F, so everything must be on one side before differentiating.

directional derivatives & the gradient

f_x and f_y are slopes in exactly two directions — parallel to the axes. The **directional derivative** is the slope of the surface walking in any direction: for a unit vector $\hat{u}=\langle u_1,u_2\rangle$,

$$D_{\hat{u}}f(a,b)=\lim_{h\to 0}\frac{f(a+hu_1,\,b+hu_2)-f(a,b)}{h}$$

No new limit work is ever needed — the partials already contain every direction:

$$D_{\hat{u}}f=f_x u_1+f_y u_2$$

An angle version of the same thing: any unit direction is $\hat{u}=\langle\cos\theta,\sin\theta\rangle$, so $D_{\hat{u}}f=f_x\cos\theta+f_y\sin\theta$.

What is being asked? — four questions

- **A** **rate in a given direction** ⇒ $D_{\hat{u}}f=\nabla f\cdot\hat{u}$. Normalize the direction first.
- **B** **steepest ascent** ⇒ direction is ∇f, and the rate is $|\nabla f|$. Steepest descent is $-\nabla f$.
- **C** **no change** ⇒ any direction perpendicular to ∇f — that is along the level curve.
- **D** **tangent plane to a level surface** ⇒ ∇F is the normal vector.

The gradient answers all four, so compute ∇f first and then read the question again. The single most common lost mark is skipping the normalization in A.

The gradient — package the partials

$$\nabla f=\langle f_x,\ f_y\rangle \qquad\Longrightarrow\qquad D_{\hat{u}}f=\nabla f\cdot\hat{u}$$

One vector now answers every direction question via a dot product. Sanity check: $\hat{u}=\langle 1,0\rangle$ gives $\nabla f\cdot\hat{u}=f_x$ — the partials are just the axis-direction special cases.

Worked: $D_{\hat{u}}f$ for $f(x,y)=x^2-xy+3y^2$ at $P(1,2)$ toward $\vec{v}=\langle 3,4\rangle$

1. PARTIALS
$f_x=2x-y \qquad f_y=-x+6y$

↓

2. GRADIENT AT THE POINT
$\nabla f(1,2)=\langle 2-2,\ -1+12\rangle=\langle 0,11\rangle$

↓

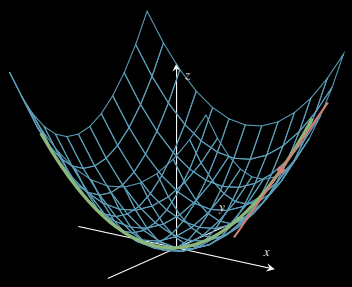
3. UNIT VECTOR — ALWAYS NORMALIZE
$|\vec{v}|=\sqrt{9+16}=5 \ \Rightarrow\ \hat{u}=\left\langle\frac{3}{5},\frac{4}{5}\right\rangle$ *(never dot with the raw $\vec{v}$)*

↓

4. DOT
$D_{\hat{u}}f(1,2)=\nabla f\cdot\hat{u}=0\cdot\frac{3}{5}+11\cdot\frac{4}{5}=\frac{44}{5}$

"Toward the point Q" variant: direction of $Q(4,6)$ from $P(1,2)$ means $\vec{v}=Q-P=\langle 3,4\rangle$ — subtract first, then normalize; it's the same problem.

$D_{\hat{u}}f(1,1)$ on $z=x^2+y^2$, $\hat{u}=\langle\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\rangle$



diagonal trace along $\hat{u}$; tangent at $(1,1,2)$, slope $D_{\hat{u}}f=\nabla f\cdot\hat{u}=\langle 2,2\rangle\cdot\langle\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\rangle=2\sqrt{2}$ — same picture as the f_x trace, tilted to a chosen direction.

Interpreting the value — why $\hat{u}$ must be a unit vector

$D_{\hat{u}}f$ carries the same units as a partial derivative — output per unit of distance travelled. That is true only because $\hat{u}$ has length 1, so one step of $\hat{u}$ is one unit of distance.

Skip the normalization and the units become meaningless, not merely scaled: you would be reporting change per step of some arbitrary length. This is the most common lost mark in the section.

Two more readings that exam questions ask for directly:

- $|\nabla f|$ is the greatest possible rate at that point. No direction beats it.
- Therefore $D_{\hat{u}}f$ always lies in $[-|\nabla f|,\ |\nabla f|]$ — a value outside that range is an arithmetic error.

Worked (in context): walking northeast on the plate

The plate from the partials section: $T(x,y)=99-x^2+xy+3y$ in °C, x,y in metres, and at $(2,1)$ we found $\nabla T=\langle -3,5\rangle$. Walk northeast.

1. BUILD $\hat{u}$ — NORMALIZE FIRST
northeast is $\vec{v}=\langle 1,1\rangle$, $\|\vec{v}\|=\sqrt{2}$, so $\hat{u}=\left\langle \frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right\rangle$

↓

2. DOT WITH THE GRADIENT
$$D_{\hat{u}}T=\nabla T\cdot\hat{u}=\frac{-3}{\sqrt{2}}+\frac{5}{\sqrt{2}}=\frac{2}{\sqrt{2}}=\sqrt{2}\approx 1.41$$

↓

3. SAY IT — WITH UNITS
walking northeast from $(2,1)$, the temperature rises about $1.41\,^\circ$C per metre.

↓

4. SANITY-CHECK AGAINST THE MAXIMUM
$|\nabla T|=\sqrt{(-3)^2+5^2}=\sqrt{9+25}=\sqrt{34}\approx 5.83$
1.41 sits inside $[-5.83,\ 5.83]$ ✓, and it is well below the maximum — northeast is not the steepest way up. That direction is $\nabla T=\langle -3,5\rangle$, which points north and west.

The aha: the two partials said "east cools, north warms". Northeast mixes them, so it lands between the two — positive, but far short of the best available climb.

What the gradient means — three directions that matter

With φ the angle between ∇f and $\hat{u}$: $D_{\hat{u}}f=\nabla f\cdot\hat{u}=|\nabla f|\cos\varphi$. Reading off the extremes of $\cos\varphi$:

- **Steepest ascent** ($\varphi=0$): f increases fastest in the direction of ∇f, at rate $|\nabla f|$.
- **Steepest descent** ($\varphi=\pi$): fastest decrease in the direction of $-\nabla f$, at rate $-|\nabla f|$.
- **No change** ($\varphi=\frac{\pi}{2}$): $D_{\hat{u}}f=0$ perpendicular to ∇f — walking along the level curve.

Consequence: ∇f is perpendicular to the level curve of f through the point — contour maps and gradients are orthogonal families.

Worked: steepest ascent on $f(x,y)=x^2+y^2$ at $(1,2)$

1. GRADIENT AT THE POINT
$\nabla f=\langle 2x,2y\rangle\ \Rightarrow\ \nabla f(1,2)=\langle 2,4\rangle$

↓

2. RATE — MAGNITUDE OF THE GRADIENT
max rate of increase $=|\nabla f(1,2)|=\sqrt{2^2+4^2}=\sqrt{20}=2\sqrt{5}$

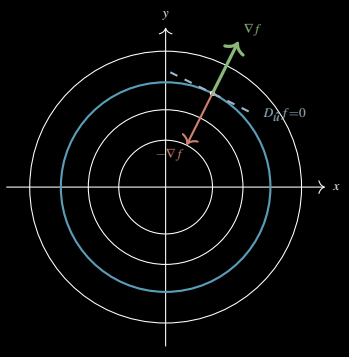
↓

3. DIRECTION — NORMALIZE THE GRADIENT
$\hat{u}=\frac{1}{2\sqrt{5}}\langle 2,4\rangle=\left\langle \frac{2}{2\sqrt{5}},\frac{4}{2\sqrt{5}}\right\rangle=\left\langle \frac{1}{\sqrt{5}},\frac{2}{\sqrt{5}}\right\rangle$

↓

4. THE OTHER TWO ANSWERS FOR FREE
steepest descent: $-\hat{u}$, rate $-2\sqrt{5}$; no change: $\pm\left\langle -\frac{2}{\sqrt{5}},\frac{1}{\sqrt{5}}\right\rangle$ *(tangent to the circle through $(1,2)$)*

∇f points radially away from the bowl's bottom — straight across the contour circles. This is the chain-rule level-curve example closing its loop: walking the circle gave $\frac{dz}{dt}=0$ because that walk is the $\varphi=\frac{\pi}{2}$ direction.



∇f crosses every contour at 90°

Three variables — nothing changes

$$\nabla f=\langle f_x,\ f_y,\ f_z\rangle \qquad D_{\hat{u}}f=\nabla f\cdot\hat{u} \quad (\hat{u}\text{ a unit vector in 3D})$$

Quick run: $f(x,y,z)=xy+z^2$ at $P(1,2,1)$ toward $\vec{v}=\langle 1,2,2\rangle$: $\nabla f=\langle y,x,2z\rangle=\langle 2,1,2\rangle$, $|\vec{v}|=3$, $D_{\hat{u}}f=\frac{2\cdot 1+1\cdot 2+2\cdot 2}{3}=\frac{8}{3}$.

Perpendicularity levels up a dimension: ∇f is normal to the level *surface* $f=c$. Instant example: $f=x^2+y^2+z^2$ has $\nabla f=2\langle x,y,z\rangle$ — radial, perpendicular to every concentric sphere, as it must be.

Tangent plane to a level surface & the normal line

The normal fact is a machine: at a point P on the surface $F(x,y,z)=c$,

tangent plane: $\nabla F(P)\cdot\langle x-x_0,\ y-y_0,\ z-z_0\rangle=0$ normal line: $P+t\,\nabla F(P)$

One gradient hands over both objects — it's the plane's normal and the line's direction.

Worked: ellipsoid $x^2+4y^2+z^2=18$ at $(1,2,1)$

1. CHECK THE POINT, TAKE ∇F
$1+16+1=18$ ✓; $\nabla F=\langle 2x,8y,2z\rangle\ \to\ \langle 2,16,2\rangle$

↓

2. TANGENT PLANE
$2(x-1)+16(y-2)+2(z-1)=0$: distribute: $2x-2+16y-32+2z-2=0$; collect: $2x+16y+2z=36$; divide by 2: $x+8y+z=18$

↓

3. NORMAL LINE — SAME VECTOR, NEW JOB
$x=1+t,\ y=2+8t,\ z=1+t$ *(scaled ∇F by $\frac{1}{2}$ — any multiple works)*

The aha: the graph case $z=f(x,y)$ is this machine on $F=f(x,y)-z$: $\nabla F=\langle f_x,f_y,-1\rangle$ — exactly the tangent-plane formula from the linear-approximation section, now explained.

Worked (reverse): recover ∇f from two directional derivatives

At P: $D_{\hat{u}}f=3$ along $\hat{u}=\langle 1,0\rangle$, and $D_{\hat{w}}f=\sqrt{2}$ along $\hat{w}=\langle \frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\rangle$. Find ∇f and the max rate.

1. EACH D IS A LINEAR EQUATION
$\nabla f\cdot\langle 1,0\rangle=f_x=3$; $\frac{1}{\sqrt{2}}(f_x+f_y)=\sqrt{2}$

↓

2. SOLVE THE SYSTEM
multiply the second by $\sqrt{2}$: $f_x+f_y=\sqrt{2}\cdot\sqrt{2}=2$; substitute $f_x=3$: $3+f_y=2\ \Rightarrow$ $f_y=-1$: $\nabla f=\langle 3,-1\rangle$

↓

3. MAX RATE FOR FREE
$|\nabla f|=\sqrt{3^2+(-1)^2}=\sqrt{10}$, in direction $\frac{1}{\sqrt{10}}\langle 3,-1\rangle$ *(two dots pin two unknowns — n directions recover an n-dim gradient)*

Common pitfalls

- Normalize the direction first — $D_{\hat{u}}f$ demands a *unit* vector. Dotting with a raw $\vec{v}$ scales the slope by $|\vec{v}|$. This is the #1 miss.
- Direction vs rate are different answers: the direction of fastest increase is ∇f (a vector); the fastest rate is $|\nabla f|$ (a number). Answer the one asked.

- ∇f lives on the floor — it is a vector in the *domain* (xy-plane), pointing where to walk. It is not a 3D arrow lying on the surface.
- $D_{\hat{u}}f$ is a scalar (a slope); ∇f is a vector. Don't hand back a vector when a slope is asked.
- "Toward Q" means $\vec{v}=Q-P$, not Q itself — subtract, then normalize.
- $\nabla f=\vec{0} \Rightarrow$ every directional derivative is 0: a **critical point** — no uphill direction exists (max/min/saddle candidates, next section's business).

max/min — critical points & the second derivative test

The gradient section's last pitfall is this section's starting gun: extrema can only happen where $\nabla f=\vec{0}$ — both partials zero *simultaneously* — or where a partial is undefined. These are the **critical points**: the spots where the tangent plane is horizontal (or missing). Finding them means solving the system $f_x=0$, $f_y=0$ — that's where the actual work lives.

Which hunt is this? — three problems

A **classify critical points** (no region given) $\Rightarrow$ solve $\nabla f=\vec{0}$, then run the D test on each.

B **absolute max/min on a closed region** $\Rightarrow$ collect three sets of candidates: interior critical points, each edge, and every corner. Compare values. No D test needed.

C **optimize with a constraint** $\Rightarrow$ substitute the constraint to remove a variable, or use Lagrange: $\nabla f=\lambda\nabla g$.

THE THREE THINGS YOU SOLVE

1. The system $f_x=0$, $f_y=0$ — both zero at the same point. Factor and enumerate cases.
2. Each boundary edge — substitute the edge equation into f to get a one-variable Calc 1 problem.
3. Which quantity is constrained — read whether the volume or the material is the given one. The roles swap. The recipe does not.

Second derivative test — the discriminant D

A critical point can be a min, a max, or a **saddle** (min in one slice, max in another). The classifier, built from the second partials at the critical point:

$$D=f_{xx}f_{yy}-\left(f_{xy}\right)^2$$

- $D>0$, $f_{xx}>0$: local min (both slices bend up — a bowl)
- $D>0$, $f_{xx}<0$: local max (both bend down — a dome)
- $D<0$: saddle point — definitive, not "test failed"
- $D=0$: inconclusive — the only case the test can't decide

Read it as: $D>0$ means the pure seconds agree and dominate the mixed term — a genuine bowl or dome, and the sign of f_{xx} says which; $D<0$ means the slices disagree — saddle.

Worked: classify all critical points of $f(x,y)=x^3+y^2-6xy$

1. SET BOTH PARTIALS TO ZERO — A SYSTEM
$f_x=3x^2-6y=0$ $\qquad$ $f_y=2y-6x=0$

↓

2. SOLVE SIMULTANEOUSLY
f_y gives $y=3x$; substitute into f_x: $3x^2-6(3x)=3x^2-18x=3x(x-6)=0 \Rightarrow x=0,6$; then $y=3x$ gives $y=0,18$: critical points $(0,0)$ and $(6,18)$

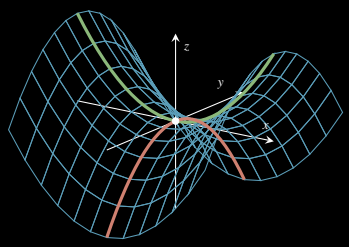
↓

3. SECOND PARTIALS & D
$f_{xx}=6x$, $\quad f_{yy}=2$, $\quad f_{xy}=-6$ $\qquad$ $D=(6x)(2)-(-6)^2=12x-36$

↓

4. CLASSIFY EACH POINT SEPARATELY
$(0,0)$: $D=-36<0 \Rightarrow$ saddle. $\quad$ $(6,18)$: $D=36>0$, $f_{xx}=36>0 \Rightarrow$ local min; value: $f(6,18)=6^3+18^2-6(6)(18)=216+324-648=-108$

$z=x^2-y^2$: $D<0$ at the origin — saddle



the x-slice bends up (min through the point), the y-slice bends down (max through it) — the slices disagree, so the point is neither: that disagreement is exactly what $D<0$ detects.

Worked: when the system factors — $f(x,y)=\left(x^2-y^2\right)e^{-x^2-y^2}$

The example above solved by substitution and gave two points. Most exam systems instead factor, and then you enumerate cases.

1. PRODUCT RULE, AND FACTOR IMMEDIATELY
write $g=e^{-x^2-y^2}$ *(so $g_x=-2xg$, $g_y=-2yg$)* and $u=x^2-y^2$:
$$f_x=2xg+u(-2xg)=2xg(1-u) \qquad f_y=-2yg+u(-2yg)=-2yg(1+u)$$

↓

2. KILL THE EXPONENTIAL — IT IS NEVER ZERO
$g=e^{-x^2-y^2}>0$ always, so it cannot supply a root and divides out:
$$f_x=0 \iff x=0 \text{ or } u=1 \qquad f_y=0 \iff y=0 \text{ or } u=-1$$
Never cancel a factor without saying why it is nonzero — that sentence is worth marks

↓

3. ENUMERATE ALL FOUR CASES
(two choices from f_x × two from f_y — every combination must be checked)
$x=0$ **and** $y=0$: $\Rightarrow (0,0)$
$x=0$ **and** $u=-1$: $u=0-y^2=-1 \Rightarrow y^2=1 \Rightarrow (0,\pm1)$
$u=1$ **and** $y=0$: $u=x^2-0=1 \Rightarrow x^2=1 \Rightarrow (\pm1,0)$
$u=1$ **and** $u=-1$: impossible — this case is empty, and saying so *is* the work
Five critical points total.

↓

4. SECOND PARTIALS — DIFFERENTIATE THE FACTORED FORMS
$$f_{xx}=2g(1-u)-4x^2g(1-u)-4x^2g \qquad f_{yy}=-2g(1+u)+4y^2g(1+u)+4y^2g$$
$$f_{xy}=2x\Big[(-2yg)(1-u)+g(2y)\Big]=4xyg\,u$$

↓

5. CLASSIFY $(0,0)$ — HERE $u=0$, $g=1$
$f_{xx}=2(1)(1)-0-0=2$, $\quad f_{yy}=-2(1)(1)+0+0=-2$, $\quad f_{xy}=4(0)(0)(1)(0)=0$
$$D=(2)(-2)-0^2=-4<0 \Rightarrow \text{saddle}$$

↓

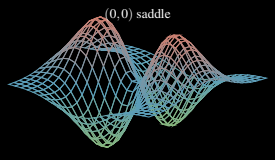
6. CLASSIFY $(1,0)$ — HERE $u=1$, $g=e^{-1}$
$f_{xx}=2g(0)-4g(0)-4g=-4g$, $\quad f_{yy}=-2g(2)+0+0=-4g$, $\quad f_{xy}=4(1)(0)g(1)=0$
$$D=(-4g)(-4g)-0=16g^2=\frac{16}{e^2}>0, \quad f_{xx}=-\frac{4}{e}<0 \Rightarrow \text{local max}, \ f=\frac{1}{e}$$

↓

7. CLASSIFY $(0,1)$ — HERE $u=-1$, $g=e^{-1}$
$f_{xx}=2g(2)-0-0=4g$, $\quad f_{yy}=-2g(0)+4g(0)+4g=4g$, $\quad f_{xy}=4(0)(1)g(-1)=0$
$$D=16g^2>0, \quad f_{xx}=\frac{4}{e}>0 \Rightarrow \text{local min}, \ f=-\frac{1}{e}$$

↓

8. THE LAST TWO BY SYMMETRY
f is unchanged by $x\to -x$ and by $y\to -y$, so $(-1,0)$ copies $(1,0)$ *(max, $\frac{1}{e}$)* and $(0,-1)$ copies $(0,1)$ *(min, $-\frac{1}{e}$)* — state the symmetry, don't redo the algebra



height is colour-coded: two red ridges at $(\pm1,0)$ *(max $\frac{1}{e}$)*, two green troughs at $(0,\pm1)$ *(min $-\frac{1}{e}$)*, and the saddle where they cross — the five points the case table found, and nothing else

The aha: when both partials factor, the system is not "solve two equations" — it is a grid of cases. Take one root from each factored equation, every

combination, and check each pairing for consistency. Cases that contradict are answers too: writing "impossible" is what stops you from inventing a sixth critical point.

Worked: classify with f never given *— only its partials*

Sometimes the problem hands you f_x and f_y and nothing else. You can still find and classify every critical point, because the test only ever uses derivatives.

$$f_x = 3\left(x^2-1\right)(y+2) \qquad\qquad f_y = x\left(x^2-3\right)$$

1. SANITY-CHECK THAT THESE CAN COME FROM ONE f
Clairaut says the mixed partials must agree:

$$\frac{\partial}{\partial y}\left[3\left(x^2-1\right)(y+2)\right] = 3x^2-3 \qquad \frac{\partial}{\partial x}\left[x^3-3x\right] = 3x^2-3 \;\checkmark$$

(if these disagreed, no f exists and the question is broken)

↓

2. BOTH ZERO — READ THE ROOTS OFF THE FACTORS
$f_x=0 \iff x=1,\ x=-1,$ or $y=-2$
$f_y=0 \iff x=0,\ x=\sqrt{3},$ or $x=-\sqrt{3}$ *($x^2-3=0 \Rightarrow x=\pm\sqrt{3}$)*

↓

3. INTERSECT THE TWO SOLUTION SETS
$x=\pm 1$ paired with $x \in \{0, \pm\sqrt{3}\}$: contradiction — x cannot be two different numbers. Empty.
$y=-2$ paired with each x: $(0,-2)$, $(\sqrt{3},-2)$, $(-\sqrt{3},-2)$
(f_y never mentions y, so it constrains x only — every critical point must sit on the line $y=-2$)

↓

4. SECOND PARTIALS — DIFFERENTIATE WHAT YOU WERE GIVEN

$$f_{xx} = \frac{\partial}{\partial x}\left[3\left(x^2-1\right)(y+2)\right] = 6x(y+2)$$

$$f_{yy} = \frac{\partial}{\partial y}\left[x^3-3x\right] = 0 \qquad\qquad f_{xy} = 3\left(x^2-1\right)$$

↓

5. THE DISCRIMINANT COLLAPSES

$$D = f_{xx}f_{yy} - (f_{xy})^2 = f_{xx}\cdot 0 - 9\left(x^2-1\right)^2 = -9\left(x^2-1\right)^2$$

a square is ≥ 0, so $D \leq 0$ everywhere — with $f_{yy} \equiv 0$, D can never be positive

↓

6. EVALUATE D AT EACH POINT
$(0,-2)$: $D=-9(0-1)^2=-9<0 \Rightarrow$ saddle
$(\pm\sqrt{3},-2)$: $D=-9(3-1)^2=-9(4)=-36<0 \Rightarrow$ saddle
All three are saddles. Report locations and types only — without f there are no heights to report, and inventing one is the trap.

The aha: the second derivative test never touches f itself — it reads f_{xx}, f_{yy}, f_{xy}, all of which are one derivative away from what you were handed. And the structure gave the answer before the arithmetic did: $f_{yy} \equiv 0$ forces $D \leq 0$, so no maximum or minimum was ever possible. Look at the shape of D before plugging in points.

Absolute extrema on a closed region — the candidate hunt

EVT levels up: f continuous on a closed, bounded region R *must* attain an absolute max and min — the hunt is the Closed Interval Method with a boundary instead of two endpoints:

1. **Interior**: find critical points inside R (solve the system as above. No need to classify — just collect values).
2. **Boundary**: substitute each edge's equation into f — the problem drops to one-variable Calc 1 on an interval. Collect its critical values *and* the endpoints (corners).
3. **Compare**: biggest collected value = abs max, smallest = abs min.

Worked: $f(x,y)=x^2-2xy+2y$ *on the rectangle* $0 \leq x \leq 3, \quad 0 \leq y \leq 2$

1. INTERIOR CRITICAL POINTS
$f_y=-2x+2=0 \Rightarrow x=1$; put $x=1$ into $f_x=2x-2y=0$: $2-2y=0 \Rightarrow y=1$. Value: $f(1,1)=1-2(1)(1)+2(1)=1$

↓

2. WALK THE FOUR EDGES — EACH IS CALC 1
$y=0$: $f=x^2$ on $[0,3]$ — increasing: $f(0,0)=0$, $f(3,0)=9$
$y=2$: $f=x^2-2x(2)+2(2)=x^2-4x+4=(x-2)^2$ — vertex at $x=2$: $f(0,2)=4$, $f(2,2)=0$, $f(3,2)=1$
$x=0$: $f=2y$ — $f(0,0)=0$ to $f(0,2)=4$, no interior critical point
$x=3$: $f=9-6y+2y=9-4y$ — $f(3,0)=9$ down to $f(3,2)=1$, none

↓

3. COMPARE ALL CANDIDATE VALUES
$\{1,\ 0,\ 9,\ 4,\ 0,\ 1\} \Rightarrow$ abs max 9 at $(3,0)$; abs min 0 at $(0,0)$ and $(2,2)$

The aha: on each edge one variable freezes at a number, and a 2-variable problem collapses to four 1-variable problems — nothing new to learn, just bookkeeping. Note the interior critical point ($f=1$) won *neither* title: the extremes lived on the boundary, which is why it can never be skipped.

Worked (sloped edge): $f=x^2+y^2-2x-2y$ *on the triangle* $(0,0),(4,0),(0,4)$

The exam version — one boundary piece isn't axis-aligned, so its substitution actually does something:

1. INTERIOR
$f_x=2x-2=0,\ f_y=2y-2=0 \Rightarrow (1,1)$, inside ✓: $f(1,1)=1+1-2-2=-2$

↓

2. TWO AXIS EDGES
$y=0$: $f=x^2-2x$, vertex $x=1$: $f=-1$; ends $0,\ 8$. $x=0$: symmetric: -1; ends $0,\ 8$

↓

3. THE SLOPED EDGE — SUBSTITUTE $y=4-x$
$f=x^2+(4-x)^2-2x-2(4-x)$; expand each piece: $(4-x)^2=16-8x+x^2$ and $-2(4-x)=-8+2x$:

$$f = x^2+16-8x+x^2-2x-8+2x = 2x^2-8x+8 = 2(x-2)^2$$

vertex $x=2$: $f=0$; corners: $2(0-2)^2=8$ and $2(4-2)^2=8$

↓

4. COMPARE ALL CANDIDATES
$\{-2,\ -1,\ -1,\ 0,\ 0,\ 8,\ 8\}$: abs min -2 at $(1,1)$; abs max 8 at $(4,0)$ and $(0,4)$

Same hunt, one new move: the sloped edge's equation substitutes in exactly like an axis edge — it just leaves more algebra behind.

Worked: applied — open-top box, volume 4*, least material*

Minimize surface area $S=xy+2xz+2yz$ (base + four sides, no top) subject to $xyz=4$.

1. OBJECTIVE + CONSTRAINT
minimize $S=xy+2xz+2yz$ with $xyz=4$

↓

2. SOLVE THE CONSTRAINT FOR z, SUBSTITUTE
$xyz=4 \Rightarrow z=\dfrac{4}{xy}$. Each z-term simplifies by cancellation:

$$S = xy+2x\cdot\frac{4}{xy}+2y\cdot\frac{4}{xy} = xy+\frac{8}{y}+\frac{8}{x}$$

(the x cancels in the first fraction, the y in the second — now an ordinary critical-point problem)

↓

3. SOLVE THE SYSTEM
$S_x=y-\dfrac{8}{x^2}=0 \Rightarrow y=\dfrac{8}{x^2}$; put that into $S_y=x-\dfrac{8}{y^2}=0$:

$$x = \frac{8}{(8/x^2)^2} = \frac{8x^4}{64} = \frac{x^4}{8} \;\Rightarrow\; 8x=x^4 \;\Rightarrow\; x^3=8 \;\Rightarrow\; x=2$$

back-substitute: $y=\dfrac{8}{2^2}=2$, $\quad z=\dfrac{4}{2\cdot 2}=1$

↓

4. CONFIRM IT'S A MIN, ANSWER THE QUESTION
$S_{xx}=\dfrac{16}{x^3}=2$, $\quad S_{yy}=2$, $\quad S_{xy}=1$: $\qquad D=(2)(2)-1^2=3>0$, $\quad S_{xx}>0$ ✓min.
$S=(2)(2)+\frac{8}{2}+\frac{8}{2}=4+4+4=12$

The applied recipe: write the objective, kill a variable with the constraint, then it's the same critical-point machine — and step 4's D-check is what

upgrades "a critical point" to "the minimum".

Worked: applied, flipped — given the material $S=12$, most volume

Same box, roles swapped: now the *surface area is the given* and volume is the objective. Maximize $V=xyz$ subject to $xy+2xz+2yz=12$.

1. OBJECTIVE + CONSTRAINT — SPOT THE SWAP
maximize $V=xyz$ with $xy+2xz+2yz=12$ *("given $S=\ldots$" ⇒ S is the constraint now)*

↓

2. SOLVE THE CONSTRAINT FOR z
z appears linearly, so isolate it: move the z-free term, factor, divide:

$$2xz+2yz=12-xy \;\Rightarrow\; 2z(x+y)=12-xy \;\Rightarrow\; z=\frac{12-xy}{2(x+y)}$$

↓

3. SUBSTITUTE INTO THE OBJECTIVE
$V=xy\cdot z=xy\cdot\dfrac{12-xy}{2(x+y)}=\dfrac{12xy-x^2y^2}{2(x+y)}$ — now a two-variable critical-point problem

↓

4. QUOTIENT RULE FOR V_x (freeze y)
top$'=12y-2xy^2$, bottom$'=2$:

$$V_x=\frac{(12y-2xy^2)\cdot 2(x+y)-(12xy-x^2y^2)\cdot 2}{[2(x+y)]^2}$$
$$=\frac{(12y-2xy^2)(x+y)-(12xy-x^2y^2)}{2(x+y)^2}$$

↓

5. SIMPLIFY THE NUMERATOR
expand: $(12y-2xy^2)(x+y)=12xy+12y^2-2x^2y^2-2xy^3$; subtract $12xy-x^2y^2$:

$$12y^2-x^2y^2-2xy^3=y^2(12-x^2-2xy) \;\Rightarrow\; V_x=\frac{y^2(12-x^2-2xy)}{2(x+y)^2}$$

↓

6. V_y BY SYMMETRY, SET BOTH TO ZERO
V is unchanged under $x\leftrightarrow y$, so $V_y=\dfrac{x^2(12-y^2-2xy)}{2(x+y)^2}$. With $x,y>0$ the y^2, x^2 factors can't vanish — the zeros live in the parentheses:

$$12-x^2-2xy=0 \qquad \text{and} \qquad 12-y^2-2xy=0$$

↓

7. SUBTRACT THE TWO EQUATIONS
$(12-x^2-2xy)-(12-y^2-2xy)=y^2-x^2=0 \;\Rightarrow\; x=y$ (both positive). Substitute $y=x$ into either: $12-x^2-2x^2=0\Rightarrow x^2=4\Rightarrow x=y=2$; then step 2's formula: $z=\dfrac{12-4}{2(2+2)}=1$

↓

8. CONFIRM IT'S A MAX — BOUNDARY ARGUMENT
$V\to 0$ whenever the box degenerates ($x\to 0$, $y\to 0$, or $xy\to 12$ so $z\to 0$); one interior critical point ⇒ it must be the max: $V=2\cdot 2\cdot 1=4$

The aha: it's the same box — $V=4$ needed $S=12$, and $S=12$ buys $V=4$. Min-material and max-volume are dual problems: the winning shape (square base, height half the base) is identical. Only which quantity is pinned changes. Working notes:

- Eliminate whichever variable the constraint is linear in.
- Symmetric equations in x,y: subtract them ($\Rightarrow x=y$) instead of grinding substitutions.
- The D-test isn't the only closer: when every boundary kills the objective, the lone critical point wins by elimination.

Worked: applied — closest point on a plane

Find the point on $x+y+z=1$ nearest the origin:

1. MINIMIZE d^2, NOT d
same minimizer, no square root: $g=x^2+y^2+z^2$

↓

2. KILL A VARIABLE
$z=1-x-y$: $g(x,y)=x^2+y^2+(1-x-y)^2$

↓

3. SOLVE THE SYSTEM
$g_x=2x-2(1-x-y)=0\Rightarrow x=1-x-y$; $g_y=2y-2(1-x-y)=0\Rightarrow y=1-x-y$. Same right side $\Rightarrow x=y$; then $x=1-x-x=1-2x \;\Rightarrow\; 3x=1 \;\Rightarrow\; x=\frac{1}{3}$, so $y=\frac{1}{3}$, $z=1-\frac{1}{3}-\frac{1}{3}=\frac{1}{3}$

↓

4. ANSWER & FORMULA CHECK
point $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$, with

$$d=\sqrt{3\cdot\left(\tfrac{1}{3}\right)^2}=\sqrt{\tfrac{1}{3}}=\tfrac{1}{\sqrt{3}}=\tfrac{\sqrt{3}}{3}$$

— the point-to-plane formula agrees: $\frac{|0+0+0-1|}{\sqrt{3}}$ ✓

Lagrange multipliers — the no-elimination route

When the constraint won't solve cleanly for a variable, don't eliminate — align the gradients: at a constrained extremum the level curve of f kisses the constraint $g=c$, so

$$\nabla f=\lambda\,\nabla g \qquad \text{plus the constraint } g=c$$

One equation per variable + the constraint; λ is bookkeeping, not part of the answer.

Worked: the box again — minimize S with $xyz=4$

1. BUILD THE SYSTEM
$\nabla S=\langle y+2z,\ x+2z,\ 2x+2y\rangle$, $\nabla g=\langle yz,\ xz,\ xy\rangle$. Component by component, plus the constraint:

$$y+2z=\lambda yz, \qquad x+2z=\lambda xz, \qquad 2x+2y=\lambda xy, \qquad xyz=4$$

↓

2. KILL λ — MULTIPLY EACH BY ITS VARIABLE
each right side becomes $\lambda xyz=4\lambda$: $x(y+2z)=y(x+2z)=z(2x+2y)=4\lambda$; expand the left sides:

$$xy+2xz = xy+2yz = 2xz+2yz$$

↓

3. READ OFF THE PROPORTIONS
first pair: cancel xy: $2xz=2yz\Rightarrow x=y$ (divide by $2z>0$). First vs third: cancel $2xz$: $xy=2yz\Rightarrow x=2z$ (divide by $y>0$)

↓

4. CONSTRAINT CLOSES IT
$x=y=2z$: $(2z)(2z)z=4z^3=4 \;\Rightarrow\; z^3=1 \;\Rightarrow\; z=1$, so $x=y=2$.
$S=(2)(2)+2(2)(1)+2(2)(1)=4+4+4=12$ — same box as substitution, no variable ever eliminated

Common pitfalls

- Both partials zero at the *same* point — solve the simultaneous system. A zero of f_x alone is nothing.
- When both partials factor, enumerate every case: one root from each factored equation, all combinations. Pairings that contradict are empty — write that down, or you will invent a critical point that does not exist.
- A never-zero factor (e^{anything}, x^2+1) cannot supply a root — divide it out, but say why it is nonzero.
- You do not need f to classify: the test reads only f_{xx}, f_{yy}, f_{xy}. Given just f_x and f_y, report locations and types — there are no heights to report.
- $D<0$ is a definitive saddle, not a failed test; $D=0$ is the only inconclusive case.
- The mixed partial is squared: $-(f_{xy})^2$ always subtracts — forgetting the square (or the sign) flips classifications.
- Consult f_{xx} only after $D>0$ — its sign means nothing at a saddle.
- Absolute extrema: never skip the boundary or its corners — the interior critical point often loses (it did above).
- Read which quantity is *given*: "volume must be 4" ⇒ V is the constraint, minimize S; "you have 12 of material" ⇒ S is the constraint, maximize V. Same recipe either way — only the roles swap.
- Answer with the value *and* the location: "max $=9$ at $(3,0)$" — coordinates alone aren't the max.
- Critical points also occur where a partial is undefined — the cone's tip $z=\sqrt{x^2+y^2}$ has a min with no horizontal tangent plane.

double integrals — rectangular regions

Calc 2's definite integral summed skinny rectangles under a curve. One dimension up: chop the rectangle $R=[a,b]\times[c,d]$ into little cells of area ΔA, sample f in each, and sum the column volumes $f\cdot\Delta A$:

$$\iint_R f(x,y)\,dA=\lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*,y_{ij}^*)\,\Delta A$$

When $f\geq 0$ this is the volume under the surface $z=f(x,y)$ over R; where $f<0$ the contribution counts negative — signed volume, the same bookkeeping as signed area.

THE PROCESS
1. Iterate — write it as two nested single integrals *(Fubini: over a rectangle the order is free)*.
2. Inner — freeze the outer variable; treat it like the constant 5.
3. Outer — now ordinary Calc 1.
4. Sanity ladder — each pass kills one variable. The answer is a pure number.

Step 1 decoded — pick the order

A f **factors** as $g(x)h(y)$ and R is a rectangle ⇒ split into a product of two singles. Fastest route.

B f does not factor ($\sin(x+y)$, $x+2y$) ⇒ nested passes, ordinary way. Exception: $e^{x+y}=e^x e^y$ does factor.

C both orders legal, one is far easier ⇒ scout ten seconds first. For xe^{xy}: dx first needs parts; dy first, $\frac{\partial}{\partial y}e^{xy}=xe^{xy}$ — the integrand is already somebody's derivative.

The only thing you solve here: "does f literally equal $g(x)$ times $h(y)$?" Everything else is Calc 1 twice.

Fubini's theorem — the workhorse

Nobody evaluates that limit. Fubini converts the double integral into two single integrals run in sequence — an **iterated integral** — and over a rectangle the order is a free choice:

$$\iint_R f\,dA=\int_a^b\int_c^d f(x,y)\,dy\,dx=\int_c^d\int_a^b f(x,y)\,dx\,dy$$

The mechanic is the partial-derivative aha run in reverse: during the inner integral the outer variable is frozen — treat it like the constant 5. Work inside out; each pass is ordinary one-variable calculus.

Worked: $\iint_R xy^2\,dA$, $R=[0,3]\times[1,2]$

1. WRITE IT ITERATED — PICK $dy\,dx$
$\int_0^3\int_1^2 xy^2\,dy\,dx$ *(inner integral matches the inner differential: y runs $1\to 2$)*

↓

2. INNER — FREEZE x
x is a constant here, so it factors out and rides along:

$$\int_1^2 xy^2\,dy=x\left[\frac{y^3}{3}\right]_1^2=x\left(\frac{8}{3}-\frac{1}{3}\right)=\frac{7}{3}x$$

↓

3. OUTER — NOW PURE CALC 1

$$\int_0^3\frac{7}{3}x\,dx=\frac{7}{3}\left[\frac{x^2}{2}\right]_0^3=\frac{7}{3}\left(\frac{9}{2}-0\right)=\frac{21}{2}$$

Note the sanity ladder: after the inner pass, y is gone. After the outer pass, x is gone — the answer is a pure number. A surviving variable means a limit never got plugged in.

Worked (dual): same integral, other order — Fubini verified

1. FLIP TO $dx\,dy$
$\int_1^2\int_0^3 xy^2\,dx\,dy$ *(limits travel with their variable: x still runs $0\to 3$, y still $1\to 2$)*

↓

2. INNER — NOW y IS FROZEN

$$\int_0^3 xy^2\,dx=y^2\left[\frac{x^2}{2}\right]_0^3=y^2\left(\frac{9}{2}-0\right)=\frac{9}{2}y^2$$

↓

3. OUTER

$$\int_1^2\frac{9}{2}y^2\,dy=\frac{9}{2}\left[\frac{y^3}{3}\right]_1^2=\frac{9}{2}\left(\frac{8}{3}-\frac{1}{3}\right)=\frac{9}{2}\cdot\frac{7}{3}=\frac{21}{2}\ \checkmark$$

The aha: both orders land on $\frac{21}{2}$ — that's Fubini in action, and re-running the other order is a built-in answer check.

Separable shortcut — when f factors over a rectangle

If $f(x,y)=g(x)\,h(y)$ *and* the region is a rectangle, the iterated integral splits into a product of two singles:

$$\iint_R g(x)\,h(y)\,dA=\left(\int_a^b g(x)\,dx\right)\left(\int_c^d h(y)\,dy\right)$$

Route three for the same example — xy^2 factors as $(x)(y^2)$:

$$\left(\int_0^3 x\,dx\right)\left(\int_1^2 y^2\,dy\right)=\frac{9}{2}\cdot\frac{7}{3}=\frac{21}{2}\ \checkmark$$

Only if it factors: $x+2y$ is a sum, not a product — no shortcut. And only over a rectangle: curved limits chain the variables together (next section).

Worked: a sum *inside the function —* $\iint_R \sin(x+y)\,dA$, $R=\left[0,\frac{\pi}{2}\right]^2$

The shortcut above needs a product. $\sin(x+y)$ has a sum *inside* a function, which is the most common way to be tempted into it.

1. CHECK THE SHORTCUT — IT FAILS
is $\sin(x+y)=g(x)h(y)$? No. And the two standard misreads:

$$\sin(x+y)\neq\sin x\sin y\qquad\sin(x+y)\neq\sin x+\sin y$$

(the real identity is $\sin x\cos y+\cos x\sin y$ — a sum of two products, still not one product)
So run the passes nested, the ordinary way.

↓

2. INNER — FREEZE x, SUBSTITUTE MENTALLY
with x frozen, $\frac{\partial}{\partial y}[-\cos(x+y)]=\sin(x+y)$:

$$\int_0^{\pi/2}\sin(x+y)\,dy=\Big[-\cos(x+y)\Big]_{y=0}^{y=\pi/2}=-\cos\left(x+\frac{\pi}{2}\right)+\cos x$$

↓

3. SIMPLIFY WITH A SHIFT IDENTITY
$\cos\left(x+\frac{\pi}{2}\right)=-\sin x$, so

$$-(-\sin x)+\cos x=\sin x+\cos x$$

(the x survived the inner pass — as it must, it was frozen, not deleted)

↓

4. OUTER

$$\int_0^{\pi/2}(\sin x+\cos x)\,dx=\Big[-\cos x+\sin x\Big]_0^{\pi/2}=(-0+1)-(-1+0)=1+1=2$$

Worked (dual): the look-alike that does *factor*

Same rectangle, same trig flavour, opposite verdict: $\iint_R \sin x\cos y\,dA$.

1. CHECK THE SHORTCUT — IT PASSES
$\sin x\cos y=g(x)h(y)$ with $g=\sin x$, $h=\cos y$ ✓, and R is a rectangle ✓⇒ split into a product of singles

↓

2. TWO INDEPENDENT ONE-VARIABLE INTEGRALS

$$\left(\int_0^{\pi/2}\sin x\,dx\right)\left(\int_0^{\pi/2}\cos y\,dy\right)=\Big[-\cos x\Big]_0^{\pi/2}\cdot\Big[\sin y\Big]_0^{\pi/2}$$
$$=\Big(0-(-1)\Big)\cdot\Big(1-0\Big)=1\cdot 1=1$$



two different surfaces over the same square: the sum runs a diagonal ridge and cannot be split. The product is one curve swept by another, so it can

The aha: the test is not "does the formula look separable" — it is "is f literally $g(x)$ times $h(y)$". A sum *inside* a function ($\sin(x+y)$, e^{x+y}, $\sqrt{x+y}$) almost never factors — though $e^{x+y}=e^x e^y$ is the one that does, because that exponent law exists and no other function has one. When in doubt, skip the shortcut: nested passes always work.

Average value over a rectangle

Same shape as Calc 2's average value, with area standing in for length:

$$f_{\text{avg}}=\frac{1}{\text{Area}(R)}\iint_R f\,dA$$

For the first worked example: $\text{Area}(R)=(3-0)(2-1)=3$, so $f_{\text{avg}}=\frac{1}{3}\cdot\frac{21}{2}=\frac{7}{2}$.

Common pitfalls

- Work inside out — the inner integral belongs to the inner differential, and its limits belong to that variable.
- The frozen variable is a constant, not zero: while integrating in y, an x factors out and survives — it doesn't vanish.
- A variable in the final answer = a limit that never got substituted — run the sanity ladder after each pass.
- The separable shortcut needs both conditions: a factored integrand *and* a rectangle. A sum *inside* a function ($\sin(x+y)$, $\sqrt{x+y}$) does not factor — except $e^{x+y}=e^x e^y$, the one function with an exponent law. Run nested passes when unsure.
- Regions where $f<0$ subtract — $\iint_R f\,dA$ is signed volume, exactly like signed area in Calc 2.

double integrals — general regions

Real regions are bounded by curves, not four straight fences. The fix: keep the outer limits numbers, and let the inner limits be functions of the outer variable — the slice length adapts as it sweeps.

THE PROCESS
1. Sketch the region. Limits come from the picture, never from the integrand.
2. Pick a slice direction — vertical (Type I) or horizontal (Type II).
3. Run one test slice: where does it enter, where does it exit? Those are the inner limits.
4. How far do slices sweep? That range is the outer limits — numbers.
5. Integrate inside out.

Iron rule: outer limits are constants. Inner limits may carry the outer variable — never the reverse.

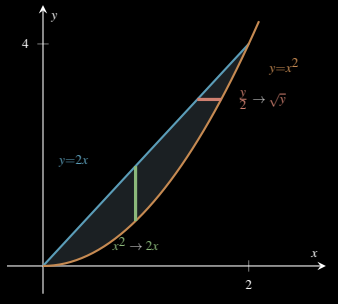
Step 2 decoded — three region shapes

A **Type I**, vertical slices, $dy\,dx$: $a\le x\le b$, $g_1(x)\le y\le g_2(x)$ — bottom curve up to top curve

B **Type II**, horizontal slices, $dx\,dy$: $c\le y\le d$, $h_1(y)\le x\le h_2(y)$ — left curve across to right curve

C **must split** — the entering or exiting curve changes formula partway. Two integrals, added.

Every region is both A and B, but rarely at the same price. Run a test slice both ways and count how many times the boundary formula changes: that count is the number of integrals.



one region, two descriptions: a vertical slice enters at $y=x^2$, exits at $y=2x$ (Type I); a horizontal slice enters at $x=\frac{y}{2}$, exits at $x=\sqrt{y}$ (Type II)

The four things you actually solve

1. WHERE DO THE CURVES MEET? — OUTER LIMITS
set them equal and solve. $x^2=2x\Rightarrow x(x-2)=0\Rightarrow x=0,2$ · $x^2=2-x^2\Rightarrow x=\pm1$

↓

2. WHICH IS TOP / WHICH IS LEFT?
Plug in one interior point. at $x=1$: $2x=2$ vs $x^2=1\Rightarrow$ the line is on top: $x^2\le y\le 2x$

↓

3. SOLVE EACH CURVE FOR THE *OTHER* VARIABLE (TYPE II)
$y=x^2\Rightarrow x=\sqrt{y}$ · $y=2x\Rightarrow x=\frac{y}{2}$ · $y=2-x^2\Rightarrow x=\pm\sqrt{2-y}$
Both roots survive if the region straddles the axis.

↓

4. DOES THE BOUNDARY CHANGE FORMULA?
run a test slice at two different heights. If they enter/exit on different curves, split there — and the split value is where the two boundary curves cross.

Worked (A → B): $\iint_D xy\,dA$, D between $y=x^2$ and $y=2x$

1–2. MEET, THEN ORDER *(things 1, 2)*
$x^2=2x\Rightarrow x=0,2$; at $x=1$ the line wins $\Rightarrow x^2\le y\le 2x$, $0\le x\le 2$

↓

3. TYPE I — INNER PLUGS THE *CURVES* INTO y

$$\int_{x^2}^{2x} xy\,dy=x\left[\frac{y^2}{2}\right]_{x^2}^{2x}=x\left(2x^2-\frac{x^4}{2}\right)=2x^3-\frac{x^5}{2}$$

$$\int_0^2\left(2x^3-\frac{x^5}{2}\right)dx=\left[\frac{x^4}{2}-\frac{x^6}{12}\right]_0^2=8-\frac{16}{3}=\frac{8}{3}$$

↓

4. TYPE II — RE-DERIVE, DON'T RE-LETTER *(thing 3)*
$x=\frac{y}{2}$ enters, $x=\sqrt{y}$ exits *(test $y=1$: $\frac{1}{2}<1$)*, y spans $0\to4$:

$$\int_0^4\int_{y/2}^{\sqrt{y}} xy\,dx\,dy=\int_0^4 y\left(\frac{y}{2}-\frac{y^2}{8}\right)dy=\left[\frac{y^3}{6}-\frac{y^4}{32}\right]_0^4=\frac{32}{3}-8=\frac{8}{3}\ \checkmark$$

The aha: the new bounds are not the old ones with letters swapped — $[0,2],[x^2,2x]$ became $[0,4],[\frac{y}{2},\sqrt{y}]$. That re-derivation is the skill.

Area: set $f=1$ and the inner pass returns top−bottom — Calc 2's area-between-curves was a double integral all along. Here $\int_0^2(2x-x^2)dx=\frac{4}{3}$.

Worked: swap the order to rescue a stuck integral — $\int_0^1\int_y^1 e^{x^2}\,dx\,dy$

1. WHY IT'S STUCK
$\int e^{x^2}dx$ has no elementary antiderivative — the inner integral cannot even start

↓

2. READ THE REGION OFF THE LIMITS
$y\le x\le 1$, $0\le y\le 1$: the triangle $(0,0),(1,0),(1,1)$ — the half of the unit square below $y=x$

↓

3. RE-SLICE VERTICALLY, THEN INTEGRATE
sweep $x:0\to1$; each slice runs $y=0$ up to $y=x$:

$$\int_0^1\int_0^x e^{x^2}\,dy\,dx=\int_0^1 e^{x^2}\left[y\right]_0^x dx=\int_0^1 x e^{x^2}dx$$

$u=x^2$, $du=2x\,dx$: $=\frac{1}{2}\left[e^u\right]_0^1=\frac{e-1}{2}$

The aha: the swap manufactured the missing x the u-sub needed. Exam signals: e^{x^2}, $\sin(x^2)$, $\cos(x^2)$ as the inner integrand ⇒ flip the order.

Worked (C): reversing the order forces a split

D between $y=x^2$ and $y=2-x^2$. One direction needs one integral, the other two.

1–2. MEET, THEN ORDER
$x^2=2-x^2\Rightarrow x=\pm1$ *(both give $y=1$)*. Type I: $x^2\le y\le 2-x^2$, $-1\le x\le 1$ — one integral, nothing to split

↓

3. TEST SLICE AT TWO HEIGHTS *(thing 4)*
at $y=\frac{1}{2}$ the slice enters and exits on $y=x^2$; at $y=\frac{3}{2}$ on $y=2-x^2\Rightarrow$ split at $y=1$

4. SOLVE EACH PIECE FOR x *(thing 3)*

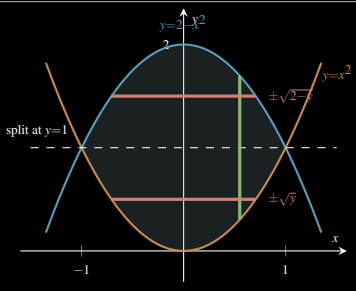
$y=x^2 \Rightarrow x=\pm\sqrt{y}$ on $0\le y\le 1$; $\quad y=2-x^2 \Rightarrow x=\pm\sqrt{2-y}$ on $1\le y\le 2$

$$\int_0^1\int_{-\sqrt{y}}^{\sqrt{y}} f\,dx\,dy+\int_1^2\int_{-\sqrt{2-y}}^{\sqrt{2-y}} f\,dx\,dy$$

5. CHECK WITH $f=1$ — BOTH ROUTES GIVE THE AREA

Type I: $\int_{-1}^{1}(2-2x^2)\,dx=\left[2x-\frac{2x^3}{3}\right]_{-1}^{1}=\frac{8}{3}$

Type II: $\int_0^1 2\sqrt{y}\,dy+\int_1^2 2\sqrt{2-y}\,dy=\frac{4}{3}+\frac{4}{3}=\frac{8}{3}$ ✓



the vertical slice runs $x^2\to 2-x^2$ everywhere — one integral. The horizontal slices change formula at $y=1$ — two integrals

Worked: roof = floor sets the shadow — in Cartesian

Below the plane $z=x+2$, above the cylinder $z=x^2$, between $y=0$ and $y=2$. Only two of the four shadow edges are stated.

1. THE SURFACES SUPPLY THE x-RANGE *(thing 1)*

$x+2=x^2 \Rightarrow x^2-x-2=0 \Rightarrow (x-2)(x+1)=0 \Rightarrow x=-1,\,2$

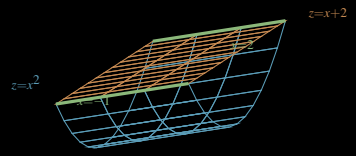
2. THE GIVEN PLANES SUPPLY THE y-RANGE

$0\le y\le 2$, handed over directly. *(The shadow is assembled from two sources — read every boundary and ask which job it does.)*

3. WHICH IS ROOF *(thing 2)*, THEN INTEGRATE

at $x=0$: plane $=2$, cylinder $=0 \Rightarrow$ height $=(x+2)-x^2=2+x-x^2$

$$V=\int_{-1}^{2}\int_0^{2}(2+x-x^2)\,dy\,dx=2\left[2x+\frac{x^2}{2}-\frac{x^3}{3}\right]_{-1}^{2}=2\left(\frac{10}{3}+\frac{7}{6}\right)=2\cdot\frac{27}{6}=9$$



the plane and the cylinder pinch shut on the two green edges — those, not the problem statement, supply the x-limits

Average value over any *region*

The rectangle version above divides by a side-length product, which only works for a rectangle. The general statement divides by the region's own measure:

$$f_{\text{avg}}=\frac{1}{\text{Area}(D)}\iint_D f\,dA \qquad f_{\text{avg}}=\frac{1}{\text{Vol}(E)}\iiint_E f\,dV$$

The denominator is itself an integral you already know: $\text{Area}(D)=\iint_D 1\,dA$ and $\text{Vol}(E)=\iiint_E 1\,dV$ — the $f=1$ aha. So this is two integrals over the same region, not a new technique.

THE PROCESS
1. **Set up the region once.** The limits serve both integrals.
2. **Integrate** f over it — the numerator.
3. **Integrate** 1 over it — the denominator.
4. **Divide.**

Worked: average of $f=xy$ over the region between $y=x^2$ and $y=2x$

The running region of this section. Both pieces are already computed above — this only assembles them.

1. NUMERATOR — WORKED ABOVE

$\iint_D xy\,dA=\frac{8}{3}$ *(the Type I / Type II example)*

2. DENOMINATOR — ALSO WORKED ABOVE

$\text{Area}(D)=\iint_D 1\,dA=\int_0^2(2x-x^2)\,dx=\frac{4}{3}$

3. DIVIDE

$$f_{\text{avg}}=\frac{8/3}{4/3}=\frac{8}{3}\cdot\frac{3}{4}=2$$

Sanity: $f=xy$ ranges from 0 up to 8 on this region *(at the corner $(2,4)$)*, so an average of 2 sits sensibly inside that span. An average outside the function's range means the denominator is wrong — usually a region measure computed as if it were a rectangle.

Common pitfalls

- Outer limits must be numbers — an outer limit carrying a variable leaves that variable alive in the "answer".
- Match limits to the differential order: $dy\,dx$ means the inner limits are y-curves $g_1(x)\to g_2(x)$.
- Swapping the order $\not=$ swapping the symbols — re-derive from the sketch every time *(thing 3)*.
- If a boundary changes formula mid-run, split and add — one curve per limit slot *(thing 4)*.
- A stuck inner integral is a hint, not a wall: e^{x^2}-type integrands are order-swap problems by design.
- Sketch first, always — every limit above came from the picture, none from the integrand.

double integrals — polar coordinates

Circular regions fight Cartesian limits ($\sqrt{a^2-x^2}$ everywhere) but describe themselves in polar. Conversion kit: $x=r\cos\theta$, $y=r\sin\theta$, $x^2+y^2=r^2$, $\tan\theta=\frac{y}{x}$. One new rule — the area element gains a factor of r:

$$\iint_R f(x,y)\,dA=\int_\alpha^\beta\int_a^b f(r\cos\theta,\,r\sin\theta)\,r\,dr\,d\theta \qquad dA=r\,dr\,d\theta$$

THE PROCESS
1. **Region** → polar: rays give θ, arcs and curves give r.
2. **Integrand** → polar: every x and y, no exceptions.
3. **Attach the r** — $dA=r\,dr\,d\theta$, even when the integrand never asked.
4. **Integrate inside out**, r first.

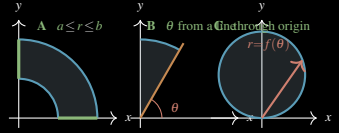
Iron rule, polar edition: outer (θ) limits are numbers; the inner (r) limit may carry θ — never the reverse.

Step 1 decoded — three region shapes

A **polar rectangle** — two arcs, two rays ⇒ all four limits constant. Disks, annuli, quarter-disks.

B **wedge cut by a line** — a boundary $y=mx$ is the ray $\theta=\arctan m$. θ comes from the line, not from habit.

C $r=f(\theta)$ — a curve through the origin (cardioid, or a circle not centred there) ⇒ the inner limit carries θ.



The four things you actually solve

1. DOES θ SURVIVE? — CONVERT FIRST, THEN LOOK
if the converted integrand still has $\cos\theta$ or $\sin\theta$, the outer pass is a real trig integral. An x, y or xy always leaves one behind; x^2+y^2 never does.

↓

2. θ FROM A LINE
$\tan\theta=\dfrac{y}{x}=m \Rightarrow \theta=\arctan m$.

$$y=x \Rightarrow \theta=\tfrac{\pi}{4} \qquad y=\sqrt{3}x \Rightarrow \theta=\tfrac{\pi}{3} \qquad x=0 \Rightarrow \theta=\tfrac{\pi}{2}$$

↓

3. OFF-CENTRE CIRCLE $\to r=f(\theta)$
substitute $x^2+y^2=r^2$ and divide by r:

$$x^2+y^2=2ay \Rightarrow r^2=2ar\sin\theta \Rightarrow r=2a\sin\theta \ (\theta:0\to\pi)$$

$$x^2+y^2=2ax \Rightarrow r=2a\cos\theta \ (\theta:-\tfrac{\pi}{2}\to\tfrac{\pi}{2})$$

(complete the square to confirm the centre: $x^2+(y-a)^2=a^2$)

↓

4. CLOSE THE SHADOW — SET ROOF = FLOOR
for a volume, the solid's edge is where the two surfaces meet:

$$2y=x^2+y^2 \Rightarrow r=2\sin\theta \qquad x^2+y^2=4 \Rightarrow r=2$$

Worked (A): volume under $z=4-x^2-y^2$ over the disk $x^2+y^2\le 4$

1–3. REGION, INTEGRAND, JACOBIAN
disk: $0\le r\le 2,\ 0\le\theta\le 2\pi$ *(all constants)*. $4-x^2-y^2=4-r^2$ *(no θ — it will die)*. So $\displaystyle\int_0^{2\pi}\int_0^{2}(4-r^2)\,r\,dr\,d\theta$

↓

4. INSIDE OUT
$(4-r^2)r=4r-r^3$: $\displaystyle\int_0^{2}(4r-r^3)\,dr=\Big[2r^2-\frac{r^4}{4}\Big]_0^{2}=8-4=4$, then $\displaystyle\int_0^{2\pi}4\,d\theta=8\pi$

In Cartesian this needed $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$ and a trig substitution. Polar made every limit a constant.

Worked (A): when θ survives — $\iint_R xy\,dA$, *R:* $1\le r\le 2,\ 0\le\theta\le\frac{\pi}{2}$

1–3. CONVERT — BOTH FACTORS CARRY A θ
$xy=(r\cos\theta)(r\sin\theta)=r^2\cos\theta\sin\theta$, times the Jacobian r gives $r^3\cos\theta\sin\theta$

↓

4. IT FACTORS — SPLIT (§5.1 SHORTCUT, IN POLAR)

$$\left(\int_1^{2}r^3\,dr\right)\left(\int_0^{\pi/2}\cos\theta\sin\theta\,d\theta\right)=\Big[\frac{r^4}{4}\Big]_1^{2}\cdot\Big[\frac{\sin^2\theta}{2}\Big]_0^{\pi/2}=\frac{15}{4}\cdot\frac{1}{2}=\frac{15}{8}$$

The aha: θ dies only when the converted integrand has no θ in it *(thing 1)*. The example above got lucky. This one didn't.

Worked (rescue): Cartesian $\to$ polar — $\displaystyle\int_{-1}^{1}\int_0^{\sqrt{1-x^2}} e^{x^2+y^2}\,dy\,dx$

1. STUCK — AND SWAPPING THE ORDER WON'T HELP
$e^{x^2+y^2}$ has no elementary antiderivative in x *or* y. Read the region off the limits: y runs to $\sqrt{1-x^2}$, so $x^2+y^2=1$; x runs $-1\to 1 \Rightarrow$ upper half disk: $0\le r\le 1$, $0\le\theta\le\pi$

↓

2. CONVERT — THE JACOBIAN IS THE u-SUB FUEL
$e^{x^2+y^2}=e^{r^2}$; $u=r^2$, $du=2r\,dr$:

$$\int_0^{\pi}\int_0^{1}e^{r^2}\,r\,dr\,d\theta=\int_0^{\pi}\frac{1}{2}\Big[e^{u}\Big]_0^{1}d\theta=\frac{e-1}{2}\cdot\pi=\frac{\pi(e-1)}{2}$$

Exam signals for "go polar": $e^{x^2+y^2}$, $\sqrt{x^2+y^2}$, $\sin(x^2+y^2)$, or a $\sqrt{a^2-x^2}$ limit.

Worked (B): read θ off a line — $\displaystyle\int_0^{1}\int_{\sqrt{3}x}^{\sqrt{4-x^2}}\frac{dy\,dx}{\sqrt{x^2+y^2}}$

1. BOUNDARIES
bottom $y=\sqrt{3}x$ *(a line)*; top $x^2+y^2=4$ *(the circle $r=2$)*; left $x=0$

↓

2. θ FROM THE LINE *(thing 2)*
$\tan\theta=\sqrt{3} \Rightarrow \theta=\frac{\pi}{3}$; the y-axis is $\theta=\frac{\pi}{2}$; r runs $0\to 2$
(corner check: $\sqrt{3}x=\sqrt{4-x^2} \Rightarrow x=1$, $y=\sqrt{3}$, and $(1,\sqrt{3})$ has $r=2$, $\theta=\frac{\pi}{3}$ ✓)

↓

3. THE JACOBIAN *CANCELS* THE INTEGRAND
$\dfrac{1}{\sqrt{x^2+y^2}}=\dfrac{1}{r}$, and $dA=r\,dr\,d\theta$, so $\dfrac{1}{r}\cdot r=1$:

$$\int_{\pi/3}^{\pi/2}\int_0^{2}1\,dr\,d\theta=2\left(\frac{\pi}{2}-\frac{\pi}{3}\right)=\frac{\pi}{3}$$

The aha: the Jacobian is never a tax — read what it does to *this* integrand. Here it deleted it.

Worked (C): area inside the cardioid $r=1+\cos\theta$

Boundary is $r=f(\theta)$, so the inner limit carries θ. Area means $f=1$.

1–2. BUILD THE INTEGRAL, INNER
$\displaystyle\int_0^{2\pi}\int_0^{1+\cos\theta}r\,dr\,d\theta$; $\displaystyle\int_0^{1+\cos\theta}r\,dr=\Big[\frac{r^2}{2}\Big]_0^{1+\cos\theta}=\frac{(1+\cos\theta)^2}{2}$

↓

3. EXPAND, POWER-REDUCE, INTEGRATE
$\frac{(1+\cos\theta)^2}{2}=\frac{1+2\cos\theta+\cos^2\theta}{2}$, and $\cos^2\theta=\frac{1+\cos2\theta}{2}$, giving $\frac{3}{4}+\cos\theta+\frac{1}{4}\cos2\theta$:

$$\int_0^{2\pi}\left(\frac{3}{4}+\cos\theta+\frac{\cos2\theta}{4}\right)d\theta=\Big[\frac{3\theta}{4}+\sin\theta+\frac{\sin2\theta}{8}\Big]_0^{2\pi}=\frac{3\pi}{2}$$

Stop after step 2 and you are holding $\int\frac{1}{2}r(\theta)^2d\theta$ — the polar area formula is a double integral with the inner pass pre-done.

Worked: volume between two *surfaces — roof minus floor*

Not every solid sits on $z=0$. Inside the cylinder $x^2+y^2=1$, above $z=y-1$, below $z=3-y$.

1. HEIGHT = ROOF − FLOOR *(thing 1: check which is on top)*
$(3-y)-(y-1)=4-2y$; roof is above when $y<2$, and $y\le 1$ on this disk ✓

↓

2. SHADOW COMES FROM THE *WALL*
the planes are lid and base. The cylinder alone sets the shadow: $0\le r\le 1, 0\le\theta\le 2\pi$

↓

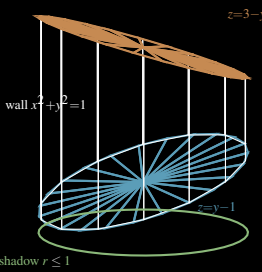
3. CONVERT AND BUILD THE INTEGRAL
$4-2y=4-2r\sin\theta$ *(a θ survives)*: $\displaystyle V=\int_0^{2\pi}\int_0^{1}(4-2r\sin\theta)\,r\,dr\,d\theta$

↓

4. INSIDE OUT — THE FULL LAP KILLS THE SINE
$\displaystyle\int_0^{1}(4r-2r^2\sin\theta)\,dr=\Big[2r^2-\frac{2r^3}{3}\sin\theta\Big]_0^{1}=2-\frac{2}{3}\sin\theta$, then

$$\int_0^{2\pi}\left(2-\frac{2}{3}\sin\theta\right)d\theta=\Big[2\theta+\frac{2}{3}\cos\theta\Big]_0^{2\pi}=4\pi+\left(\frac{2}{3}-\frac{2}{3}\right)=4\pi$$

Sanity: the tilts oppose, so mean height $=4$; times the disk's area π gives 4π ✓ Over half a lap the sine does *not* cancel: $\int_0^{\pi}$ gives $2\pi-\frac{4}{3}$.



the floor and the roof both tilt with y; the wall alone fixes the shadow. Height = roof − floor, integrated over that shadow

Worked (C): find the shadow by setting roof = floor

Above the paraboloid $z=x^2+y^2$, below the plane $z=2y$. Nobody hands you the shadow.

1. SET THEM EQUAL *(thing 4)*, **THEN COMPLETE THE SQUARE**
$2y=x^2+y^2 \Rightarrow x^2+y^2-2y=0 \Rightarrow x^2+(y-1)^2=1$ — radius 1, centred $(0,1)$, not the origin

↓

2. OFF-CENTRE CIRCLE → $r=f(\theta)$ *(thing 3)*

$r^2=2r\sin\theta \Rightarrow r=2\sin\theta$, traced once over $0\le\theta\le\pi$

↓

3. HEIGHT, BUILD THE INTEGRAL

$2y-(x^2+y^2)=2r\sin\theta-r^2$: $\quad V=\displaystyle\int_0^{\pi}\int_0^{2\sin\theta}\left(2r\sin\theta-r^2\right)r\,dr\,d\theta$

↓

4. INNER

$$\int_0^{2\sin\theta}\left(2r^2\sin\theta-r^3\right)dr=\left[\frac{2r^3}{3}\sin\theta-\frac{r^4}{4}\right]_0^{2\sin\theta}=\frac{16}{3}\sin^4\theta-4\sin^4\theta=\frac{4}{3}\sin^4\theta$$

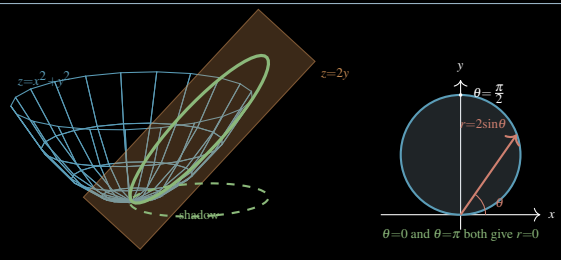
↓

5. OUTER — POWER-REDUCE TWICE

$\sin^4\theta=\left(\frac{1-\cos2\theta}{2}\right)^2=\frac{3}{8}-\frac{1}{2}\cos2\theta+\frac{1}{8}\cos4\theta$, so

$$\int_0^{\pi}\sin^4\theta\,d\theta=\left[\frac{3\theta}{8}-\frac{\sin2\theta}{4}+\frac{\sin4\theta}{32}\right]_0^{\pi}=\frac{3\pi}{8}$$

$$V=\frac{4}{3}\cdot\frac{3\pi}{8}=\frac{\pi}{2}$$



left: the plane slices the paraboloid bowl. Right: their intersection, projected down, is the off-centre disk $r=2\sin\theta$

Worked: fill a bowl to depth h — the unknown sits in the limits

Both bowls are one-liners once you see the pattern. a is a steepness dial, not part of the method — most problems use $a=1$.

bowl	waterline	water to depth h
cone $z=a\sqrt{x^2+y^2}$	$r=\dfrac{h}{a}$	$V=\dfrac{\pi h^3}{3a^2}$
paraboloid $z=a(x^2+y^2)$	$r=\sqrt{\dfrac{h}{a}}$	$V=\dfrac{\pi h^2}{2a}$

DERIVING THE CONE ROW ($a=1$)

waterline: $r=h$. Height of a column: $h-r$.

$$V=\int_0^{2\pi}\int_0^{h}(h-r)\,r\,dr\,d\theta=2\pi\left[\frac{hr^2}{2}-\frac{r^3}{3}\right]_0^{h}=2\pi\left(\frac{h^3}{2}-\frac{h^3}{3}\right)=2\pi\cdot\frac{h^3}{6}=\frac{\pi h^3}{3}$$

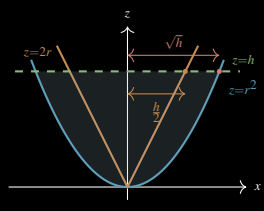
The check worth memorising: compare each to the cylinder it sits in (radius = waterline, height $=h$). A cone holds $\frac{1}{3}$ of its cylinder. A paraboloid holds $\frac{1}{2}$. Both ratios are independent of a — which is exactly why the coefficient is not part of the method.

REVERSE — GIVEN THE VOLUME, FIND THE DEPTH

the cone bowl $z=\sqrt{x^2+y^2}$ holds 9π:

$$\frac{\pi h^3}{3}=9\pi \Rightarrow h^3=27 \Rightarrow h=3$$

Integrate with h as a symbol all the way to a formula, *then* solve. Substituting a number early destroys the only thing you can solve for. *(Two bowls holding equal water: set the two formulas equal and solve the same way.)*



cross-section through the axis: one waterline, two different waterline radii — $\sqrt{h}$ for the bowl, $\frac{h}{2}$ for the cone

Common pitfalls (polar)

- Forgetting the r in $dA=r\,dr\,d\theta$ is THE classic lost point — the substitution never pays without it.
- θ's range comes from the region, not habit: full disk $0\to2\pi$, upper half $0\to\pi$, right half $-\frac{\pi}{2}\to\frac{\pi}{2}$.
- Convert the whole integrand — every x and y becomes r and θ before you integrate anything.
- "The trig term drops out" is a full-lap privilege: $\int_0^{2\pi}\sin\theta\,d\theta=0$, but over $0\to\pi$ it does not. Evaluate the bracket.
- A circle not centred at the origin is still polar — complete the square. It just gives $r=f(\theta)$ instead of a constant *(thing 3)*.
- A $\sqrt{a^2-x^2}$ limit or an x^2+y^2 integrand is a polar invitation — read the region off the limits first, then switch.

triple integrals — rectangular coordinates

Three nested Calc 1 integrals. Each pass deletes one variable. After three you have a number. With $f=1$ you get volume; with $f=\delta$ you get mass. The integrating is never the hard part — finding the limits is the whole problem, so everything below is about limits.

$$\iiint_E f\,dV=\int_{\text{outer}}\int_{\text{middle}}\int_{\text{inner}} f\;\;d(\text{in})\,d(\text{mid})\,d(\text{out})$$

THE PROCESS — EVERY TRIPLE INTEGRAL, NO EXCEPTIONS

1. **Inner variable** — pick the one trapped between two surfaces.
2. **Solve** both of those surfaces for that variable ⇒ inner limits.
3. **Project** the solid onto the plane of the other two ⇒ the **shadow**.
4. **Read the shadow** as a §5.2 Type I/II region ⇒ middle & outer limits.
5. **Integrate inside out** — each pass kills one variable.

Limit ladder: inner may carry both outer variables · middle only the outermost · outer must be numbers. A variable alive in your answer means you broke this.

Step 1 decoded — which variable goes inside?

Read the list of bounding surfaces and match one of three shapes:

A two surfaces are $z=f(x,y)$ — a roof and a floor ⇒ **inner** $=z$, project onto xy

B a surface never mentions one variable ($x=z^2$ has no y) ⇒ that variable is free: it is **inner**, project onto the plane of the other two

C a vertical wall $x^2+y^2=a^2$ ⇒ **inner** $=z$, shadow is a disk — go **cylindrical** (next section)

(A is the default. B is the one people miss — it is why "project onto xy" is a habit, not a rule.)

The four things you actually solve

Every limit in every problem comes from one of these. Nothing else.

1. WHICH SURFACE IS ON TOP?

Plug in one interior point and compare. No algebra — and skipping it flips your sign.

for example, $z=x^2$ vs $z=x+2$ at $x=0$: $0<2$ ⇒ $x^2\le z\le x+2$

↓

2. SOLVE A SURFACE FOR THE INNER VARIABLE

Rearrange one equation. A curved face needs a real inversion.

$$x+y+z=1 \Rightarrow z=1-x-y \qquad x=z^2 \Rightarrow z=\sqrt{x}$$
$$z=1-x^2 \Rightarrow x=\sqrt{1-z} \qquad x^2+y^2+z^2=2z \Rightarrow \rho=2\cos\varphi$$

↓

3. CLOSE THE SHADOW — SET TWO BOUNDARIES EQUAL

When the stated fences don't close the region, the surfaces themselves close it. Set them equal, solve, and those roots are outer limits.

$$x^2=2-x \Rightarrow x=1,-2 \qquad z^2=3-2z \Rightarrow z=1 \text{ (reject } z=-3)$$

↓

4. NON-EMPTY CHECK — IT *IS* THE NEXT LIMIT

If the inner run is $g\le x\le h$, demanding $h\ge g$ produces the middle limit. Easy to skip. Skipping it loses a whole limit.

$$\sqrt{1-z}\ge y \Rightarrow 1-z\ge y^2 \Rightarrow z\le 1-y^2$$

Seeing the solid — four moves before any integral

Most lost time here is spent trying to draw one good 3D picture. Do not. Draw flat pictures instead, in this order:

1. SPOT THE EXTRUDED SURFACES — THE BIGGEST SIMPLIFICATION

A surface missing a variable is a **cylinder**: it extrudes along that axis, unchanged.

$z=1-x^2$ has no y ⇒ a parabolic sheet running the length of the y-axis. $\quad y=1-x$ has no z ⇒ a slanted wall running straight up.

Two extruded surfaces missing *different* variables is what makes a solid hard.

↓

2. DRAW THE THREE TRACES
Set $x=0$, then $y=0$, then $z=0$. Each gives a flat curve or region in a coordinate plane. Three flat pictures beat one bad 3D sketch.

↓

3. FIND THE CORNERS
Solve the surfaces in pairs. Corners pin the solid down and catch a mis-drawn face immediately.

↓

4. GET ALL THREE SHADOWS AT ONCE
Project onto xy, xz and yz. Every one of the six orderings reads its outer and middle limits off one of these three pictures. Compute them once and the reordering drill becomes bookkeeping.

Worked (A, warm-up): $\iiint_B xyz\,dV$, $B=[0,1]\times[0,2]\times[0,3]$

A box: all six limits are constants, so steps 1–4 are free.

1. ITERATE, THEN WORK INSIDE OUT
$\int_0^1\int_0^2\int_0^3 xyz\,dz\,dy\,dx$: $\int_0^3 xyz\,dz = xy\left[\frac{z^2}{2}\right]_0^3=\frac{9}{2}xy$, then $\int_0^2 \frac{9}{2}xy\,dy=\frac{9}{2}x\left[\frac{y^2}{2}\right]_0^2=9x$, then $\int_0^1 9x\,dx=9\left[\frac{x^2}{2}\right]_0^1=\frac{9}{2}$

Separable check (§5.1, three factors — xyz factors, box region): $\left(\int_0^1 x\,dx\right)\left(\int_0^2 y\,dy\right)\left(\int_0^3 z\,dz\right)=\frac{1}{2}\cdot 2\cdot\frac{9}{2}=\frac{9}{2}$ ✓

Worked (A): volume of the tetrahedron under $x+y+z=1$

The solid cut from the first octant by the plane $x+y+z=1$.

1–2. INNER $=z$, SO SOLVE THE PLANE FOR z
(thing 2) $x+y+z=1 \Rightarrow z=1-x-y$; floor $z=0$. Inner: $0\le z\le 1-x-y$

↓

3. PROJECT — SET $z=0$ IN THE ROOF
the shadow's edge is where the roof hits the floor *(thing 3)*: $x+y=1$. Shadow = triangle

↓

4. READ THE SHADOW (§5.2)
$0\le y\le 1-x$, $0\le x\le 1$ — outer limits are numbers ✓

↓

5. INTEGRATE INSIDE OUT

$$\int_0^{1-x-y} 1\,dz = 1-x-y \quad \textit{(roof − floor)}$$

$$\int_0^{1-x}(1-x-y)\,dy=\left[(1-x)y-\frac{y^2}{2}\right]_0^{1-x}=(1-x)^2-\frac{(1-x)^2}{2}=\frac{(1-x)^2}{2}$$

$$\int_0^1 \frac{(1-x)^2}{2}\,dx=\left[-\frac{(1-x)^3}{6}\right]_0^1=0-\left(-\frac{1}{6}\right)=\frac{1}{6}$$

Geometry check: $V=\frac{1}{3}(\text{base})(\text{height})=\frac{1}{3}\cdot\frac{1}{2}\cdot 1=\frac{1}{6}$ ✓

Worked (B): first octant, cylinder $x=z^2$*, planes* $x=3-2z$ *and* $y=4$

The case that breaks the "project onto xy" habit. Scan the surface list for a missing letter first.

1. INNER VARIABLE — WHICH LETTER IS MISSING?
$x=z^2$ and $x=3-2z$ mention only x and z. y appears once, as the fence $y=4 \Rightarrow y$ is free: **inner** $=y$, project onto xz *(variant B)*

↓

2. INNER LIMITS
y runs floor to fence: $0\le y\le 4$ *(constants — nothing to solve)*

↓

3. CLOSE THE xz SHADOW *(thing 3)*
the two curved/slanted boundaries meet where

$$z^2=3-2z \Rightarrow z^2+2z-3=0 \Rightarrow (z+3)(z-1)=0 \Rightarrow z=1$$

($z=-3$ is outside the first octant — reject it, and say so)

↓

4. WHICH IS LEFT, WHICH IS RIGHT *(thing 1)*
at $z=\frac{1}{2}$: $z^2=\frac{1}{4}$ vs $3-2z=2 \Rightarrow z^2\le x\le 3-2z$, with $0\le z\le 1$

↓

5. ASSEMBLE AND INTEGRATE

$$V=\int_0^1\int_{z^2}^{3-2z}\int_0^4 1\,dy\,dx\,dz=\int_0^1\int_{z^2}^{3-2z}4\,dx\,dz=4\int_0^1\left[(3-2z)-z^2\right]dz$$
$$=4\left[3z-z^2-\frac{z^3}{3}\right]_0^1=4\left(3-1-\frac{1}{3}\right)=4\cdot\frac{5}{3}=\frac{20}{3}$$

The aha: the projection plane is a choice, and the surfaces tell you which. Here the same solid could be done projecting onto xy (shadow = the rectangle $0\le x\le 3$, $0\le y\le 4$) — but then z is trapped between $z=\sqrt{x}$ and $z=\frac{3-x}{2}$, both inversions, for more work and the same $\frac{20}{3}$.

Re-ordering: run the process again, don't swap letters

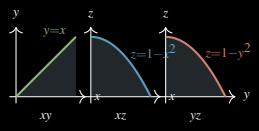
Asked for a different order, re-run steps 1–4 with a new inner variable. Nothing carries over. E: first octant under $z=1-x^2$, beside $y=x$ — asymmetric, so no ordering is a relabelling of another.

order	outer	middle	inner
$dz\,dy\,dx$	$0\le x\le 1$	$0\le y\le x$	$0\le z\le 1-x^2$
$dz\,dx\,dy$	$0\le y\le 1$	$y\le x\le 1$	$0\le z\le 1-x^2$
$dy\,dz\,dx$	$0\le x\le 1$	$0\le z\le 1-x^2$	$0\le y\le x$
$dy\,dx\,dz$	$0\le z\le 1$	$0\le x\le \sqrt{1-z}$	$0\le y\le x$
$dx\,dz\,dy$	$0\le y\le 1$	$0\le z\le 1-y^2$	$y\le x\le \sqrt{1-z}$
$dx\,dy\,dz$	$0\le z\le 1$	$0\le y\le \sqrt{1-z}$	$y\le x\le \sqrt{1-z}$

The red limit appears in no original equation. Here is where it comes from:

INNER $=x \Rightarrow$ SOLVE FOR x *(thing 2)*, THEN CHECK NON-EMPTY *(thing 4)*
$z=1-x^2 \Rightarrow x=\sqrt{1-z}$; the wall gives $x=y$. Inner: $y\le x\le\sqrt{1-z}$.
That run has positive length only when $\sqrt{1-z}\ge y \Rightarrow 1-z\ge y^2 \Rightarrow z\le 1-y^2$ — which is the middle limit, and is also the yz-shadow's upper edge.

Check ($f=1$): $\int_0^1\int_0^x\left(1-x^2\right)dy\,dx=\int_0^1\left(x-x^3\right)dx=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}$, and the $dx\,dy\,dz$ order gives $\int_0^1\frac{1-z}{2}\,dz=\frac{1}{4}$ ✓



the three shadows — one per choice of inner variable. The yz one is derived, not stated, and it is exactly the red limit above

Worked (the hard kind): two surfaces, each missing a **different** *variable*

E in the first octant, under $z=1-x^2$ and beside $y=1-x$. Both are extruded, along different axes — the case that costs the most time.

MOVE 1. THE EXTRUSIONS
$z=1-x^2$ (no y): parabolic sheet along y. $y=1-x$ (no z): slanted wall straight up. The floor is $z=0$, the back is $x=0$.

↓

MOVE 2. THE TRACES
$y=0$: the parabola $z=1-x^2$. $z=0$: the line $y=1-x$. $x=0$: the unit square $0\le y\le 1$, $0\le z\le 1$.

↓

MOVE 3. THE CORNERS
$(0,0,0)$, $(0,1,0)$, $(0,0,1)$, $(0,1,1)$, and $(1,0,0)$ — at $x=1$ both $1-x$ and $1-x^2$ vanish, so the solid pinches to a single point.

↓

MOVE 4. THE THREE SHADOWS
xy: the triangle $0\le y\le 1-x$. xz: the region under $z=1-x^2$. yz: the full unit square *(at $x=0$ every (y,z) in it is reached)*.

Four orderings now fall straight out of the shadows:

order	outer	middle	inner
$dz\,dy\,dx$	$0\le x\le 1$	$0\le y\le 1-x$	$0\le z\le 1-x^2$
$dz\,dx\,dy$	$0\le y\le 1$	$0\le x\le 1-y$	$0\le z\le 1-x^2$
$dy\,dz\,dx$	$0\le x\le 1$	$0\le z\le 1-x^2$	$0\le y\le 1-x$
$dy\,dx\,dz$	$0\le z\le 1$	$0\le x\le \sqrt{1-z}$	$0\le y\le 1-x$

THE LAST TWO ORDERINGS SPLIT — AND HERE IS WHY

put x innermost. It is capped by both walls at once:

$$x \le 1-y \quad \text{and} \quad x \le \sqrt{1-z} \qquad \Rightarrow \qquad x \le \min\left(1-y,\ \sqrt{1-z}\right)$$

A *min* in a limit always means a split. Find where the two caps trade places:

$$\sqrt{1-z}=1-y \Rightarrow 1-z=(1-y)^2 \Rightarrow z=2y-y^2$$

That curve cuts the yz square in two.

↓

SO $dx\,dz\,dy$ IS TWO INTEGRALS

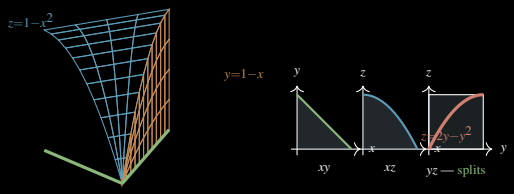
$$\int_0^1\int_0^{2y-y^2}\int_0^{1-y} f\,dx\,dz\,dy + \int_0^1\int_{2y-y^2}^{1}\int_0^{\sqrt{1-z}} f\,dx\,dz\,dy$$

below the curve the plane is the binding cap; above it the parabola is.

Check with $f=1$:

$$\int_0^1 (1-x)\left(1-x^2\right)dx = \int_0^1 \left(1-x-x^2+x^3\right)dx = 1-\frac{1}{2}-\frac{1}{3}+\frac{1}{4} = \frac{5}{12}$$

and the split route gives $\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$ ✓



the solid, then its three shadows. The red curve on the yz square is where the two caps trade places — that is the split, and it appears in no original equation

Common pitfalls (triple)

- Match the ladder to the differential order — $dz\,dy\,dx$ means z-limits innermost, x-limits outermost, outer a number.
- Re-ordering is re-running the process, not exchanging letters. A symmetric solid (the tetrahedron) makes the swap *look* legal. An asymmetric one exposes it.
- A *min* or *max* in a limit means the shadow splits — two surfaces cap the inner variable, and the split curve is where they trade places. Solve them equal to find it.
- An inner limit pair must not be empty — that check is the next limit out *(thing 4)*. Skip it and a middle limit vanishes.
- The xy-plane is not the only shadow — a surface that never mentions a variable marks it free *(variant B)*.
- Volume is $f=1$, not $f=z$ — roof-minus-floor appears by *integrating* $1\,dz$; don't also make it the integrand.

coordinate conversions — the full table

Three systems, six directions between them. **Rectangular** (x,y,z), **cylindrical** (r,θ,z), **spherical** (ρ,θ,φ).

This document's convention: ρ is the distance from the origin, φ is the angle down from the $+z$ axis ($0 \le \varphi \le \pi$), and θ is the same floor angle in all three systems. Physics texts swap θ and φ — trust the definition, never the letter.

THE ONLY THREE FACTS TO MEMORIZE

1. $r=\rho\sin\varphi$ — the shadow of ρ on the floor.
2. $z=\rho\cos\varphi$ — the height of ρ.
3. θ **never changes** — it is the same angle in cylindrical and spherical.
Everything below follows from these three plus $x=r\cos\theta$, $y=r\sin\theta$.

All six directions

direction	formulas
rect → cyl	$r=\sqrt{x^2+y^2}$, $\theta=\arctan\left(\frac{y}{x}\right)$, $z=z$
cyl → rect	$x=r\cos\theta$, $y=r\sin\theta$, $z=z$
rect → sph	$\rho=\sqrt{x^2+y^2+z^2}$, $\theta=\arctan\left(\frac{y}{x}\right)$, $\varphi=\arccos\left(\frac{z}{\rho}\right)$
sph → rect	$x=\rho\sin\varphi\cos\theta$, $y=\rho\sin\varphi\sin\theta$, $z=\rho\cos\varphi$
cyl → sph	$\rho=\sqrt{r^2+z^2}$, $\theta=\theta$, $\varphi=\arccos\left(\frac{z}{\sqrt{r^2+z^2}}\right)$
sph → cyl	$r=\rho\sin\varphi$, $\theta=\theta$, $z=\rho\cos\varphi$
rect → polar	$r=\sqrt{x^2+y^2}$, $\theta=\arctan\left(\frac{y}{x}\right)$ *(2D — same quadrant fix)*
polar → rect	$x=r\cos\theta$, $y=r\sin\theta$

The 2D pair is cylindrical with the z dropped — polar is cylindrical, flattened. A worked quadrant fix lives in the Trigonometry section; the same arctan trap below applies. The cyl/sph pair is the easiest — cylindrical and spherical differ only in how they name the floor distance and the height.

Volume elements: $dV=dz\,dy\,dx$, $dV=r\,dz\,dr\,d\theta$, $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$.

TWO TRAPS IN THE ANGLES

arctan returns only QI and QIV. If $x<0$, add π to θ. Sketch the point and check the quadrant every time.

Use arccos for φ, not $\arctan\left(\frac{r}{z}\right)$. With $r=3$, $z=-4$: $\arccos\left(\frac{-4}{5}\right)=143°$ ✓, but $\arctan\left(\frac{3}{-4}\right)=-37°$ fails — it lands below the xy-plane, where φ can never go.

Converting surfaces — *the row you will use most*

Points are rare on an exam. Surfaces are the real job: you convert a boundary to read a limit off it.

rectangular	cylindrical	spherical	use
x^2+y^2	r^2	$\rho^2\sin^2\varphi$	
$x^2+y^2+z^2$	r^2+z^2	ρ^2	
z	z	$\rho\cos\varphi$	
cylinder $x^2+y^2=a^2$	$r=a$	$\rho\sin\varphi=a$	cyl
plane $z=c$	$z=c$	$\rho\cos\varphi=c$	either
cone $z=a\sqrt{x^2+y^2}$	$z=ar$	$\varphi=\arctan\frac{1}{a}$	sph
sphere $x^2+y^2+z^2=a^2$	$r^2+z^2=a^2$	$\rho=a$	sph
sphere $x^2+y^2+z^2=2az$	$r^2+z^2=2az$	$\rho=2a\cos\varphi$	sph
paraboloid $z=a(x^2+y^2)$	$z=ar^2$	$\rho=\frac{\cos\varphi}{a\sin^2\varphi}$	cyl

The use column names the system where that surface is simplest — integrate there. A cone is one *number* in spherical (φ is constant). A paraboloid stays ugly in spherical and is clean in cylindrical.

The 45° cone: $a=1$ gives $\varphi=\arctan 1=\frac{\pi}{4}$. Steeper cone $z=2r$: $\varphi=\arctan\frac{1}{2}\approx 26.6°$ — not a nice angle, which is why textbook cones are almost always $z=\sqrt{x^2+y^2}$.

Worked: one point, all three systems

Convert $(x,y,z)=\left(-1,\sqrt{3},2\right)$ to cylindrical, then to spherical.

1. RECT → CYL — AND MIND THE QUADRANT

$r=\sqrt{(-1)^2+(\sqrt{3})^2}=\sqrt{1+3}=2$

$\arctan\left(\frac{\sqrt{3}}{-1}\right)=\arctan(-\sqrt{3})=-\frac{\pi}{3}$ *(a QIV angle)*, but $x<0$ and $y>0$ put the point in QII:

$$\theta=-\frac{\pi}{3}+\pi=\frac{2\pi}{3}, \qquad z=2 \qquad \Rightarrow \qquad \left(2,\ \tfrac{2\pi}{3},\ 2\right)$$

↓

2. CYL → SPH — REUSE r AND z

$\rho=\sqrt{r^2+z^2}=\sqrt{4+4}=2\sqrt{2}$

$$\varphi=\arccos\left(\frac{z}{\rho}\right)=\arccos\left(\frac{2}{2\sqrt{2}}\right)=\arccos\left(\frac{1}{\sqrt{2}}\right)=\frac{\pi}{4}$$

θ is unchanged: $\left(2\sqrt{2},\ \frac{2\pi}{3},\ \frac{\pi}{4}\right)$

↓

3. CHECK — CONVERT BACK

$x=\rho\sin\varphi\cos\theta=2\sqrt{2}\cdot\frac{1}{\sqrt{2}}\cdot\left(-\frac{1}{2}\right)=2\cdot\left(-\frac{1}{2}\right)=-1$ ✓

$y=\rho\sin\varphi\sin\theta=2\cdot\frac{\sqrt{3}}{2}=\sqrt{3}$ ✓ $\qquad z=\rho\cos\varphi=2\sqrt{2}\cdot\frac{1}{\sqrt{2}}=2$ ✓

Always route through cylindrical. Going rectangular → spherical directly means computing ρ and φ from three coordinates at once. Finding r first reduces it to a 2D problem you have done since trigonometry.

triple integrals — cylindrical coordinates

Cylindrical = polar + z: (r,θ,z) with $x=r\cos\theta$, $y=r\sin\theta$, $z=z$, $x^2+y^2=r^2$. The volume element inherits polar's r:

$$dV=r\,dz\,dr\,d\theta$$

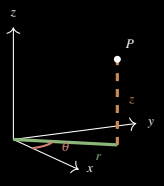
Symmetry cue: solids built around an axis — cylinders, cones, paraboloids (the quadric gallery) — where cross-sections are circles.

THE PROCESS — IT IS THE TRIPLE PROCESS, SHADOW IN POLAR

1. Inner $=z$ — roof and floor, solved for z *(variant A or C of the triple table)*.
2. Project onto xy ⇒ the shadow, then read it in polar (§5.3's three shapes).
3. Attach the r — $dV=r\,dz\,dr\,d\theta$.
4. Integrate inside out: z, then r, then θ.

A shadow is a **full disk** ⇒ $0\le\theta\le2\pi$ and odd trig terms cancel
B shadow is a **part lap** (half-disk, wedge) ⇒ θ from the bounding rays; trig terms survive
C shadow is $r=f(\theta)$ ⇒ the r-limit carries θ (set roof = floor to find it)

Two different r's can appear: one from the integrand ($x^2+y^2=r^2$), one from the Jacobian. Both are mandatory.



cylindrical = polar in the floor plus an elevator: r and θ locate the shadow of P, then z lifts it straight up

Worked: volume between $z=x^2+y^2$ and $z=4$

1. FIND THE SHADOW
roof meets floor where $x^2+y^2=4\Rightarrow r=2$: shadow = disk, $0\le r\le2$, $0\le\theta\le2\pi$

↓

2. CONVERT THE z LIMITS
floor: paraboloid $z=x^2+y^2=r^2$; roof: $z=4$. Inner: $r^2\le z\le4$ *(inner limits may carry r — the ladder rule, cylindrical edition)*

↓

3. BUILD THE INTEGRAL WITH $f=1$

$$\int_0^{2\pi}\int_0^{2}\int_{r^2}^{4} r\,dz\,dr\,d\theta \quad (dV=r\,dz\,dr\,d\theta)$$

↓

4. INNER = COLUMN HEIGHT, TIMES r

$$\int_{r^2}^{4} r\,dz=r\Big[z\Big]_{r^2}^{4}=r\left(4-r^2\right)=4r-r^3$$

↓

5. MIDDLE

$$\int_0^{2}\left(4r-r^3\right)dr=\Big[2r^2-\frac{r^4}{4}\Big]_0^{2}=(8-4)-0=4$$

↓

6. OUTER

$$\int_0^{2\pi}4\,d\theta=4\Big[\theta\Big]_0^{2\pi}=8\pi$$

The aha: steps 4–6 are literally the polar worked example (8π, same passes) — that double integral was this triple integral with the z-pass pre-done. Area→ $\iint 1$, polar area→ inner pass, and now volume→ $\iiint 1$: every formula below is an integral with early passes baked in.

Worked (dual): $\iiint_E (x^2+y^2)\,dV$, E: cylinder $x^2+y^2\le1$, $0\le z\le2$

1. CONVERT REGION AND INTEGRAND
region: $0\le r\le1$, $0\le\theta\le2\pi$, $0\le z\le2$; integrand: $x^2+y^2=r^2$

↓

2. BUILD THE INTEGRAL — TWO DIFFERENT r'S

$$\int_0^{2\pi}\int_0^{1}\int_0^{2}r^2\cdot r\,dz\,dr\,d\theta=\int_0^{2\pi}\int_0^{1}\int_0^{2}r^3\,dz\,dr\,d\theta$$

r^2 is the integrand, r is the Jacobian — both mandatory, and they merge into r^3

↓

3. INNER

$$\int_0^{2}r^3\,dz=r^3\Big[z\Big]_0^{2}=2r^3$$

↓

4. MIDDLE

$$\int_0^{1}2r^3\,dr=2\Big[\frac{r^4}{4}\Big]_0^{1}=2\left(\frac{1}{4}-0\right)=\frac{1}{2}$$

↓

5. OUTER

$$\int_0^{2\pi}\frac{1}{2}\,d\theta=\frac{1}{2}\Big[\theta\Big]_0^{2\pi}=\pi$$

(This integral is the moment of inertia of a uniform cylinder about its axis — the x^2+y^2 weight counts mass by its distance^2 from the axis.)

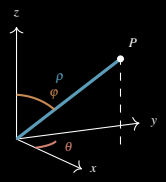
triple integrals — spherical coordinates

(ρ,θ,φ): $\rho=$ distance from the origin, $\theta=$ the same polar angle as always, $\varphi=$ angle down from the $+z$ axis, $0\le\varphi\le\pi$.

$$x=\rho\sin\varphi\cos\theta \qquad y=\rho\sin\varphi\sin\theta \qquad z=\rho\cos\varphi \qquad x^2+y^2+z^2=\rho^2$$

$$dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$$

Physics texts swap the letters θ and φ — trust the definition (which angle drops from the z-axis), never the letter. Symmetry cue: spheres, $x^2+y^2+z^2$, cones through the origin.



spherical: ρ is the straight-line distance from the origin to P; φ is measured down from the $+z$ axis (not up from the floor); θ is the same floor angle as always

THE PROCESS
1. Convert every surface to ρ,φ,θ — use $x^2+y^2+z^2=\rho^2$ and $z=\rho\cos\varphi$.
2. ρ **innermost**, φ middle, θ outer — almost always.
3. Attach $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$.
4. Integrate inside out. If all six limits are constants, the integral separates into three.

Step 1 decoded — three region shapes

A **ball / spherical shell** ⇒ ρ between constants, $\varphi:0\to\pi$, $\theta:0\to2\pi$. Fully separable.
B **cone** $z=a\sqrt{x^2+y^2}$ ⇒ it is just a φ limit: $\tan\varphi=\frac{1}{a}$. The 45° cone $z=\sqrt{x^2+y^2}$ is $\varphi=\frac{\pi}{4}$.
C **sphere through the origin** $x^2+y^2+z^2=2az\Rightarrow\rho=2a\cos\varphi$ — the ρ-limit carries φ, and the integral no longer separates.

Worked: volume of a sphere of radius a — derive the grade-school formula

1. DESCRIBE THE BALL
$0 \le \rho \le a$, $0 \le \varphi \le \pi$ *(pole to pole)*, $0 \le \theta \le 2\pi$ *(full lap)* — all constants

↓

2. BUILD THE INTEGRAL — FULLY SEPARABLE

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta = \left(\int_0^{a} \rho^2 \, d\rho\right)\left(\int_0^{\pi} \sin\varphi \, d\varphi\right)\left(\int_0^{2\pi} d\theta\right)$$

(the §5.1 separable shortcut, three factors: integrand factors, limits all constants)

↓

3. THE ρ FACTOR

$$\int_0^{a} \rho^2 \, d\rho = \left[\frac{\rho^3}{3}\right]_0^{a} = \frac{a^3}{3} - 0 = \frac{a^3}{3}$$

↓

4. THE φ FACTOR

$$\int_0^{\pi} \sin\varphi \, d\varphi = \Big[-\cos\varphi\Big]_0^{\pi} = -(-1) - (-1) = 2$$

↓

5. THE θ FACTOR, THEN MULTIPLY

$\int_0^{2\pi} d\theta = \Big[\theta\Big]_0^{2\pi} = 2\pi$, so

$$V = \frac{a^3}{3} \cdot 2 \cdot 2\pi = \frac{4}{3}\pi a^3$$

The aha: the formula memorized since grade school just fell out of a coordinate system — three one-line integrals.

Worked (dual): $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} \, dV$ over the unit ball — the rescue

1. WHY SPHERICAL
Cartesian and cylindrical are both hopeless. But $x^2+y^2+z^2 = \rho^2$, so $(x^2+y^2+z^2)^{3/2} = (\rho^2)^{3/2} = \rho^3$: the integrand collapses to e^{ρ^3}. Ball: $\rho : 0 \to 1$, $\varphi : 0 \to \pi$, $\theta : 0 \to 2\pi$

↓

2. BUILD THE INTEGRAL — SEPARABLE AGAIN

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{1} e^{\rho^3} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta = \left(\int_0^{1} e^{\rho^3} \rho^2 \, d\rho\right)\left(\int_0^{\pi} \sin\varphi \, d\varphi\right)\left(\int_0^{2\pi} d\theta\right)$$

↓

3. THE ρ^2 IS THE u-SUB FUEL
$u = \rho^3$, $du = 3\rho^2 \, d\rho \Rightarrow \rho^2 \, d\rho = \frac{du}{3}$; bounds: $\rho = 0 \to u = 0$, $\rho = 1 \to u = 1$:

$$\int_0^{1} e^{\rho^3} \rho^2 \, d\rho = \frac{1}{3}\int_0^{1} e^{u} \, du = \frac{1}{3}\Big[e^{u}\Big]_0^{1} = \frac{e-1}{3}$$

↓

4. ANGULAR FACTORS — SAME BRACKETS AS THE SPHERE

$\Big[-\cos\varphi\Big]_0^{\pi} = 2$, $\Big[\theta\Big]_0^{2\pi} = 2\pi$

↓

5. MULTIPLY

$$\frac{e-1}{3} \cdot 2 \cdot 2\pi = \frac{4\pi(e-1)}{3}$$

The aha — the manufactured-factor thread, complete: the order swap manufactured the x that e^{x^2} needed (§5.2), the polar r fueled e^{r^2} (§5.3), and here ρ^2 fuels e^{ρ^3}. The Jacobian is never a tax — it's the missing piece.

Worked: the ice-cream cone — φ as a region limit

The solid inside the cone $\varphi = \frac{\pi}{3}$, capped by the sphere $\rho = 1$:

1. THE CONE IS JUST A φ RANGE
$\varphi = \frac{\pi}{3}$ names every ray $60°$ off the $+z$ axis — "inside the cone" means $0 \le \varphi \le \frac{\pi}{3}$; cap: $0 \le \rho \le 1$; θ: full lap

↓

2. BUILD THE INTEGRAL

$$\int_0^{2\pi}\int_0^{\pi/3}\int_0^{1} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$$

↓

3. EVALUATE THE FACTORS

$$\left[\frac{\rho^3}{3}\right]_0^{1} = \frac{1}{3}, \qquad \Big[-\cos\varphi\Big]_0^{\pi/3} = -\frac{1}{2} - (-1) = \frac{1}{2}, \qquad \Big[\theta\Big]_0^{2\pi} = 2\pi$$

↓

4. MULTIPLY

$$V = \frac{1}{3} \cdot \frac{1}{2} \cdot 2\pi = \frac{\pi}{3}$$

In Cartesian this solid is a nightmare. In spherical it's a box — three constant ranges. That's the whole game: pick coordinates that make the region a box.

Worked: when the ρ limit is a function of φ

Every spherical example above had three constant ranges. Here is the general case: the solid inside the cone $\varphi = \frac{\pi}{4}$ and inside the sphere $x^2+y^2+z^2 = 2z$.

1. CONVERT THE SPHERE
$x^2+y^2+z^2 = \rho^2$ and $z = \rho\cos\varphi$, so

$$\rho^2 = 2\rho\cos\varphi$$

↓

2. DIVIDE BY ρ
legal away from the origin:

$$\rho = 2\cos\varphi$$

(completing the square in Cartesian confirms it: $x^2+y^2+(z-1)^2 = 1$ — radius 1, centred at $(0,0,1)$, so the sphere sits on the origin)

↓

3. THE CONE IS JUST A φ RANGE
"inside the cone $\varphi = \frac{\pi}{4}$" means $0 \le \varphi \le \frac{\pi}{4}$; θ runs a full lap $0 \to 2\pi$; ρ runs $0 \to 2\cos\varphi$ the inner limit carries φ — the ladder rule, spherical edition

↓

4. BUILD THE INTEGRAL

$$V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{2\cos\varphi} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$$

Not separable this time — the ρ-limit mentions φ, so the §5.1 shortcut is off the table

↓

5. INNER

$$\int_0^{2\cos\varphi} \rho^2 \, d\rho = \left[\frac{\rho^3}{3}\right]_0^{2\cos\varphi} = \frac{(2\cos\varphi)^3}{3} - 0 = \frac{8\cos^3\varphi}{3}$$

carry the $\sin\varphi$ along: the middle integrand is $\frac{8}{3}\cos^3\varphi\sin\varphi$

↓

6. MIDDLE — THE $\sin\varphi$ IS THE u-SUB FUEL
$u = \cos\varphi$, $du = -\sin\varphi \, d\varphi$; bounds: $\varphi = 0 \to u = 1$, $\varphi = \frac{\pi}{4} \to u = \frac{\sqrt{2}}{2}$:

$$\frac{8}{3}\int_0^{\pi/4} \cos^3\varphi \sin\varphi \, d\varphi = -\frac{8}{3}\int_1^{\sqrt{2}/2} u^3 \, du = \frac{8}{3}\int_{\sqrt{2}/2}^{1} u^3 \, du = \frac{8}{3}\left[\frac{u^4}{4}\right]_{\sqrt{2}/2}^{1}$$

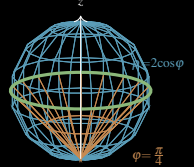
$$= \frac{8}{3} \cdot \frac{1}{4}\left(1 - \frac{1}{4}\right) = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2} \qquad \left(\left(\tfrac{\sqrt{2}}{2}\right)^4 = \tfrac{4}{16} = \tfrac{1}{4}\right)$$

↓

7. OUTER

$$\int_0^{2\pi} \frac{1}{2} \, d\theta = \frac{1}{2}\Big[\theta\Big]_0^{2\pi} = \frac{1}{2} \cdot 2\pi = \pi$$

Sanity: the whole sphere has radius 1, so volume $\frac{4}{3}\pi$; the cone keeps only a piece, and $\pi < \frac{4}{3}\pi$ ✓



the sphere rests on the origin, so its radius ρ shrinks as φ grows; the cone caps φ; they meet on the circle $r = 1$, $z = 1$

The aha: this is $r = 2\sin\theta$ one dimension up. A sphere through the origin gives $\rho = f(\varphi)$ exactly as a circle through the origin gives $r = f(\theta)$ — and the fix is the same: let the innermost limit depend on the next variable out.

Which coordinate system? — the decision guide

- **Cartesian**: boxes, planes, tetrahedra — straight-edged regions.
- **Polar / cylindrical**: circles in 2D; axis symmetry in 3D — cylinders, cones, paraboloids. The tell is x^2+y^2.

- **Spherical**: point symmetry — spheres, balls, cones through the origin. The tell is $x^2+y^2+z^2$.
- **Your own u,v**: none of the above fits, but the integrand names a combination — $e^{(x-y)/(x+y)}$, $(x+2y)^{10}$, a slanted parallelogram. Build the map, take the Jacobian.
- The question also gets a vote: "find the mass" means the integrand is a density, not 1 — same region, same coordinates, different f.
- The integrand gets a vote: $e^{x^2+y^2}$ forces polar even over an ugly region — a rescue-worthy integrand outranks a pretty region.

Common pitfalls (cylindrical & spherical)

- Cylindrical Jacobian is one r, same as polar: $dV=r\,dz\,dr\,d\theta$. An integrand's r^2 doesn't excuse it.
- Spherical Jacobian is $\rho^2\sin\varphi$, never ρ^2 alone — and it's $\sin\varphi$, not $\sin\theta$.
- φ stops at π: a full sphere is $\varphi:0\to\pi$ with $\theta:0\to2\pi$. Running φ to 2π covers the ball twice.
- Forgetting a Jacobian entirely turns every one of these problems into a wrong answer with perfect calculus after step 1 — write dV out before integrating anything.

mass, center of mass & moments

Swap the integrand $f=1$ for a **density** and the very same integral stops counting volume and starts counting mass. Nothing else in the machinery changes.

$$M=\iiint_E \delta\,dV \qquad\qquad M=\iint_D \delta\,dA \ \text{(a flat \textbf{lamina})}$$

The **moments** weight each point by how far out it sits, and dividing by the mass gives the balance point:

$$M_{yz}=\iiint_E x\,\delta\,dV \quad M_{xz}=\iiint_E y\,\delta\,dV \quad M_{xy}=\iiint_E z\,\delta\,dV$$

$$\bar{x}=\frac{M_{yz}}{M} \qquad \bar{y}=\frac{M_{xz}}{M} \qquad \bar{z}=\frac{M_{xy}}{M}$$

THE PROCESS — NOTHING NEW, ONE NEW INTEGRAND
1. Set up the region exactly as before — same shadow, same limits, same coordinates.
2. Swap the integrand: $f=1$ gives volume, $f=\delta$ gives mass, $f=x\delta$ gives a moment.
3. Use symmetry first — if δ never mentions a variable and the region is symmetric in it, that coordinate of the balance point is the geometric middle, for free.
4. Divide by the mass — never by the volume.

Sanity check that catches almost everything: denser at the top must put $\bar{z}$ above the geometric centre. If it doesn't, the error is upstream. Notation clash: most texts write density as ρ, which is also the spherical radius. This section uses δ for density so the two never collide — but on an exam, read which one the problem means. With δ constant the density cancels and $(\bar{x},\bar{y},\bar{z})$ is the purely geometric **centroid**.

Worked: mass of a box with a vertical density gradient

$B=[0,2]\times[0,1]\times[0,3]$, density $\delta(x,y,z)=1+z$ — the material packs tighter toward the top.

1. BUILD THE INTEGRAL — THE DENSITY IS THE INTEGRAND

$$M=\int_0^2\int_0^1\int_0^3 (1+z)\,dz\,dy\,dx$$

↓

2. INNER — FREEZE x AND y (NEITHER APPEARS)

$$\int_0^3 (1+z)\,dz=\Big[z+\frac{z^2}{2}\Big]_0^3=\Big(3+\frac{9}{2}\Big)-0=\frac{15}{2}$$

↓

3. MIDDLE

$$\int_0^1 \frac{15}{2}\,dy=\frac{15}{2}\Big[y\Big]_0^1=\frac{15}{2}(1-0)=\frac{15}{2}$$

↓

4. OUTER

$$\int_0^2 \frac{15}{2}\,dx=\frac{15}{2}\Big[x\Big]_0^2=\frac{15}{2}\cdot2=15$$

Sanity: δ runs linearly from 1 to 4, so the mean density is $\frac{1+4}{2}=\frac{5}{2}$; the volume is $2\cdot1\cdot3=6$; and $\frac{5}{2}\cdot6=15$ ✓

Worked (dual): where does that box balance?

1. TWO COORDINATES ARE FREE — USE SYMMETRY
$\delta=1+z$ never mentions x or y, so in those directions the box is uniform and balances at its geometric middle:

$$\bar{x}=\frac{0+2}{2}=1 \qquad \bar{y}=\frac{0+1}{2}=\frac{1}{2}$$

↓

2. THE z MOMENT — WEIGHT EACH POINT BY ITS HEIGHT

$$M_{xy}=\int_0^2\int_0^1\int_0^3 z(1+z)\,dz\,dy\,dx, \qquad z(1+z)=z+z^2$$

↓

3. INNER

$$\int_0^3 (z+z^2)\,dz=\Big[\frac{z^2}{2}+\frac{z^3}{3}\Big]_0^3=\Big(\frac{9}{2}+9\Big)-0=\frac{27}{2}$$

↓

4. MIDDLE AND OUTER — BOTH JUST MULTIPLY

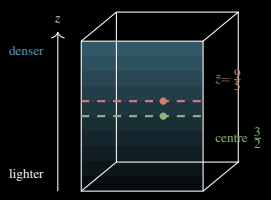
$$\int_0^1 \frac{27}{2}\,dy=\frac{27}{2}, \qquad \int_0^2 \frac{27}{2}\,dx=\frac{27}{2}\cdot2=27$$

↓

5. DIVIDE BY THE MASS

$$\bar{z}=\frac{M_{xy}}{M}=\frac{27}{15}=\frac{9}{5}$$

Sanity — and this is the whole point: $\frac{9}{5}=1.8$ sits above the geometric centre $\frac{3}{2}=1.5$, because the box is heavier at the top. A $\bar{z}$ on the wrong side of the middle means a sign, a limit, or the moment integrand is wrong. Always run this check — it catches errors no amount of re-reading the algebra will.



denser bands toward the top pull the center of mass up off the geometric centre

Worked: a lamina, in polar

The quarter disk $r\le2$ in the first quadrant, with density $\delta=\sqrt{x^2+y^2}$ — mass grows with distance from the origin.

1. CONVERT BOTH REGION AND DENSITY
region: $0\le r\le2,\ \ 0\le\theta\le\frac{\pi}{2}$; density: $\delta=\sqrt{x^2+y^2}=r$

↓

2. BUILD THE INTEGRAL — WATCH THE TWO r'S

$$M=\int_0^{\pi/2}\int_0^2 \underbrace{r}_{\text{density}}\cdot\underbrace{r}_{\text{Jacobian}}\,dr\,d\theta=\int_0^{\pi/2}\int_0^2 r^2\,dr\,d\theta$$

Both are mandatory and they merge into r^2 — dropping either one is the classic lost point

↓

3. INNER

$$\int_0^2 r^2\,dr=\Big[\frac{r^3}{3}\Big]_0^2=\frac{8}{3}-0=\frac{8}{3}$$

↓

4. OUTER

$$\int_0^{\pi/2}\frac{8}{3}\,d\theta=\frac{8}{3}\Big[\theta\Big]_0^{\pi/2}=\frac{8}{3}\cdot\frac{\pi}{2}=\frac{4\pi}{3}$$

↓

5. THE x MOMENT — NOW THREE r'S

$x=r\cos\theta$, so $x\,\delta\,dA=(r\cos\theta)(r)(r)\,dr\,d\theta=r^3\cos\theta\,dr\,d\theta$, and it factors:

$$M_y=\left(\int_0^{2} r^3\,dr\right)\left(\int_0^{\pi/2}\cos\theta\,d\theta\right)=\left[\frac{r^4}{4}\right]_0^{2}\cdot\Big[\sin\theta\Big]_0^{\pi/2}=4\cdot1=4$$

↓

6. DIVIDE

$$\bar{x}=\frac{M_y}{M}=\frac{4}{\frac{4\pi}{3}}=\frac{4\cdot3}{4\pi}=\frac{3}{\pi},\qquad \bar{y}=\frac{3}{\pi}\quad \textit{(by the } y{=}x \textit{ symmetry of region and density)}$$

Sanity: $\frac{3}{\pi}\approx0.95$ — inside the quarter disk, and pushed outward from the origin, which is where a density that grows with r should put it ✓

Common pitfalls (mass & moments)

- The moment integrand is $x\,\delta$, not x — and not δ alone. Three factors ride in a polar moment: the coordinate, the density, and the Jacobian.
- Divide by the mass, never by the volume — $\bar{z}=\frac{M_{xy}}{M}$. Dividing by V silently gives the centroid of a uniform solid instead.
- M_{xy} pairs with $\bar{z}$, not with $\bar{x}$: the subscript names the plane the moment is taken about, so the surviving coordinate is the missing letter.
- Use symmetry before integrating: if δ never mentions a variable and the region is symmetric in it, that coordinate of the balance point is the geometric middle — for free.
- Sanity-check the direction: denser at the top must give $\bar{z}$ above the middle. If it does not, the error is upstream.

change of variables — where every Jacobian comes from

Polar, cylindrical and spherical have all arrived as asserted formulas: $dA=r\,dr\,d\theta$, $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$. They are not conventions to memorise. They are determinants, and the machine that produces them works for substitutions you invent yourself.

$$\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}x_u & x_v\\ y_u & y_v\end{vmatrix}\qquad dA=\left|\frac{\partial(x,y)}{\partial(u,v)}\right|du\,dv$$

$$\iint_R f(x,y)\,dA=\iint_S f\big(x(u,v),y(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right|du\,dv$$

where S is the region R redrawn in the uv-plane. In 3D the matrix is 3×3 and dV picks up its determinant the same way.

THE PROCESS

1. **Let the integrand choose u and v** — name the combinations that appear in it.
2. **Invert**: solve for x and y in terms of u,v. The Jacobian needs them this way round.
3. **Map every boundary edge** into the uv-plane ⇒ the new region S.
4. **Take the determinant**, then its absolute value.
5. **Integrate** over S with the extra factor.

When to reach for this: no stock coordinate system fits, but the integrand names a combination — $e^{(x-y)/(x+y)}$, $(x+2y)^{10}$, a slanted parallelogram. Polar and spherical are just this process with u,v pre-chosen.

Worked (reverse): re-derive $dA=r\,dr\,d\theta$

1. NAME THE MAP

$u=r,\quad v=\theta$, and the substitution is $x=r\cos\theta,\quad y=r\sin\theta$

↓

2. THE FOUR PARTIALS

$x_r=\cos\theta,\quad x_\theta=-r\sin\theta,\quad y_r=\sin\theta,\quad y_\theta=r\cos\theta$

↓

3. THE DETERMINANT — WRITE IT BEFORE EXPANDING

$$\frac{\partial(x,y)}{\partial(r,\theta)}=\begin{vmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{vmatrix}=(\cos\theta)(r\cos\theta)-(-r\sin\theta)(\sin\theta)$$

↓

4. SIMPLIFY, THEN PYTHAGORAS

$$=r\cos^2\theta+r\sin^2\theta=r\left(\cos^2\theta+\sin^2\theta\right)=r\cdot1=r$$

↓

5. TAKE THE ABSOLUTE VALUE

$r\geq0$ always, so $|r|=r$ and

$$dA=r\,dr\,d\theta\ \blacksquare$$

Worked (reverse, 3D): re-derive $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$

1. BUILD THE 3×3 — COLUMNS ARE $\partial/\partial\rho,\ \partial/\partial\varphi,\ \partial/\partial\theta$

from $x=\rho\sin\varphi\cos\theta,\quad y=\rho\sin\varphi\sin\theta,\quad z=\rho\cos\varphi$:

$$\begin{vmatrix}\sin\varphi\cos\theta & \rho\cos\varphi\cos\theta & -\rho\sin\varphi\sin\theta\\ \sin\varphi\sin\theta & \rho\cos\varphi\sin\theta & \rho\sin\varphi\cos\theta\\ \cos\varphi & -\rho\sin\varphi & 0\end{vmatrix}$$

↓

2. EXPAND ALONG ROW 3 — IT CARRIES THE ZERO

$$=\cos\varphi\cdot M_1-(-\rho\sin\varphi)\cdot M_2+0\cdot M_3$$

(signs $+,-,+$ along the row. The third term dies on the 0)

↓

3. THE FIRST MINOR

$$M_1=\begin{vmatrix}\rho\cos\varphi\cos\theta & -\rho\sin\varphi\sin\theta\\ \rho\cos\varphi\sin\theta & \rho\sin\varphi\cos\theta\end{vmatrix}=\rho^2\sin\varphi\cos\varphi\cos^2\theta+\rho^2\sin\varphi\cos\varphi\sin^2\theta$$
$$=\rho^2\sin\varphi\cos\varphi\left(\cos^2\theta+\sin^2\theta\right)=\rho^2\sin\varphi\cos\varphi$$

↓

4. THE SECOND MINOR

$$M_2=\begin{vmatrix}\sin\varphi\cos\theta & -\rho\sin\varphi\sin\theta\\ \sin\varphi\sin\theta & \rho\sin\varphi\cos\theta\end{vmatrix}=\rho\sin^2\varphi\cos^2\theta+\rho\sin^2\varphi\sin^2\theta=\rho\sin^2\varphi$$

↓

5. ASSEMBLE

$$\cos\varphi\left(\rho^2\sin\varphi\cos\varphi\right)+\rho\sin\varphi\left(\rho\sin^2\varphi\right)=\rho^2\sin\varphi\cos^2\varphi+\rho^2\sin^3\varphi$$

↓

6. FACTOR, THEN PYTHAGORAS AGAIN

$$=\rho^2\sin\varphi\left(\cos^2\varphi+\sin^2\varphi\right)=\rho^2\sin\varphi$$

and $\sin\varphi\geq0$ on $0\leq\varphi\leq\pi$, so the absolute value changes nothing ■

Worked (forward): invent your own substitution — $\iint_T e^{\frac{x-y}{x+y}}\,dA$

T is the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.

1. WHY EVERY EARLIER TOOL FAILS

$e^{(x-y)/(x+y)}$ has no antiderivative in x or in y, so neither order starts. The order swap (§5.2) hits the same wall. The region is a triangle, so polar is useless. New move needed.

↓

2. LET THE INTEGRAND CHOOSE THE VARIABLES

the exponent is $\frac{x-y}{x+y}$, so set

$$u=x-y,\qquad v=x+y\qquad\Rightarrow\qquad \text{integrand}=e^{u/v}$$

↓

3. INVERT THE MAP — THE JACOBIAN NEEDS x,y IN TERMS OF u,v

add: $u+v=(x-y)+(x+y)=2x\Rightarrow x=\frac{u+v}{2}$

subtract: $v-u=(x+y)-(x-y)=2y\Rightarrow y=\frac{v-u}{2}$

↓

4. MAP THE BOUNDARY — ONE EDGE AT A TIME

edge $x=0$ *(from $(0,0)$ to $(0,1)$)*: $u=-y,\ v=y\Rightarrow u=-v$

edge $y=0$ *(from $(0,0)$ to $(1,0)$)*: $u=x,\ v=x\Rightarrow u=v$

edge $x+y=1$ *(the hypotenuse)*: $v=1$

so S is the triangle $0\leq v\leq1,\quad -v\leq u\leq v$ — Type I in v

↓

5. THE JACOBIAN

$x_u=\frac12,\quad x_v=\frac12,\quad y_u=-\frac12,\quad y_v=\frac12$:

$$\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}\frac12 & \frac12\\ -\frac12 & \frac12\end{vmatrix}=\frac12\cdot\frac12-\frac12\cdot\left(-\frac12\right)=\frac14+\frac14=\frac12$$

so $dA=\frac12\,du\,dv$

↓

6. BUILD THE INTEGRAL IN uv

$$\int_0^1 \int_{-v}^{v} e^{u/v} \cdot \frac{1}{2}\, du\, dv$$

↓

7. INNER — v IS FROZEN, SO $\frac{1}{v}$ IS A CONSTANT

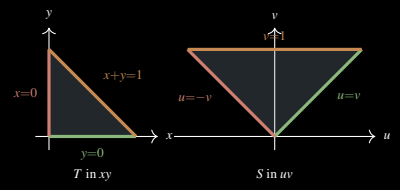
$\frac{d}{du} e^{u/v} = \frac{1}{v} e^{u/v}$, so the antiderivative of $e^{u/v}$ is $v\,e^{u/v}$:

$$\frac{1}{2}\int_{-v}^{v} e^{u/v}\, du = \frac{1}{2}\left[v\,e^{u/v}\right]_{-v}^{v} = \frac{v}{2}\left(e^1 - e^{-1}\right)$$

↓

8. OUTER

$$\frac{e-e^{-1}}{2}\int_0^1 v\, dv = \frac{e-e^{-1}}{2}\left[\frac{v^2}{2}\right]_0^1 = \frac{e-e^{-1}}{2}\cdot\frac{1}{2} = \frac{e-\frac{1}{e}}{4}$$



the same three edges, colour for colour: the map straightens the awkward hypotenuse into $v=1$ and turns the impossible exponent into $e^{u/v}$

The aha — the whole thread closes here. Polar and spherical were never special coordinate systems. They are two substitutions somebody already chose for you, and their r and $\rho^2\sin\varphi$ are just this determinant, evaluated once and memorised. When no stock system fits, read the integrand and invent the map. The Jacobian is the fee, and — as every rescue above showed — usually the payoff too.

Common pitfalls (change of variables)

- Take the absolute value of the determinant. A negative Jacobian only means the map flips orientation. Area and volume are never negative.
- Direction matters: $\frac{\partial(x,y)}{\partial(u,v)}$ converts $du\,dv$ into dA. If you built $\frac{\partial(u,v)}{\partial(x,y)}$ instead, you need its reciprocal — they multiply to 1.
- Convert the region, not just the integrand — map every boundary edge into the uv-plane. A perfect Jacobian over the wrong region is still the wrong answer.
- Invert the map before differentiating: the matrix wants x and y *as functions of* u and v. Solving $u=x-y$, $v=x+y$ for x and y is step zero, not an optional tidy-up.
- The determinant may still carry variables (r, $\rho^2\sin\varphi$) — it is a function, not a number, and it rides inside the integral.

Legend

Color & notes
functions $f, g, \sin$
variables x, y, n
numbers 0–9
operators $+ - =$
delimiters $( )\ [ ]$
titles in white
labels · results · pitfalls
$\boxed{u}$ / $\boxed{du}$ substitution

Greek
constants (fixed values):
$\pi \approx 3.14159$
γ fluid wt density
ρ weight per length
variable & operators:
θ angle (varies)
Δ "change in"
Σ sum of terms
notation, not values:
$\Theta\,\Omega\,\omega$ growth rate, like O

Calculus & relations
$\int$ integral
$\sum$ sum
lim limit
∞ infinity
$\to$ approaches
$\Rightarrow$ implies
$\Leftrightarrow$ iff
$\sqrt{\ }$ root
$\approx$ approx
$\neq\ <\ \leq\ >\ \geq$ compare
$\pm\ \mp$ plus / minus
$\cdot\ \times\ \div$ mult / div
$\circ$ compose

Vectors
$\langle\,\rangle$ components
$\|\vec{v}\|$ magnitude
$\vec{v}$, $\hat{v}$ vector, unit
$\mathbf{i}\,\mathbf{j}\,\mathbf{k}$ basis
$\vec{u}\cdot\vec{v}$ dot product
proj projection

Logic & sets
$\forall$ for all
$\exists$ exists
$\neg$ not
$\wedge\ \vee$ and / or
$\oplus$ xor / sym. diff
$\in\ \notin$ in / not in
$\subseteq\ \subset$ subset
$\cup\ \cap$ union / inter.
$\setminus$ set minus
$\equiv$ equiv / congruent
$|$ divides / $|x|$ / s.t.
$\lceil\ \rceil$ ceiling
$\binom{n}{k}$ choose
deg degree
mod modulo